Momentum in Financial Markets

A first-principles tutorial for quantitative practitioners

Abstract

Prices absorb news slowly, so an asset that has risen for several months tends to rise a little further. This chapter explains why momentum exists, how to measure it, which signals capture it, and how to test it so that chance results are not mistaken for real ones.

Momentum has no physical cause. Good news raises an asset's value, and the price reaches that value in steps. Investors learn the news at different times, and large funds take weeks to trade. If investors as a group overshoot, the price reverses later. One mechanism explains both continuation and reversal. Part of the profit may compensate for crash risk, and the size of that part is unknown. Researchers have established the effect itself across two centuries of data, dozens of countries and every major asset class. The horizon sets the sign. Returns over days to a month tend to reverse. Returns over 2 to 12 months tend to continue. Returns over 3 to 5 years tend to reverse again. The standard stock signal therefore uses the past 12 months and skips the most recent one. The variance ratio measures whether a market trends at a given horizon. A trend follower's profit is roughly proportional to it.

Most trend indicators are the same kind of quantity. Moving averages, crossovers, MACD and regression slopes are all weighted sums of past returns. At matched look-backs they agree closely. Three choices matter more than the formula: the look-back, the benchmark (zero, peers or a factor model), and scaling by volatility. Trend following wins only 30–45% of the time. It pays off in crises. Cross-sectional momentum crashes when markets rebound after a collapse. It lost about 90% in two months in 1932 and about 70% in 2009. Scaling positions down when volatility is high is the most effective protection against this.

The main risk in research is self-deception. The best of a thousand worthless strategies shows a Sharpe ratio near 1.2 by chance. A 10-year Sharpe ratio of 1.0 is consistent with any true value from 0.35 to 1.65. A hit rate of 51.6% is good, and 70% signals a bug, usually look-ahead. A random walk crosses its starting level only about 13 times in 250 days, so random prices look like trends. The chapter concludes that a realistic, fully costed program earns a Sharpe ratio of 0.4–0.8. A backtest above 2 indicates an error. Diversify across many markets, scale down when volatility rises, and count every configuration tested.

How to read this chapter. Sections 1–3 are conceptual and historical. §1 builds the model: momentum as incomplete adjustment of price to news, and the variance ratio as the statistic that measures it. §2 traces how three communities discovered the effect. §3 is the annotated reading list. Section 4 is the technical core. It surveys every serious way to *measure* momentum and describes each method with the same fields in the same order. Sections 5–7 cover engineering. §5 reduces the measures to one master form with four design choices. §6 covers implementation, and §7 covers evaluation. Sections 8 and 9 give the synthesis: current practice, a decision tree, and a staged roadmap for building a strategy.

Different readers can start in different places.

- Readers who want the model and its consequences should read §1, §5 and §9.1.
- Readers building a signal should read §4.0, §4.3, §5.2 and §6.
- Readers judging a backtest should read §6.10 and §7.

Appendix A defines every advanced concept that the main text uses briefly, starting from first principles. A.11 is the full glossary of symbols. Appendix B lists the works cited in the text beyond the reading list of §3.

Objectives. After this chapter, you should be able to:

- explain momentum as incomplete price adjustment, and derive momentum and reversal from one impulse response
- read a variance-ratio profile, and use it to say whether and at what horizon a market trends
- write any common momentum measure as a kernel, benchmark, normalizer and transform, and say which choices matter most
- separate momentum from trend, drift and volatility, and specify tests that do not confuse them
- normalize, size and combine momentum signals, and track the powers of volatility correctly
- evaluate a momentum backtest honestly, with corrections for overlapping data, multiple testing and costs.

Epistemic tags. The chapters in this collection flag claims by status:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement.
- **[Contested]** — documented, but with active disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. The evidence may be private or absent. A [Practice] claim is not a debunked claim.

Tags appear only where the status changes what a reader should do. A tag governs the sentence, list item or heading it opens. A few claims carry a qualified or compound tag, such as **[Fact, single study]** or **[Fact for the mechanism, Hypothesis for the magnitude]**, where a single tag would mislead.

Notation. The table lists the symbols that recur in the chapter, with the section that defines or first uses each. Symbols used in a single section are defined where they appear. Appendix A.11 gives the full glossary. Bold symbols are cross-sectional vectors over N assets.

Symbol	Meaning	Defined in
P_t; p_t	Price at time t; log price, $p_t = \ln P_t$	§1.2, §4.1
R_t; r_t	Simple return, $R_t = P_t/P_{t-1} - 1$; log return, $r_t = p_t - p_{t-1}$	§4.1.1, §4.1.2
$r_{a:b}$	**Windowed return**: the cumulative log return over the half-open window $(a, b]$, $r_{a:b} = \sum_{i=a+1}^{b} r_i = p_b - p_a$. So $r_{t-L:t}$ spans the L bars ending at t, and $r_{t:t+H}$ is the forward return over the next H bars	§4.1
μ; σ	Unconditional mean return; volatility	§1.1, §4.0
$\hat{\sigma}_t$	**Estimate**, formed from information available at t, of the *per-bar* return volatility (estimates carry hats)	§4.3.1
σ^*	Volatility target	§4.6.1, §6.7
ρ_k	Lag-k autocorrelation, Corr(r_t, r_{t-k})	§1.1, A.1
v_t; δ_t	Log of fundamental value; innovation to it	§1.2
Ψ_j; ψ_j	Cumulative impulse response; per-period impulse response, $\psi_j = \Psi_j - \Psi_{j-1}$	§1.2
$\mathcal{F}_t$	Information set available at t	§1.1
L; S; H	Lookback; skip; holding period, all in bars	§4.1.1
q; VR(q)	Aggregation horizon; variance ratio at that horizon	§1.2
N; T; A	Number of assets; bars of history; bars per year (252 daily, 12 monthly)	§4.0, §4.3.2
$s_{i,t}$; $w_{i,t}$	Signal; portfolio weight	§4.0
h_k; $b_{i,t}$; $\mathcal{N}_{i,t}$; $g(\cdot)$	Kernel, benchmark, normalizer and transform of the master form	§5.2
Hi$_t$; Lo$_t$	Bar high; bar low	§4.3.1, §4.4
IC; IR; TC	Information coefficient; information ratio; transfer coefficient	§7.3.1
M	Number of strategy configurations tried	§7.6
Q; V; Y	Order size; daily volume; impact coefficient	§1.3, §6.8
$Z(\cdot)$; $Z^{-1}(\cdot)$	Standard normal CDF; its inverse	§7.6.2
$\mathbb{1}\{\cdot\}$	Indicator	§4.2.3

The chapter writes bar extremes as Hi$_t$ and Lo$_t$ rather than H_t and L_t, because H and L already denote the holding period and the lookback. A few symbols have two meanings, because each use is standard in its own literature:

- α is an EMA smoothing constant in §4 and Jensen's alpha in §7.
- β is a regression slope in §4.2 and a factor loading in §4.6.4 and §7.4.4.

- λ is an EWMA decay in §4 and Kyle's price-impact coefficient in §2.6.
- $H(\omega)$ is a filter's transfer function in §4.8, and H is the observation matrix of a state-space model in §4.7.1.
- γ is a risk-aversion constant in §4.0 and the Euler–Mascheroni constant in §7.6.2. γ_k is an autocovariance, and γ_3, γ_4 are skewness and kurtosis.

Each section states which meaning applies where the symbol appears.

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Appendix A is a self-contained reference for every advanced concept the main text uses briefly. Each entry gives a plain-language definition first and a mathematical definition second. It also says which sections use the concept.

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1. What momentum is

1.1 The wrong intuition, and why it matters

The word "momentum" comes from mechanics, and the physical meaning misleads. A physical body in motion has a conserved quantity, mass times velocity, which persists unless a force acts on it. Prices have no such conserved quantity. They have no mass and no inertia. Nothing makes a price that rose yesterday keep rising today. Models built on the physical analogy assume persistence, and they fail at the first regime change.

The correct starting point is informational. Momentum reduces to one condition:

The conditional expectation of the next return is a function of past returns.

$$\mathbb{E}[r_{t+1} \mid r_t, r_{t-1}, ...] \neq \mathbb{E}[r_{t+1}]$$

Momentum is the claim that this function *increases* with past returns over some horizon. The claim concerns only the **joint distribution of returns at different lags**. It makes no claim about a force acting on price.

Two clarifications about that claim matter later. First, the claim concerns a *conditional first* moment, which is an expected return. Second, the *second*-moment structure of the process makes that conditional mean vary. The best linear forecast from one lag is $\mu + \rho_1 (r_t - \mu)$, so the autocovariances carry the predictability. Every estimator in §4 reads autocovariances to infer a conditional mean. Keeping the two moments distinct prevents a common error. The error is testing for one moment and claiming the other.

1.2 The conceptual model: incomplete adjustment

The model below generates almost all of momentum's observed behavior from one assumption. This chapter treats it as the right model to use throughout.

Work in logs throughout, so that returns add. Let v_t be the log of the "true" value of an asset. The true value is the price that would hold if every participant instantly knew everything and could trade at scale without friction. Suppose v follows a random walk with innovations δ_t:

$$v_t = v_{t-1} + \delta_t, \qquad \delta_t \sim \text{iid}(0, \sigma_\delta^2)$$

Under the strict Efficient Market Hypothesis, $p_t = v_t$, and returns are unpredictable. Now relax that assumption as little as possible. Let the price adjust only *partially* to each innovation, with the remainder arriving over later periods. Define the **cumulative impulse response** Ψ_j as the fraction of a value innovation that the price has impounded j periods after it arrives, with $\Psi_{-1} = 0$. The log price is then a distributed lag on value innovations:

$$p_t \;=\; c \;+\; \sum_{j\geq 0} \Psi_j \, \delta_{t-j}, \qquad \lim_{j\to\infty} \Psi_j = 1$$

Here c is a constant of integration, the level from which the innovations accumulate. The terminal condition $\Psi_\infty = 1$ says the price *eventually* reaches value, so no mispricing is permanent. If $\Psi_0 < 1$, the price reaches value gradually. The value process has the same form, $v_t = c + \sum_{j\geq 0} \delta_{t-j}$, with the same constant. That gives an equivalent and more intuitive form for the price:

$$p_t \;=\; \underbrace{v_t}_{\text{fundamental value}} \;-\; \underbrace{\sum_{j\geq 0} \left(1 - \Psi_j\right) \delta_{t-j}}_{\text{news not yet impounded}}$$

Price equals value minus the news that the price has not yet impounded. That second term is stationary, even though p_t and v_t are not. Stationarity needs $1 - \Psi_j \to 0$ fast enough to be square-summable. Any finite adjustment window satisfies this. So the model is well behaved: the *mispricing* is stationary, and the *price* is a random walk plus a stationary correction. Differencing gives the return as a moving average of the same innovations. The weights are the **per-period** impulse response $\psi_j \equiv \Psi_j - \Psi_{j-1}$:

$$r_t \;=\; p_t - p_{t-1} \;=\; \sum_{j\geq 0} \left(\Psi_j - \Psi_{j-1}\right) \delta_{t-j} \;=\; \sum_{j\geq 0} \psi_j \, \delta_{t-j}, \qquad \sum_{j\geq 0} \psi_j = 1$$

This is an MA(∞) process in δ, so its autocorrelations follow directly:

$$\rho_k \;=\; \frac{\sum_{j\geq 0} \psi_j \, \psi_{j+k}}{\sum_{j\geq 0} \psi_j^2}$$

The *sign pattern* of ψ determines the result. Pure under-reaction means the price only ever moves toward value and never past it, so $\psi_j \geq 0$ for all j. Every term in the numerator above is then non-negative. So $\rho_k \geq 0$ at every lag, and $\rho_k = 0$ beyond the adjustment period. The inequality is strict at every lag inside the adjustment window whenever ψ has contiguous support, which is the realistic case. A ψ with gaps, such as $(\frac{1}{2}, 0, \frac{1}{2})$, gives $\rho_1 = 0$ with $\rho_2 = \frac{1}{2}$. **A single serially uncorrelated shock to value, impounded gradually, produces autocorrelated returns.** Momentum is the *autocorrelation that incomplete adjustment produces*.

The mapping between adjustment and autocorrelation is exact. If adjustment is instantaneous ($\psi_0 = 1$, all other $\psi_j = 0$), then $\rho_k = 0$ for all k, and there is no momentum. Suppose instead that half the shock arrives immediately and half the next period ($\psi_0 = \psi_1 = \frac{1}{2}$). Then $\sum_j \psi_j^2 = \frac{1}{2}$ and $\sum_j \psi_j \psi_{j+1} = \frac{1}{4}$, which gives $\rho_1 = 0.5$ and $\rho_k = 0$ for $k \geq 2$. **The speed of adjustment and the return autocorrelation describe the same fact.** The mapping holds in general, but the matching numbers do not. The shock share and the first-lag autocorrelation coincide here only because $\psi_0 = \psi_1$ is exactly the configuration that maximizes ρ_1 for an MA(1). If 70% of the shock arrives immediately and 30% the next period, then $\rho_1 = 0.21/0.58 \approx 0.36$, not 0.7.

Two corollaries follow that most intuitions miss.

1. **Momentum must decay, but reversal needs more than that.** One might argue that the terminal condition $\Psi_\infty = 1$ forces reversal. It does not. It only forces momentum to be *transient*. If $\psi_j \geq 0$ throughout, the price rises toward value and never passes it. Then $\rho_k \geq 0$ out to the adjustment window and zero beyond, with no reversal at any horizon. Reversal requires a separate, empirically relevant assumption: **overshoot**, $\Psi_K > 1$ for some intermediate K. The terminal condition then forces the price back down, which puts $\psi_j < 0$ at longer lags. Overshoot explains less than it seems to, for two reasons. First, it does not make ρ_k negative lag by lag. Each ρ_k is a *sum of products* $\sum_j \psi_j \psi_{j+k}$,

so weights of mixed sign can still leave $\rho_k > 0$ at the very lags where the negative ψ's occur. Second, overshoot is necessary but not sufficient for *net* reversal. The variance-ratio identity in the next subsection makes this exact: net reversal means $\sum_j \psi_j^2 > 1$, and mild overshoot does not reach that level. Momentum and long-horizon reversal are two features of one impulse-response function. **[Fact]** Datasets that show 3–12 month momentum in equities generally also show 3–5 year reversal (De Bondt & Thaler, 1985).

2. **Momentum is horizon-specific by construction.** The sign of the autocorrelation depends on the horizon at which one samples the impulse response. So the same asset can be mean-reverting at 1 day, trending at 6 months, and mean-reverting at 4 years, with no contradiction.

The one diagnostic that unifies the chapter: the variance ratio

Every estimator in Section 4 ultimately estimates this quantity. Define the q-period **variance ratio** (Lo & MacKinlay, 1988):

$$\mathrm{VR}(q) \;=\; \frac{\mathrm{Var}(p_t - p_{t-q})}{q \cdot \mathrm{Var}(p_t - p_{t-1})} \;=\; 1 + 2\sum_{k=1}^{q-1}\left(1 - \frac{k}{q}\right)\rho_k$$

The second equality is the usual Bartlett-weighted expansion. The impulse response also gives the variance ratio directly, without the autocorrelations. Adopt $\Psi_j \equiv 0$ for $j < 0$. Then $p_t - p_{t-q} = \sum_{j\geq 0}\left(\Psi_j - \Psi_{j-q}\right)\delta_{t-j}$, and

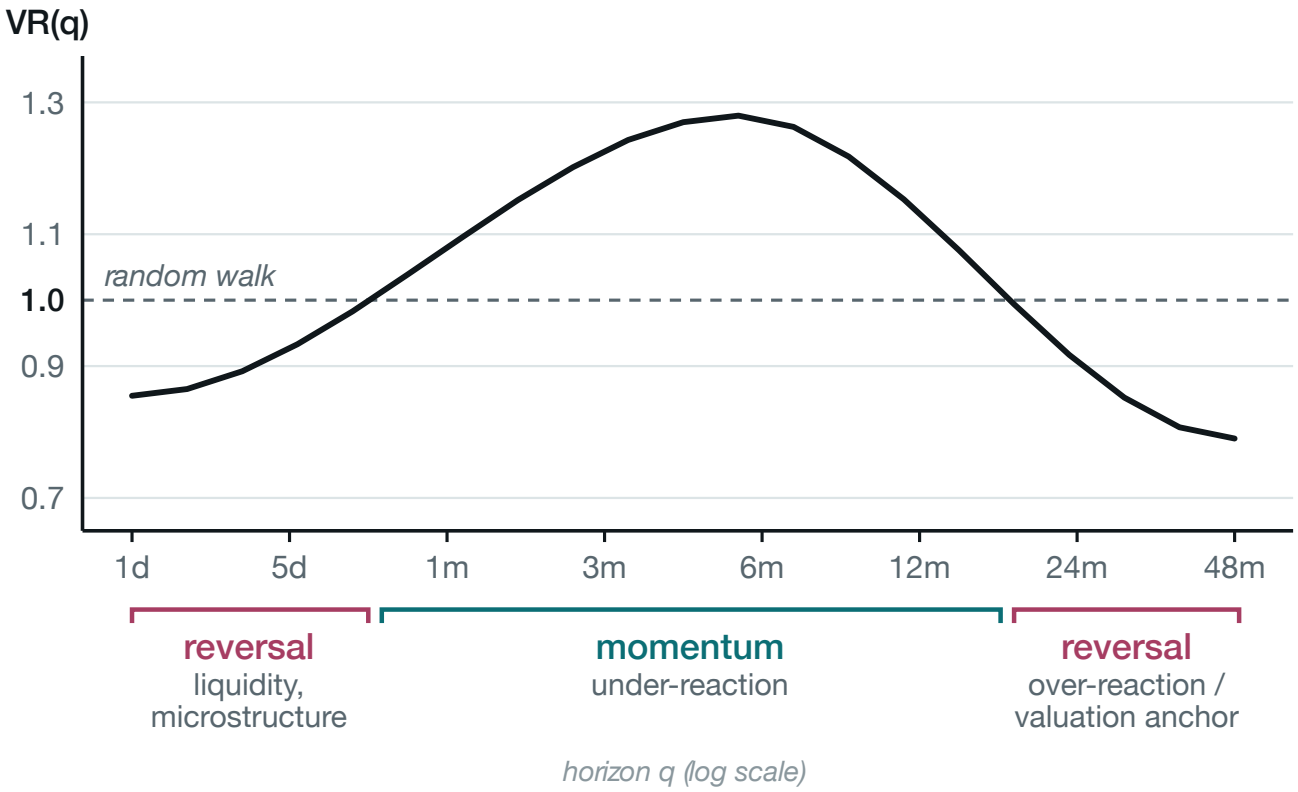
$$\mathrm{VR}(q) \;=\; \frac{\sum_{j\geq 0}\left(\Psi_j - \Psi_{j-q}\right)^2}{q\sum_{j\geq 0}\psi_j^2}, \qquad \mathrm{VR}(\infty) \;=\; \frac{1}{\sum_{j\geq 0}\psi_j^2}$$

The variance-ratio profile summarizes Ψ. The limiting case settles corollary 1 above. Under pure under-reaction, $\psi_j \geq 0$ and $\sum_j \psi_j = 1$. Together these give $\sum_j \psi_j^2 \leq \left(\max_j \psi_j\right)\sum_j \psi_j \leq 1$, hence $\mathrm{VR}(\infty) \geq 1$. **No amount of merely gradual adjustment can produce net reversal.** Only overshoot severe enough to push $\sum_j \psi_j^2$ above 1 can. The equality case $\sum_j \psi_j^2 = 1$ forces ψ to have a single nonzero value. The whole shock arrives in one period, possibly after a lag. That case is the random walk.

Under a random walk, $\mathrm{VR}(q) = 1$ for all q, because variance scales linearly with time. Departures from 1 have these meanings:

- $\mathrm{VR}(q) > 1$: the q-scale is **trending**. Prices diffuse faster than a random walk, and the net autocorrelation out to lag $q - 1$ is positive.
- $\mathrm{VR}(q) < 1$: the q-scale is **mean-reverting**.

The function $q \mapsto \mathrm{VR}(q)$, the *variance-ratio profile*, is the single most informative descriptive statistic about momentum in a series. It shows *at what horizon* the asset trends, and that horizon directly sets the lookback. [Practice] **Recommendation: when evaluating an unfamiliar market, study its variance-ratio profile before any backtest equity curve.** The figure shows a schematic profile.



Schematic variance-ratio profile for a typical liquid equity. ***[Fact]*** *The three-regime shape is robust across US equities, international equities, and many futures markets: short-horizon reversal, intermediate-horizon trend, and long-horizon reversal. The crossover points differ by asset class and era.*

The variance ratio also gives the cleanest available statement of what a trend-following P&L is. To leading order, the expected P&L of a canonical trend rule on a single asset is proportional to a difference of two variances. Dao, Nguyen, Deremble, Lempérière, Bouchaud & Potters (2017) show this result. One is the asset's variance measured at the trend's timescale. The other is its variance measured at the rebalancing timescale:

$$\mathbb{E}[\text{trend P\&L}] \;\propto\; \underbrace{\sigma^2_{\text{long horizon}} - \sigma^2_{\text{short horizon}}}_{\propto\, \mathrm{VR}(q)-1}$$

Both variances here are **per unit of time**: $\sigma^2_{\text{long horizon}} = \mathrm{Var}(p_t - p_{t-q})/q$ and $\sigma^2_{\text{short horizon}} = \mathrm{Var}(p_t - p_{t-1})$. Written this way, the link to the variance ratio is immediate, since $\mathrm{VR}(q) - 1 = (\sigma^2_{\text{long horizon}} - \sigma^2_{\text{short horizon}})/\sigma^2_{\text{short horizon}}$. Comparing raw, unnormalized variances at two horizons would be uninformative, because the longer one is mechanically larger.

The underbrace needs care. Rearranging the identity gives $\sigma^2_{\text{long horizon}} - \sigma^2_{\text{short horizon}} = \sigma^2_{\text{short horizon}}\,(\mathrm{VR}(q) - 1)$. So the P&L of a *raw* trend rule tracks $\mathrm{VR}(q) - 1$ only at fixed $\sigma^2_{\text{short horizon}}$. The proportionality fails across assets, and across time for one asset whose volatility changes. The normalization that every practitioner already applies removes the extra factor. The raw rule carries two powers of $\sigma_{\text{short horizon}}$. One is in the signal, which is a past return. The other is in the return that the signal multiplies. Standardizing the signal by its own volatility *and* sizing the position at $1/\sigma_{\text{short horizon}}$ divides out both powers. The result is $\mathbb{E}[\text{P\&L}] \propto \mathrm{VR}(q) - 1$ itself. So the **fully volatility-scaled** trend rule, not the raw rule, is long the variance ratio with no extra factor. That is the version of the statement to use.

A trend follower is structurally long the variance ratio. This sentence explains three facts. It explains the convexity of CTA returns. It explains why trend does well in dispersive crises and badly in choppy ranges. It also explains why "trend following is a long straddle" (Fung & Hsieh, 2001) is more than an analogy.

1.3 Why momentum exists despite the Efficient Market Hypothesis

A first clarification resolves much of the confusion in this debate. Fama (1970) states the EMH as the claim that prices reflect information *given a model of equilibrium expected returns.* He does not state it as the claim that returns are unpredictable.

Any test of efficiency is therefore a **joint test** of (a) efficiency and (b) the assumed asset-pricing model. If momentum earns positive average returns, either markets are inefficient or the model of risk is wrong. Return data alone cannot tell which. This is the **joint hypothesis problem**. It explains why the momentum debate has run for 30 years without resolution.

A second clarification concerns the version of the EMH that says no one earns excess returns without bearing risk or cost. That version says competition should eliminate risk-adjusted excess returns net of costs. It never claimed that they vanish *instantly* or *completely*. The claim concerns limits. Grossman & Stiglitz (1980) made this precise. If prices were fully revealing, no one would pay to gather information. So in equilibrium, prices must be *slightly* inefficient, by exactly enough to compensate information gathering. Momentum is one form of that inefficiency.

Four families of explanation compete within this framing. They are not mutually exclusive. The defensible position is that all four contribute, in proportions that no one has measured.

(A) Slow information diffusion — [Hypothesis, well-formalized]

Hong & Stein (1999) build a model with two types of agent. "Newswatchers" trade on private fundamental signals that diffuse gradually across the population. "Momentum traders" condition only on past prices. Gradual diffusion alone generates under-reaction, and hence momentum. Momentum traders who arbitrage the under-reaction then necessarily generate *over*-reaction at longer horizons, and hence reversal. One friction produces the full momentum-then-reversal impulse response.

[Fact] Momentum is stronger in stocks with low analyst coverage and small size (Hong, Lim & Stein, 2000). This supports the explanation. Post-earnings-announcement drift (Bernard & Thomas, 1989, 1990) is a clean case in which the price impounds information over weeks after a *public* announcement. Chan, Jegadeesh & Lakonishok (1996) show that momentum and PEAD are related but not identical.

(B) Behavioral biases — [Hypothesis, contested]

Three canonical models, all published in 1998–1999, produce momentum and reversal from different psychological mechanisms:

- **Barberis, Shleifer & Vishny (1998)**: conservatism, which is under-reaction to individual signals, plus the representativeness heuristic, which is over-extrapolation of streaks. Investors are slow to update, and then they over-extrapolate.
- **Daniel, Hirshleifer & Subrahmanyam (1998)**: overconfidence in private signals plus biased self-attribution. Confirming public news inflates confidence more than disconfirming news deflates it. This drives continued over-reaction, which is momentum, followed by correction.
- **Grinblatt & Han (2005)** and **Frazzini (2006)**: the **disposition effect**. Investors holding gains sell too early, and investors holding losses hold too long. This creates a supply overhang above the reference price, the aggregate cost basis, which *slows* upward adjustment to good news. Momentum should then be predictable from unrealized capital gains, and these papers find exactly that. This chapter treats this model as the strongest of the three, because it makes a sharp auxiliary prediction that the data confirm independently of returns.

The standard critique of behavioral explanations is fair. They are flexible enough to explain almost any pattern after the fact, and the profession has not agreed on which bias dominates.

(C) Order flow and market microstructure — [Fact for the mechanism, Hypothesis for the magnitude]

This explanation appeals most to engineers, because it needs no psychology at all, only the mechanics of execution.

Institutions cannot execute a large position instantaneously without paying ruinous impact. So they slice a **metaorder** into child orders executed over hours, days or weeks. Two robust microstructure facts follow:

1. **Order flow has long memory.** The sign sequence of market orders is positively autocorrelated. Its autocorrelation function decays slowly, as a power law, out to thousands of trades (Lillo & Farmer, 2004; Bouchaud, Gefen, Potters & Wyart, 2004). This is direct evidence of order splitting.
2. **The square-root law of impact.** Consider a metaorder of size Q in a market with daily volume V and volatility σ. Its expected price impact is approximately

$$\Delta p \;\approx\; Y\sigma\sqrt{Q/V}, \qquad Y = O(1)$$

   The units are easy to get wrong. Q and V use the *same* units, both in shares or both in currency, so Q/V is the dimensionless participation fraction. σ is the *daily* return volatility. So Δp is a **relative** price move, not a currency amount. Empirically, $Y \approx 0.5$–1. This concave, roughly universal relation holds across markets, asset classes and decades. Bouchaud, Bonart, Donier & Gould (2018) give the synthesis. The relation reappears in §6.8 as the binding constraint on capacity.

Together, these facts produce momentum. A persistent, one-directional flow that takes weeks to complete pushes the price persistently in one direction. Anyone who detects that flow earns momentum returns. Part of the impact also **decays** after the metaorder finishes. This again produces momentum followed by partial reversal.

A second microstructure channel is *mechanical flow*. Index inclusion, the rebalancing of risk-parity and volatility-target funds, option dealer hedging (gamma imbalance), and trend followers themselves all generate order flow that depends on price. Such flow is a positive feedback loop by construction.

(D) Risk-based / rational explanations — [Contested, and improving]

If momentum portfolios are riskier in a way that standard models miss, momentum is not a puzzle. Early versions of this argument were weak. Momentum's CAPM beta is near zero, and momentum survives the Fama–French three-factor adjustment. That is why Carhart (1997) added momentum as a fourth factor rather than explaining it with existing factors. The modern versions are much stronger:

- **Conditional betas.** Kelly, Moskowitz & Pruitt (2021, *JFE* 140) use instrumented principal components. They show that past-return characteristics predict future *realized betas*, and that time-varying conditional risk exposures explain a sizable fraction of momentum and long-term reversal returns. This is currently the most serious rational challenge.
- **Momentum's dynamic beta.** Daniel & Moskowitz (2016) and Geczy & Samonov (2016) document that the momentum portfolio's market beta varies systematically with the market state. It turns sharply negative after bear markets. The momentum premium therefore partly compensates for a *conditional* crash exposure, not an unconditional one.
- **Real options / growth-rate risk.** Berk, Green & Naik (1999) and Johnson (2002) show that firms whose expected growth rate has risen mechanically have both higher past returns and higher risk. That combination generates momentum in equilibrium.

The honest summary

[Fact] Momentum returns exist out of sample, across asset classes, countries and two centuries. **[Contested]** Researchers have not settled why they exist. [Practice] **Recommendation: treat the microstructure and flow explanation as the primary mechanism, because it is directly observable.** Treat slow diffusion and the disposition effect as strong secondary channels. Treat the risk-based explanations as a warning that some apparent "alpha" is a conditional beta that has not been measured. The unmeasured conditional beta is the reason momentum crashes.

1.4 Momentum is not: trend, drift, volatility, mean reversion, acceleration, or regime

Writers conflate these concepts constantly, including in published work. The table separates them.

Concept	Formal object	What it is about	Confusable with momentum because…
Momentum	$\mathbb{E}[r_{t+1} \mid r_{t-L:t}]$ increasing in past returns; for a linear forecast this is $\rho_k > 0$	*Conditional first moment*, driven by second-moment (autocovariance) structure	—
Trend	A low-frequency component τ_t in a decomposition $p_t = \tau_t + c_t + \varepsilon_t$	A property of the *price path*, an unobserved state	Trend-followers profit from both, and the P&L does not distinguish them
Drift	$\mu = \mathbb{E}[r_t]$, unconditional	A constant, not a prediction	A positive μ makes naive momentum tests look significant
Volatility	$\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$	*Conditional second moment*	Both are persistent, and volatility clustering is far stronger than return autocorrelation
Mean reversion	$\rho_k < 0$ at the relevant k	Same object, opposite sign	Literally the same statistic at a different horizon
Acceleration	$\partial^2 p / \partial t^2$; change in momentum	Second derivative of the level	"Momentum of momentum"; very low signal-to-noise
Regime	Latent state $S_t \in \{1..K\}$ modulating parameters	A *parameter* of the above, not a quantity	Regimes make momentum appear and disappear

Three of these distinctions deserve more space, because confusing them causes real losses.

Trend vs. momentum: the sharpest distinction in this document

Consider a price process that is a deterministic upward line plus iid noise:

$$p_t = \mu t + \varepsilon_t, \qquad \varepsilon_t \sim \text{iid}(0, \sigma_\varepsilon^2)$$

This series has a **perfect trend** and **negative** return autocorrelation. The return is $r_t = \mu + \varepsilon_t - \varepsilon_{t-1}$, an MA(1) with a unit negative coefficient. Its autocorrelation is $\rho_1 = -\sigma_\varepsilon^2 / 2\sigma_\varepsilon^2 = -1/2$, with $\rho_k = 0$ beyond. So the series has a trend and *anti*-momentum: an up move predicts a down move. Conversely, a process $r_t = \phi r_{t-1} + u_t$ with $\phi > 0$ and zero mean has momentum but no trend in the level sense. Its level has no systematic direction.

Trend concerns the level, and momentum concerns the returns. A trend-following *rule*, such as "long when $P_t > \mathrm{MA}_L(P)_t$", does not distinguish them. It profits from drift, trend *and* return autocorrelation alike. That is fine for making money and bad for research. A profitable trend backtest does not show which of the three is present, and the three have completely different stability properties. Drift is stable and carries little information. Return autocorrelation is fragile and carries a lot.

This distinction has a precise consequence in Section 7. **[Contested]** Huang, Li, Wang & Zhou (2020, *JFE* 135) argue that the unconditional mean confounds the canonical time-series momentum test. The test regresses r_{t+1} on $\mathrm{sign}(r_{t-12:t})$. But $\mathbb{E}[r_{t+1} \cdot \mathrm{sign}(r_{t-12:t})]$ is positive whenever $\mu > 0$, even with *zero* predictability. After controlling for the unconditional mean, they find little evidence of an absolute time-series momentum effect in their sample. Moskowitz, Ooi & Pedersen (2012) and later replications dispute the strength of this critique. **The disagreement is active and important. It determines how researchers must specify any new test.**

Volatility vs. momentum

Decompose $r_t = \mu_t + \sigma_t \epsilon_t$, with ϵ_t standardized. Momentum is a claim about μ_t. Volatility clustering is a claim about σ_t. Empirically, **[Fact]** the persistence in σ_t is an order of magnitude stronger and more reliable than the persistence in μ_t. Daily $|r_t|$ has an autocorrelation of 0.2–0.4 at lag 1, decaying slowly, while daily r_t has an autocorrelation near zero. Two consequences follow:

1. Volatility is the *easy* prediction problem. Every momentum measure that divides by an estimate of σ uses the easy problem to improve the hard one.
2. A raw return signal mixes direction with volatility. A large past return may indicate direction, which is momentum, or merely high volatility. Normalizing separates the two. This is the single highest-value transformation in Section 4.

Regime as a parameter, not a phenomenon

"Momentum works in trending regimes" is nearly a tautology. The useful version says that the *parameters* are state-dependent: the sign and size of ρ_k, the optimal lookback, and the level of volatility. Cooper, Gutierrez & Hameed (2004) show **[Fact]** that momentum profits in US equities are concentrated after positive market states, and are near zero or negative after negative ones. Daniel & Moskowitz (2016) refine this: momentum crashes occur in panic states, with high volatility and a rebounding market.

In practice, regime conditioning is not optional. It separates a strategy with a −70% drawdown from one without.

1.5 Momentum across time scales

Momentum takes different forms at different horizons. Each horizon has its own sign, cause, capacity and decay rate. The table lists them.

Horizon	Dominant effect	Mechanism	Capacity	Notes
Sub-second to minutes	**Momentum** in order flow; price near-efficient	Order splitting, queue dynamics, latency arbitrage	Very low	Flow is predictable, but *price* is much less predictable because market makers offset it
Minutes to hours	Mixed; intraday momentum at specific times	Metaorder execution, VWAP/close flows	Low	**[Fact]** "Intraday momentum": the first half-hour return predicts the last half-hour return (Gao, Han, Li & Zhou, 2018)
1 day – 1 month	**Reversal** (cross-sectional)	Compensation for liquidity provision; bid-ask bounce	Medium	Jegadeesh (1990), Lehmann (1990). The classic momentum signal *skips* this month for exactly this reason
2 – 12 months	**Momentum** — the classic effect	Under-reaction, flow, disposition effect	High	Jegadeesh & Titman (1993). The 12-2 or 12-1 signal is the canonical form
1 – 3 years	Weak / transition	—	—	Signal largely absent
3 – 5 years	**Reversal**	Over-reaction correction, valuation anchoring	High	De Bondt & Thaler (1985); this is where the value effect operates

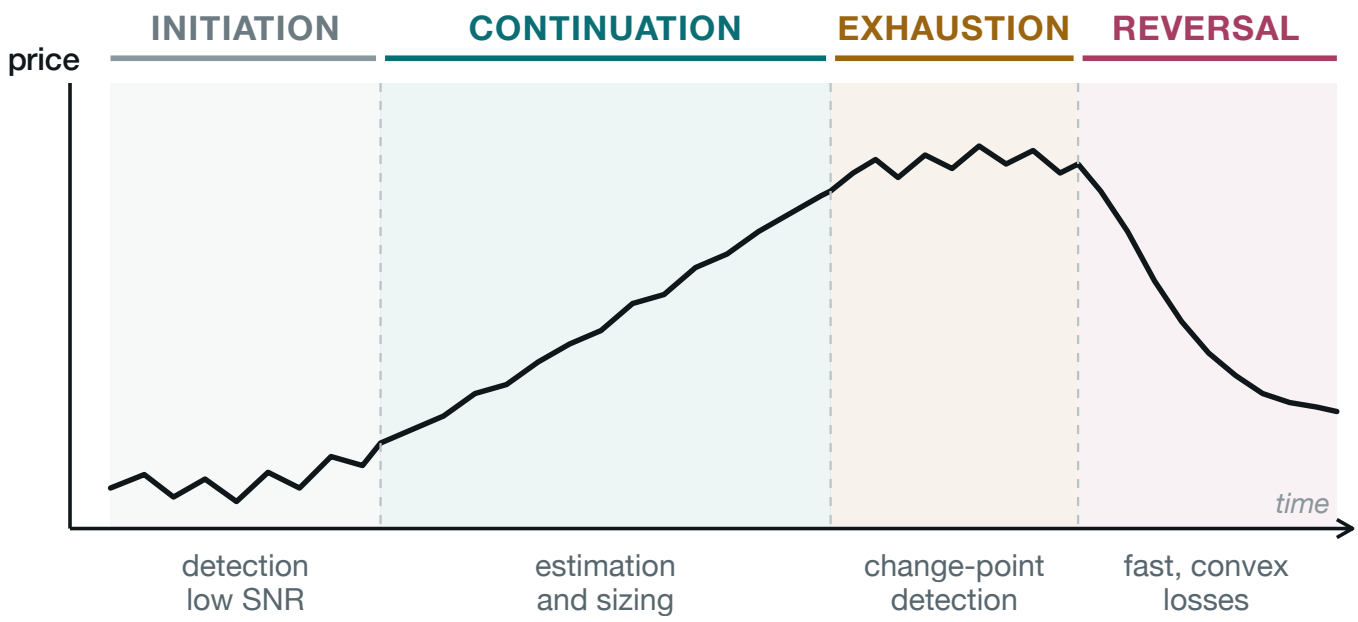
Two refinements matter.

Intermediate-horizon momentum, or "echo". [Contested] Novy-Marx (2012, *JFE* 103) shows that in US equities, returns from $t-12$ to $t-7$ predict future returns *better* than returns from $t-6$ to $t-2$. On this evidence, momentum is an "echo" and does not decay smoothly. The finding fits poorly with any simple under-reaction explanation, and it is not uniformly robust across markets and later samples. Treat it as a real feature of US equity data. Its generality is unsettled.

The time scale depends on the asset class. [Fact] Futures trend following works well at lookbacks of 1–12 months (Moskowitz, Ooi & Pedersen, 2012; Hurst, Ooi & Pedersen, 2017). Currency momentum is weaker and more prone to crashes (Menkhoff, Sarno, Schmeling & Schrimpf, 2012). Commodity momentum interacts strongly with the term structure and carry (Erb & Harvey, 2006). Do not port an equity lookback to a futures book without re-deriving it.

1.6 The life cycle of a trend

This section carries the label **[Practice / Hypothesis]** on purpose. The four-phase description below is a useful organizing device with partial empirical support. It is not an established taxonomy. The description is worth stating precisely because it shows *which statistical problem each phase poses.* The figure and table set out the four phases.



Phase	Signal quality	Volatility	What to do
Initiation	weak, ambiguous; many false positives	rising	small size, wide stops
Continuation	strong, stable; $\rho_k > 0$	moderate, clustered	full size, vol-target
Exhaustion	strong but decaying; acceleration < 0	rising	reduce, tighten vol scaling
Reversal	inverts abruptly	spikes	be flat or reversed

Initiation. The statistical problem is *detection*, which means distinguishing the onset of a persistent drift from noise. Signal-to-noise is lowest here, because by construction the new state has few observations. Any detector trades off Type I against Type II error, and the lookback length sets that trade-off. Short lookbacks detect early, with many false positives. Long lookbacks detect late, with few. No parameter resolves the trade-off. Only the trader's cost structure can say which point on the trade-off curve to choose. **[Fact]** The distribution of trend-following trade P&L is heavily right-skewed, with a low hit rate, typically 30–45%. Most detections are false, and the strategy stops them out cheaply. A minority of trades produce the profit that covers the losses of all the others.

Continuation. The statistical problem is *estimation and sizing*. Given that a trend exists, the questions are how large the drift is relative to volatility and how much risk the position should take. Autocorrelation is positive, and volatility clusters and is comparatively predictable. Volatility-scaled position sizing (§4.3.1, §6.7) contributes most in this phase.

Exhaustion. The statistical problem is *change-point detection*. It is the hardest of the four. Candidate observable markers **[Practice, weakly supported]** are:

- rising volatility with flat or declining absolute price progress, which means a deteriorating "efficiency ratio"
- negative acceleration while momentum remains positive
- divergence between price extremes and oscillator extremes
- crowding measures, such as positioning data, factor-return correlation and dealer gamma.

None of these is a robust standalone signal in the published evidence. Treat exhaustion detection as risk *reduction*, not as a reversal trade.

Reversal. Momentum's losses are not symmetric with its gains. **[Fact]** Daniel & Moskowitz (2016) document "momentum crashes". In panic states, following market declines and with high volatility, the momentum portfolio's conditional beta turns sharply negative. The "loser" leg concentrates in high-beta distressed names. When the market rebounds, the prices of the short leg rise sharply. The canonical episodes are July–August 1932, when the momentum strategy lost roughly 90% in two months, and March–May 2009, when US equity momentum lost roughly 70% or more. The payoff structure resembles being **short a call option on the market, conditional on being in a panic state**.

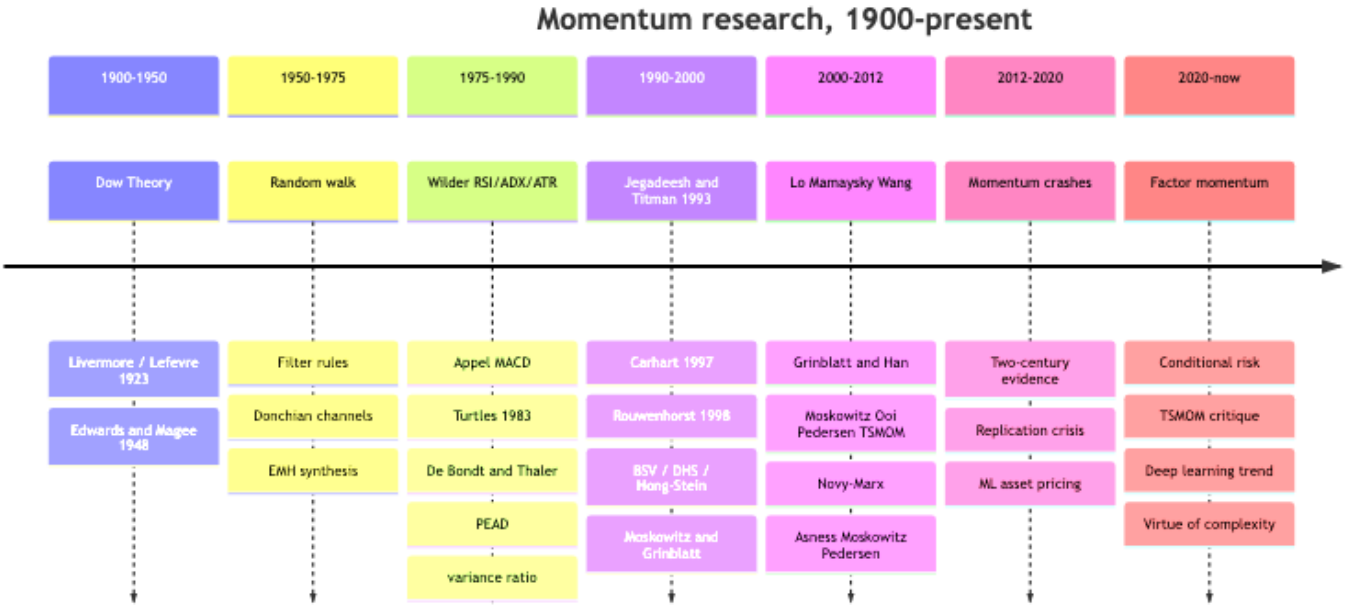
This is why the modern treatment of momentum cannot separate it from risk management. Risk management is part of the strategy, and it changes the strategy's fundamental character.

§1 Key takeaways

1. Momentum is a statement about the **conditional distribution of returns given past returns**, not a force. Do not use the physics analogy.
2. Momentum arises from **incomplete price adjustment**. One assumption, that prices impound information gradually, generates momentum, later reversal and horizon dependence together.
3. The **variance-ratio profile** $\mathrm{VR}(q)$ is the unifying diagnostic. Trend-following P&L is structurally long $\mathrm{VR}(q) - 1$.
4. The EMH does not forbid momentum. Tests of efficiency are joint tests with an asset-pricing model, and Grossman–Stiglitz guarantees a residual inefficiency. Four families of explanation have support: slow diffusion, behavioral bias, order flow and conditional risk. None is decisive.
5. **Trend ≠ momentum ≠ drift.** A trend rule profits from all three, so a profitable backtest is not evidence of predictability. Specify tests that separate them (see the critique by Huang et al.).
6. The sign changes with horizon: reversal at days, momentum at 2–12 months, reversal at 3–5 years. Skipping the most recent month avoids a known effect of opposite sign.
7. Momentum's loss distribution is **conditionally crash-prone**, driven by a beta that changes sign in panic states. Risk management is part of the strategy.

2. Historical evolution of momentum research

Three communities discovered momentum independently and had little contact with each other: chart-reading speculators, systematic futures traders, and academic financial economists. Their work converged only in the 2010s. The history is worth reading in order, for two reasons. Each community found something the others missed, and most "new" ideas in this space are rediscoveries. The timeline gives the overview.



2.1 Era I — Chart reading and the birth of the trend concept (1900–1950)

Dow Theory (Charles Dow's editorials 1899–1902; codified by Hamilton, 1922; Rhea, 1932)

Contribution. Dow Theory was the first systematic description of markets as having *nested trends* at several time scales: primary (years), secondary (weeks to months) and minor (days). It also stated the first explicit continuation principle: the

theory assumes a trend remains in force until a definite reversal signal appears. It also introduced *confirmation*: a signal in one index (the Industrials) counted only when another index (the Rails) corroborated it.

What changed. It reframed price behavior from "a series of unrelated quotations" to "a state that persists." That framing of trend as a latent regime, with rules for detection and confirmation, is the direct precursor of every regime-switching model in §4.7.2.

Limitations. It is purely qualitative, no test can falsify it as stated, and it depends on the interpreter. Rhea's codification was retrospective.

Lasting influence. The influence is substantial and, unusually for technical analysis, partly supported by evidence. Brown, Goetzmann & Kumar (1998, *Journal of Finance*) reconstructed Hamilton's actual *Wall Street Journal* market calls from 1902–1929. They found that the calls generated positive risk-adjusted returns relative to a buy-and-hold benchmark. **[Fact, single study]** The decomposition into several time scales remains standard practice.

Jesse Livermore / Edwin Lefèvre, *Reminiscences of a Stock Operator* (1923); Livermore, *How to Trade in Stocks* (1940)

Contribution. These books state the practitioner's rules. Add to positions as they gain ("pyramiding into strength"). Cut losses fast. Follow the "line of least resistance", which means trading breakouts from consolidation ranges. Livermore also tied position size to confirmation.

What changed. They established the discipline of an *asymmetric* payoff, in which rare, large gains pay for small, frequent losses. That asymmetry later became the defining statistical property of trend following. It is the practitioners' discovery of positive skew, 40 years before anyone measured it.

Limitations. The evidence is anecdotal and survivorship-biased. Livermore's own record ended in catastrophe, and the book is a novelization.

Lasting influence. The cultural influence is enormous. The rules for cutting losses and pyramiding are essentially a discrete approximation to volatility-scaled position sizing.

Edwards & Magee, *Technical Analysis of Stock Trends* (1948)

Contribution. The book is a systematic catalogue of chart patterns, support and resistance, and trendline construction. It was the first attempt to make chart reading reproducible.

What changed. It made technical analysis *teachable*, and so scalable to a profession.

Limitations. It has no statistical validation. The pattern definitions are a clear case of fitting patterns after the fact, with enormous researcher degrees of freedom. Most of the specific patterns have never survived rigorous testing.

Lasting influence. The influence is mixed. Researchers have largely discredited the *pattern catalogue* as a source of edge. Some of the *concepts* have partial modern support: support and resistance as reference-price effects, consolidation followed by breakout, and volume confirmation. These connect to the disposition effect and to the liquidity structure of the order book. Lo, Mamaysky & Wang (2000, *JF*) later gave the pattern-recognition program its only serious statistical treatment (see §2.4).

2.2 Era II — The random walk and the first quantitative tests (1950–1975)

Kendall (1953), Osborne (1959), Samuelson (1965), Fama (1965)

Contribution. This work showed empirically that stock price changes are approximately serially uncorrelated. Samuelson added a theoretical proof that *properly anticipated prices fluctuate randomly*. On his account, unpredictability is a *consequence* of rational forecasting, not an accident.

What changed. The academy declared technical analysis worthless, and the practitioner and academic communities split for 30 years.

Limitations. The work had two limitations, and both had consequences. First, the tests had low power. They mostly examined the lag-1 autocorrelation of daily returns, which is exactly the lag at which momentum does not appear. A predictable component of 5% a year is economically enormous and statistically almost invisible at daily lag 1. Second, Samuelson's theorem says that returns are unpredictable *relative to the correct risk-adjusted discount rate*. That differs from returns being unpredictable outright.

Lasting influence. The random walk remains the correct null hypothesis, and this is its real legacy. It forced researchers to state every later claim as the rejection of a specific null with a specific test statistic.

Alexander (1961) filter rules; Fama & Blume (1966)

Contribution. These were the first rigorous backtests. Alexander's "x% filter" buys after a rise of x% from a low and sells after a fall of x% from a high. It is a trend-following rule, and Alexander found it profitable. Fama and Blume re-examined it with correct handling of transaction costs and dividends, and the profits vanished.

What changed. Practitioners learned, or should have learned, that **costs, not signals, determine viability**. Academics learned that they could test mechanical rules. The sequence from Alexander to Fama and Blume is the first instance of the field's dominant recurring pattern. The pattern has three steps: a promising rule, a correction for costs and methodology, and a much smaller residual.

Limitations. The work covers a single market and a short sample, with no multiple-testing control on the filter width.

Lasting influence. The methodological influence is enormous. Every backtest since has taken its structure from this exchange.

Fama (1970), "Efficient Capital Markets"

Contribution. Fama introduced the weak, semi-strong and strong taxonomy. More importantly, he stated the **joint hypothesis** problem.

Lasting influence. The paper still defines the terms of the argument. Any claim that "momentum is an anomaly" really says that "momentum is unexplained by the models tried so far."

2.3 Era III — Trend following becomes an industry (1950–1990)

This strand developed almost entirely outside the academy, in commodity futures.

Richard Donchian (1950s–1970s)

Contribution. Donchian published the first *mechanical* trend systems: the 5/20 dual moving-average crossover and the N-day channel breakout, now called the "Donchian channel". He ran what most observers regard as the first publicly offered managed futures fund (1949).

What changed. Trend following became a *rule* rather than a judgment. A rule can be fully specified and audited, so an institution can run it at scale.

Limitations. The systems had no risk model, no portfolio construction, no cost analysis and no statistical validation. Donchian chose parameters by inspection.

Lasting influence. The moving-average crossover and the channel breakout remain the two most-used trend primitives in the world. §4.1.5 and §4.4.3 formalize them.

The Turtle experiment (Dennis & Eckhardt, 1983)

Contribution. The experiment was a natural test of whether anyone could teach trading. Richard Dennis taught novices a fully specified breakout system, with Donchian-style entries, position sizing based on ATR (average true range), and limits on total portfolio risk. Several "turtles" produced strong multi-year records.

What changed. The experiment showed that the edge came from **the system and the risk sizing**, not from the trader. The Turtle rules were also the first widely disseminated system that set position size explicitly $\propto 1/\text{ATR}$. That is volatility-scaled sizing, a decade before it appeared in the academic literature.

Limitations. The experiment had no control group, a small n, and survivorship among the reported results. It ran in the 1980s commodity markets, which were unusually favorable to trend. Later decades were far less favorable.

Lasting influence. The influence on practice is very large. Sizing normalized by ATR and per-position risk budgeting are now universal.

Welles Wilder, *New Concepts in Technical Trading Systems* (1978); Gerald Appel, MACD (late 1970s)

Contribution. Wilder introduced RSI, ADX/DMI, ATR and Parabolic SAR in one book. Few books have introduced so many durable primitives. Appel introduced MACD.

What changed. These tools gave practitioners a set of bounded, comparable, normalized quantities. ATR, a volatility estimate, and RSI, a bounded oscillator, were early solutions to problems that the academy later formalized as volatility scaling and cross-sectional standardization.

Limitations. Wilder and Appel chose all parameters by inspection, with no validation. The smoothing constants, such as Wilder's 14 periods and MACD's 12/26/9, are arbitrary, and two generations of retail traders have overfit them. Wilder's own theoretical justifications are ad hoc.

Lasting influence. Academics underrate these tools, and retail traders overrate them. Without the folklore around them, several are respectable statistics. ADX is a normalized measure of directional consistency. ATR is a range-based volatility estimator, and range estimators are genuinely more efficient than close-to-close ones (Parkinson, 1980; Garman & Klass, 1980; Yang & Zhang, 2000). MACD is a band-pass filter. See §4.

Fung & Hsieh (2001), "The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers"

Contribution. Fung and Hsieh showed that portfolios of **lookback straddles**, options that pay the maximum price range over a period, replicate CTA returns well.

What changed. Fung and Hsieh correctly reclassified trend following as an *option-like payoff* rather than a return-predicting strategy. This explains its positive skew, its convexity relative to equities, and its "crisis alpha".

Lasting influence. The paper is foundational. It underlies the modern framing of trend as a portfolio hedge, and it anticipates the variance-difference decomposition of Dao et al. (2017).

2.4 Era IV — The academic momentum revolution (1985–2000)

De Bondt & Thaler (1985), "Does the Stock Market Overreact?"

Contribution. De Bondt and Thaler documented that losers over 3–5 years subsequently outperform winners over 3–5 years. This is long-horizon **reversal**.

What changed. It was the first credible modern rejection of weak-form efficiency using past prices alone, and it launched behavioral finance.

Limitations. Chan (1988) and Ball & Kothari (1989) contest its risk adjustment and size effects.

Lasting influence. It established the *long* end of the momentum-reversal spectrum. It also framed the question that Jegadeesh and Titman answered at the short end.

Jegadeesh & Titman (1993) — the canonical paper

> *"Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency," Journal of Finance 48(1), 65–91.*

Contribution. Jegadeesh and Titman sorted US stocks on past 3–12 month returns. Buying the top decile and shorting the bottom decile earned roughly 1% per month over 1965–1989. Market risk, size and the systematic risk exposures known at the time did *not* explain the effect. They also documented that the profits partly reverse over the following two years.

What changed. This paper changed the field. Momentum quickly went from "chartist superstition" to "the most robust anomaly in asset pricing". The paper legitimized past-price predictors as an object of research.

Limitations. It covers a single market and a single sample, with no transaction costs. As the authors carefully noted, the strategy's short leg concentrates in small, illiquid names, where costs are largest.

Lasting influence. It defines the standard construction still in use. Rank on cumulative return from $t-12$ to $t-2$, skipping the most recent month to avoid short-term reversal and bid-ask bounce. Hold for 1–6 months, use decile or tercile sorts, and rebalance monthly. Jegadeesh & Titman (2001) confirmed that the effect persisted out of sample in the 1990s. [Practice] **Recommendation: if reading only one paper, read this one.**

Carhart (1997), "On Persistence in Mutual Fund Performance"

Contribution. Carhart added momentum (UMD/WML) as a fourth factor to the Fama–French three-factor model. He showed that most apparent persistence in mutual-fund skill is passive momentum exposure.

What changed. The paper made momentum a standard *factor*. Every later performance attribution had to control for it. The paper also reduced the value attributed to active management, and the industry is still adjusting to that conclusion.

Limitations. Carhart added momentum without theory, as an empirical control rather than a risk factor with an economic explanation. Fama & French (1996) had already conceded that their three-factor model could not explain momentum. Their five-factor model (2015) still does not include it, and Fama & French (2016) acknowledge the model's continued failure on momentum.

Lasting influence. UMD is a standard data series in Kenneth French's library and the default control in empirical finance.

Rouwenhorst (1998); Asness, Liew & Stevens (1997); Moskowitz & Grinblatt (1999)

Contribution. These papers generalized momentum out of sample. Rouwenhorst found momentum in 12 European markets. Asness, Liew and Stevens found it across country indices. Moskowitz and Grinblatt found that **industry** momentum accounts for much of individual-stock momentum.

What changed. Momentum stopped looking like an artifact of mining US data. The industry result also suggested that momentum might be a *group-level* phenomenon rather than a stock-level one. That line of work leads directly to factor momentum in the 2020s.

Limitations. The samples overlap, the markets are correlated, and a common global data-snooping bias is possible, because all researchers studied the same 1926–1995 window.

The 1998–1999 behavioral trilogy: BSV, DHS, Hong & Stein

Contribution. Three internally consistent models generate momentum followed by reversal from psychological primitives. See §1.3(B).

What changed. Momentum acquired *theories*, which made it respectable. Hong and Stein in particular gave a mechanism, gradual information diffusion, with testable cross-sectional implications. Hong, Lim & Stein (2000) confirmed them.

Limitations. All three models are flexible. No test has decisively confirmed or rejected any of them against the others, and that remains true today.

Lasting influence. Their terms, such as under-reaction, over-reaction, gradual diffusion and self-attribution, is now standard, even among people who reject the models.

The data-snooping counter-attack: Brock, Lakonishok & LeBaron (1992) → Sullivan, Timmermann & White (1999)

Contribution. Brock, Lakonishok and LeBaron tested 26 classic moving-average and trading-range-breakout rules on the Dow from 1897–1986, and found significant predictive power. Sullivan, Timmermann and White re-examined the *same* rules with White's (2000) Reality Check. The Reality Check corrects for selection, because those 26 rules survived decades of collective search. The test evaluated a universe of about 8,000 rules. The best rules were still significant in the original sample, but the effect had largely disappeared in data after 1986.

What changed. This exchange teaches the most important methodological lesson in the field. **The universe of rules searched, not the rule reported, determines significance.** All of §7.6 builds on it.

Lasting influence. The influence is decisive. Treat a momentum backtest presented today without a multiple-testing adjustment as uninformative.

Lo, Mamaysky & Wang (2000), "Foundations of Technical Analysis"

Contribution. The authors used nonparametric kernel regression to define chart patterns, such as head-and-shoulders, by algorithm. They then tested whether the conditional return distribution differs from the unconditional one. For several patterns it does, with statistical significance.

What changed. The paper gave the *only* rigorous treatment of the Edwards–Magee program, and its conclusion was mixed. The patterns carry incremental information, but not obviously enough to be profitable after costs.

Lasting influence. It serves as a methodological template for testing shape-based signals. In practice, it showed that the question is legitimate without endorsing a particular answer.

2.5 Era V — Consolidation, generalization, risk management (2000–2016)

Grinblatt & Han (2005); Frazzini (2006) — the disposition effect

Contribution. These papers built a measure of aggregate **unrealized capital gains** from a turnover-weighted historical cost basis. The measure subsumes much of the momentum effect. Momentum is also stronger where the disposition effect should apply most strongly, for example around mutual-fund holdings with large embedded gains or losses.

What changed. For the first time, a behavioral explanation made an *independent prediction that did not use returns*, and the data confirmed the prediction. This is the strongest behavioral evidence in the momentum literature.

Limitations. Constructing the reference price requires strong assumptions, and the results are sensitive to them.

Moskowitz, Ooi & Pedersen (2012), "Time Series Momentum" (*JFE*)

Contribution. The paper established that momentum operates on an asset's *own* past returns, as an absolute, sign-based signal, and not only relative to peers. The evidence covers 58 futures markets in equities, bonds, currencies and commodities over 1965–2009. A diversified, volatility-scaled TSMOM portfolio delivered a high Sharpe ratio, with positive skew and low correlation to traditional assets.

What changed. This paper connected the academy and the CTA industry. It gave the trend-following business peer-reviewed standing and a canonical construction. The construction takes the sign of the trailing 12-month return, scales positions to a constant volatility target per asset, and aggregates across a diversified futures universe.

Limitations. [Contested] Huang, Li, Wang & Zhou (2020, *JFE* 135, 774–794) argue that the core test conflates predictability with a positive unconditional mean. After appropriate controls, they find little evidence of an absolute TSMOM effect. Goyal & Jegadeesh (2018, *RFS*) also show that the difference between time-series and cross-sectional momentum is largely a *net long position* in the market, not a difference in predictability. Authors affiliated with AQR respond that the effect is real. The debate is unresolved. Read both sides.

Lasting influence. The influence is very large. Volatility-scaled TSMOM on a diversified futures universe is the standard academic benchmark for trend following.

Asness, Moskowitz & Pedersen (2013), "Value and Momentum Everywhere" (*JF*)

Contribution. The paper showed that value and momentum work in *eight* markets and asset classes. The key result is that momentum is **positively correlated across asset classes** and **negatively correlated with value**. That pattern suggests a common global factor structure rather than eight independent anomalies.

What changed. Researchers stopped asking whether momentum exists in a given market and started asking what common factor drives it. The combination of value and momentum became the default diversified factor portfolio. The authors proposed **liquidity risk and funding constraints** as the common driver.

Limitations. The common-factor interpretation is not uniquely identified. Correlated anomalies could reflect correlated data mining or correlated flows.

Barroso & Santa-Clara (2015); Daniel & Moskowitz (2016) — momentum crashes

Contribution. Both papers address momentum's catastrophic tail. Daniel and Moskowitz characterize the option-like conditional beta (see §1.6) and propose a dynamically hedged and scaled momentum. Barroso and Santa-Clara scale momentum exposure by the *inverse of its own recent realized volatility*. This "risk-managed momentum" nearly doubles the Sharpe ratio and greatly reduces the crash.

What changed. Volatility scaling of the *strategy*, not just of the assets, became standard. This is arguably the most valuable practical result of the last 15 years, and it generalizes. Moreira & Muir (2017, *JF*) show that volatility management improves the Sharpe ratio of many factors.

Limitations. [Contested] Cederburg, O'Doherty, Wang & Yan (2020) and others question how much of the improvement from volatility management survives out of sample and after realistic costs, given the high turnover. The improvement is largest exactly where turnover is largest.

Lasting influence. The influence was very large and immediate. Almost every practitioner momentum book now scales by volatility.

Long-history validations: Lempérière et al. (2014); Geczy & Samonov (2016); Hurst, Ooi & Pedersen (2017)

Contribution. These studies independently extended the evidence for momentum and trend far outside the original samples.

- Lempérière, Deremble, Seager, Potters and Bouchaud reconstructed trend-following returns back to 1800 for commodities and indices. After removing the assets' upward drift, they report a t-statistic of roughly 10 since 1800, and about 5 since 1960.
- Geczy and Samonov built a database of US security prices from 1801. They found momentum profits before 1927 that were positive and significant.
- Hurst, Ooi and Pedersen documented a century of positive trend-following returns across 67 markets.

What changed. These studies effectively closed the objection that momentum is "just data mining on the CRSP sample". They are the strongest available evidence that momentum is a structural feature of markets.

Limitations. Pre-modern data quality is poor, historical price series contain survivorship, and no realistic cost model exists for 19th-century markets. More subtly, the authors knew the modern result in advance.

2.6 Market microstructure and order flow

This strand ran in parallel and only recently merged with the momentum literature.

Kyle (1985); Glosten & Milgrom (1985). Contribution: formal models in which prices move because *order flow reveals information*, with a linear structure (Kyle's λ) or Bayesian updating. What changed: price impact stopped being a friction and became the *mechanism of price formation*. Lasting influence: every impact model derives from these.

Lillo & Farmer (2004); Bouchaud, Gefen, Potters & Wyart (2004). Contribution: the empirical discovery that the signs of order flow have **long memory**, while prices remain close to a martingale. The autocorrelation follows a power law, with a Hurst exponent typically around 0.6–0.8. The **propagator model** of Bouchaud et al. resolves the apparent paradox. The impact of each trade decays over time in exactly the way needed to offset the predictable flow. So the market is "statistically efficient" despite predictable order flow. What changed: the predictability of *flow* differs from the predictability of *price*, and the response of liquidity providers enforces efficiency. Limitations: the model is descriptive and needs heavy calibration. Lasting influence: it is the theoretical basis of modern execution and of high-frequency momentum.

The square-root law of market impact. Contribution: the empirical regularity $\Delta p \approx Y\sigma\sqrt{Q/V}$, with $Y \approx 0.5{-}1$, documented across markets and decades. What changed: it makes momentum's *capacity* computable rather than a matter of opinion. It also makes the causal link between institutional metaorders and multi-day price drift quantitative. Limitations: the exponent is not exactly 1/2 in all datasets, and researchers still debate the theoretical justification, latent liquidity or a locally linear order book. Lasting influence: it is the single most important formula for anyone sizing a momentum strategy.

Order-flow imbalance at high frequency (Cont, Kukanov & Stoikov, 2014; Sirignano & Cont, 2019). Contribution: *order-flow imbalance* explains short-horizon price changes overwhelmingly, through a linear relation with a high R^2 at the sub-minute scale. A deep network trained on limit-order-book data learns a nearly universal mapping from flow to price that transfers across stocks. What changed: intraday "momentum" was correctly re-identified as *flow prediction*. Lasting influence: this is what high-frequency momentum actually is, and it has little to do with the 12-month effect.

The synthesis reference for this strand is Bouchaud, Bonart, Donier & Gould, *Trades, Quotes and Prices* (2018).

2.7 Modern quantitative approaches (2016–present)

The replication crisis arrives

Harvey, Liu & Zhu (2016), "…and the Cross-Section of Expected Returns" (*RFS*) catalogued more than 300 published factors. They argued that, given the intensity of the search, a newly claimed factor needs a t-statistic of at least about **3.0**, not 2.0. Harvey & Liu (2015, 2020) developed procedures for reducing reported backtest performance and for multiple testing. Hou, Xue & Zhang (2020, *RFS*) replicated about 450 anomalies and found most of them insignificant under equal weighting with microcap controls.

Momentum's record in the replication crisis is unusually good. [Fact] It is among the small set of anomalies that survive essentially every replication protocol, across countries, asset classes and centuries. That is the strongest argument for building on it.

Factor momentum

Three studies independently established that **factors themselves exhibit momentum**. They are **Gupta & Kelly (2019, *JPM* 45(3), 13–36)**, **Ehsani & Linnainmaa (2022, *JF* 77(3), 1877–1919)** and **Arnott, Clements, Kalesnik & Linnainmaa (2023, *RFS***

36(8), 3034–3070). A factor's own recent return predicts its next return. These studies also suggest that individual-stock momentum may result from factor momentum rather than being an independent phenomenon. Ehsani and Linnainmaa argue that the momentum factor is essentially the aggregate of the autocorrelation in other factors' returns.

What changed. This work reframes momentum as a property of *risk-factor time series* rather than of individual securities. It connects momentum to the older industry-momentum result of Moskowitz and Grinblatt. **[Contested]** Researchers dispute how completely factor momentum subsumes stock momentum.

Machine learning

Gu, Kelly & Xiu (2020, *RFS*) benchmarked machine-learning methods for return prediction. Tree ensembles and neural networks materially outperformed linear models, and **predictors from the momentum family were consistently among the most important features**. **Lim, Zohren & Roberts (2019)** introduced "Deep Momentum Networks", which optimize the Sharpe ratio directly with LSTMs over trend features. **Wood, Giegerich, Roberts & Zohren (2021)** extended this with attention and Transformers. **López de Prado (2018)** contributed the essential methods: triple-barrier labeling, meta-labeling, purged and embargoed cross-validation, and the Deflated Sharpe Ratio.

Kelly, Malamud & Zhou (2024, *JF*), "The Virtue of Complexity in Return Prediction" argues against decades of orthodoxy that favored parsimony. It argues that heavily over-parameterized models with appropriate ridge regularization can outperform and exhibit "double descent". **[Contested]** The claim is important if true.

Limitations. Machine learning has improved momentum's *combination and conditioning* far more than its *core prediction*. The improvements concern how to blend horizons, when to turn the signal off, and how to size. **[Practice]** Among people who have actually deployed these models, the consensus is that raw ML alpha over a well-built, volatility-scaled, multi-horizon trend baseline is real but modest. The risk of overfitting is severe.

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§2 Key takeaways

1. Three groups found the same effect: chartists (as trend), CTAs (as a mechanical rule) and academics (as a factor). Each community contributed something. Dow contributed **multi-scale structure and confirmation**. Livermore, Donchian and the Turtles contributed **loss-cutting and volatility-based sizing**. Jegadeesh & Titman onward contributed **statistical validation with risk adjustment**.
2. **Jegadeesh & Titman (1993)** changed the field. The 12-2 construction it established is still the default.
3. **Sullivan, Timmermann & White (1999)** is the key methodological paper. It showed that the size of the search space, not the reported rule, determines significance.
4. **Moskowitz, Ooi & Pedersen (2012)** legitimized time-series momentum and connected the academy to the CTA industry. **Huang et al. (2020)** issued the most serious challenge to it. Read both.
5. **Daniel & Moskowitz (2016)** and **Barroso & Santa-Clara (2015)** established that momentum's tail is a conditional-beta phenomenon and that volatility scaling largely fixes it. This is the highest-value practical result of the modern era.
6. Microstructure has supplied the most *mechanistic* explanation: metaorder splitting plus square-root impact produces multi-day persistent drift. It also supplies the capacity formula.
7. Momentum is one of the few anomalies that **survived the replication crisis** cleanly.
8. The current frontier has three parts. The first is momentum as a property of *factors* rather than securities. The second is conditional-risk explanations. The third is machine learning used for conditioning and combination rather than for raw prediction.

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3. Foundational references

This section is a curated reading list rather than a bibliography. Each category lists works roughly in the recommended reading order. Items marked ★ are essential for a quantitative practitioner working on momentum, in this chapter's view.

3.1 Classic books

Work	Why it matters	Read it for
Lefèvre, E. (1923). *Reminiscences of a Stock Operator.*	The founding text of speculative discipline; a novelization of Jesse Livermore	The asymmetric payoff mindset: small losses, large wins, pyramiding into strength. Read as literature and psychology, not method
Rhea, R. (1932). *The Dow Theory.*	Codification of Dow/Hamilton	Multi-timescale trend structure and confirmation logic
Edwards, R. D. & Magee, J. (1948). *Technical Analysis of Stock Trends.*	The standard pattern catalogue	Historical literacy: what practitioners believed. Do not adopt the patterns uncritically
Wilder, J. W. (1978). *New Concepts in Technical Trading Systems.*	Source of RSI, ADX/DMI, ATR, Parabolic SAR	The original definitions, which differ from most software implementations. Essential for anyone implementing these indicators
Graham, B. & Dodd, D. (1934). *Security Analysis.*	The opposing intellectual tradition	The value case; value and momentum are the two halves of a well-diversified factor book
Schwager, J. (1989, 1992). *Market Wizards* / *The New Market Wizards.*	Interviews with practitioners, including many trend followers	Practitioner intuition about risk sizing and drawdown tolerance, unavailable in journals. Heavily survivorship-biased; read accordingly

3.2 Modern books

Work	Why it matters
★ Grinold, R. C. & Kahn, R. N. (1999). *Active Portfolio Management*, 2nd ed.	The framework for turning any signal into a portfolio: the Fundamental Law, information coefficients, transfer coefficients, risk budgeting. Required reading before building momentum portfolios
★ Bouchaud, J.-P., Bonart, J., Donier, J. & Gould, M. (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope.*	The definitive modern microstructure text: long memory in order flow, propagator models, the square-root impact law. This book covers both momentum's *mechanism* and its *capacity limit*
★ López de Prado, M. (2018). *Advances in Financial Machine Learning.*	Purged and embargoed cross-validation, triple-barrier labeling, meta-labeling, fractional differentiation, backtest overfitting. Opinionated and occasionally overreaching, but the cross-validation methodology alone justifies it
Ilmanen, A. (2011). *Expected Returns.*	The best single synthesis of the empirical evidence on all major return sources, momentum included. Encyclopedic and even-handed
Antonacci, G. (2014). *Dual Momentum Investing.*	The clearest practitioner account of combining absolute (time-series) and relative (cross-sectional) momentum. Simple by design
Clenow, A. (2013). *Following the Trend.*	An honest, implementable account of how a diversified futures trend program is actually built: universe, sizing, rebalancing, and realistic expectations
Chan, E. (2013). *Algorithmic Trading: Winning Strategies and Their Rationale.*	A practical treatment of momentum versus mean reversion, with runnable code and a sensible discussion of regime
Harvey, C. R., Rattray, S. & Van Hemert, O. (2021). *Strategic Risk Management.*	A modern treatment of defensive strategies, drawdown control, and where trend fits in a portfolio
Satchell, S. & Grant, A., eds. (2020). *Market Momentum: Theory and Practice.*	Edited volume; includes the published version of Baltas & Kosowski's TSMOM implementation study
Tsay, R. S. (2010). *Analysis of Financial Time Series*, 3rd ed.	The econometrics foundation: ARMA, GARCH, state space, regime switching
Durbin, J. & Koopman, S. J. (2012). *Time Series Analysis by State Space Methods*, 2nd ed.	The reference for Kalman and state-space trend extraction (§4.7.1)

3.3 Foundational academic papers

The core momentum papers, in reading order:

1. ★ **Jegadeesh, N. & Titman, S. (1993).** "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency." *Journal of Finance* 48(1), 65–91. — The founding momentum paper.
2. **De Bondt, W. & Thaler, R. (1985).** "Does the Stock Market Overreact?" *Journal of Finance* 40(3), 793–805. — The long-horizon reversal counterpart.
3. **Jegadeesh, N. & Titman, S. (2001).** "Profitability of Momentum Strategies: An Evaluation of Alternative Explanations." *Journal of Finance* 56(2), 699–720. — The out-of-sample confirmation and the reversal evidence.
4. **Carhart, M. (1997).** "On Persistence in Mutual Fund Performance." *Journal of Finance* 52(1), 57–82. — Momentum as a factor.
5. ★ **Moskowitz, T., Ooi, Y. H. & Pedersen, L. H. (2012).** "Time Series Momentum." *Journal of Financial Economics* 104(2), 228–250.
6. ★ **Asness, C., Moskowitz, T. & Pedersen, L. H. (2013).** "Value and Momentum Everywhere." *Journal of Finance* 68(3), 929–985.
7. ★ **Daniel, K. & Moskowitz, T. (2016).** "Momentum Crashes." *Journal of Financial Economics* 122(2), 221–247.
8. **Barroso, P. & Santa-Clara, P. (2015).** "Momentum Has Its Moments." *Journal of Financial Economics* 116(1), 111–120.

Statistical foundations:

9. ★ **Lo, A. W. & MacKinlay, A. C. (1988).** "Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test." *Review of Financial Studies* 1(1), 41–66. — The variance ratio.
10. ★ **Lo, A. W. & MacKinlay, A. C. (1990).** "When Are Contrarian Profits Due to Stock Market Overreaction?" *Review of Financial Studies* 3(2), 175–205. — Decomposes cross-sectional profits into autocovariance, cross-serial covariance and dispersion in means. It shows which of these terms a strategy actually earns.
11. **Lewellen, J. (2002).** "Momentum and Autocorrelation in Stock Returns." *Review of Financial Studies* 15(2), 533–564. — Shows that momentum in size and book-to-market portfolios arises largely from negative cross-serial correlation rather than own-autocorrelation.
12. **Jegadeesh, N. (1990).** "Evidence of Predictable Behavior of Security Returns." *Journal of Finance* 45(3), 881–898; and **Lehmann, B. (1990),** "Fads, Martingales, and Market Efficiency," *QJE* 105(1), 1–28. — Short-horizon reversal, and the reason momentum signals skip the last month.

Mechanism papers:

13. **Barberis, N., Shleifer, A. & Vishny, R. (1998).** "A Model of Investor Sentiment." *JFE* 49(3), 307–343.
14. **Daniel, K., Hirshleifer, D. & Subrahmanyam, A. (1998).** "Investor Psychology and Security Market Under- and Overreactions." *Journal of Finance* 53(6), 1839–1885.
15. ★ **Hong, H. & Stein, J. (1999).** "A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets." *Journal of Finance* 54(6), 2143–2184.
16. **Hong, H., Lim, T. & Stein, J. (2000).** "Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies." *Journal of Finance* 55(1), 265–295.
17. ★ **Grinblatt, M. & Han, B. (2005).** "Prospect Theory, Mental Accounting, and Momentum." *JFE* 78(2), 311–339.
18. **Vayanos, D. & Woolley, P. (2013).** "An Institutional Theory of Momentum and Reversal." *RFS* 26(5), 1087–1145. — Momentum from the fund flows of delegated management. It is a rational, flow-based alternative to the behavioral explanations.
19. ★ **Kelly, B., Moskowitz, T. & Pruitt, S. (2021).** "Understanding Momentum and Reversal." *JFE* 140(3), 726–743. — The strongest modern conditional-risk explanation.

Microstructure:

20. **Kyle, A. (1985).** "Continuous Auctions and Insider Trading." *Econometrica* 53(6), 1315–1335.
21. **Glosten, L. & Milgrom, P. (1985).** "Bid, Ask and Transaction Prices in a Specialist Market…" *JFE* 14(1), 71–100.
22. ★ **Lillo, F. & Farmer, J. D. (2004).** "The Long Memory of the Efficient Market." *Studies in Nonlinear Dynamics & Econometrics* 8(3).
23. **Bouchaud, J.-P., Gefen, Y., Potters, M. & Wyart, M. (2004).** "Fluctuations and Response in Financial Markets: The Subtle Nature of 'Random' Price Changes." *Quantitative Finance* 4(2), 176–190. — The propagator model.
24. **Cont, R., Kukanov, A. & Stoikov, S. (2014).** "The Price Impact of Order Book Events." *Journal of Financial Econometrics* 12(1), 47–88.

Refinements and variants:

25. **Moskowitz, T. & Grinblatt, M. (1999).** "Do Industries Explain Momentum?" *Journal of Finance* 54(4), 1249–1290.
26. **George, T. & Hwang, C.-Y. (2004).** "The 52-Week High and Momentum Investing." *Journal of Finance* 59(5), 2145–2176.

27. **Novy-Marx, R. (2012).** "Is Momentum Really Momentum?" *JFE* 103(3), 429–453. — Echo momentum.
28. **Blitz, D., Huij, J. & Martens, M. (2011).** "Residual Momentum." *Journal of Empirical Finance* 18(3), 506–518.
29. **Rouwenhorst, K. G. (1998).** "International Momentum Strategies." *Journal of Finance* 53(1), 267–284.
30. **Menkhoff, L., Sarno, L., Schmeling, M. & Schrimpf, A. (2012).** "Currency Momentum Strategies." *JFE* 106(3), 660–684.
31. **Ehsani, S. & Linnainmaa, J. (2022).** "Factor Momentum and the Momentum Factor." *Journal of Finance* 77(3), 1877–1919.
32. **Arnott, R., Clements, M., Kalesnik, V. & Linnainmaa, J. (2023).** "Factor Momentum." *RFS* 36(8), 3034–3070.

Critiques to read alongside the claims:

33. ★ **Huang, D., Li, J., Wang, L. & Zhou, G. (2020).** "Time Series Momentum: Is It There?" *JFE* 135(3), 774–794.
34. **Goyal, A. & Jegadeesh, N. (2018).** "Cross-Sectional and Time-Series Tests of Return Predictability: What Is the Difference?" *RFS* 31(5), 1784–1824.
35. ★ **Sullivan, R., Timmermann, A. & White, H. (1999).** "Data-Snooping, Technical Trading Rule Performance, and the Bootstrap." *Journal of Finance* 54(5), 1647–1691.
36. ★ **Harvey, C., Liu, Y. & Zhu, H. (2016).** "…and the Cross-Section of Expected Returns." *RFS* 29(1), 5–68.
37. **Hou, K., Xue, C. & Zhang, L. (2020).** "Replicating Anomalies." *RFS* 33(5), 2019–2133.
38. **Novy-Marx, R. & Velikov, M. (2016).** "A Taxonomy of Anomalies and Their Trading Costs." *RFS* 29(1), 104–147.

Methodology and evaluation:

39. ★ **White, H. (2000).** "A Reality Check for Data Snooping." *Econometrica* 68(5), 1097–1126.
40. **Hansen, P. R. (2005).** "A Test for Superior Predictive Ability." *Journal of Business & Economic Statistics* 23(4), 365–380.
41. **Romano, J. & Wolf, M. (2005).** "Stepwise Multiple Testing as Formalized Data Snooping." *Econometrica* 73(4), 1237–1282.
42. **Politis, D. & Romano, J. (1994).** "The Stationary Bootstrap." *JASA* 89(428), 1303–1313.
43. ★ **Bailey, D. & López de Prado, M. (2014).** "The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality." *Journal of Portfolio Management* 40(5), 94–107.
44. **Lo, A. W. (2002).** "The Statistics of Sharpe Ratios." *Financial Analysts Journal* 58(4), 36–52.
45. **Newey, W. & West, K. (1987).** "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica* 55(3), 703–708.
46. **Benjamini, Y. & Hochberg, Y. (1995).** "Controlling the False Discovery Rate." *JRSS-B* 57(1), 289–300.

3.4 Review and survey papers

- ★ **Jegadeesh, N. & Titman, S. (2011).** "Momentum." *Annual Review of Financial Economics* 3, 493–509. — The authors' own retrospective, and the best short survey.
- **Asness, C., Frazzini, A., Israel, R. & Moskowitz, T. (2014).** "Fact, Fiction, and Momentum Investing." *Journal of Portfolio Management* 40(5), 75–92. — Addresses the 10 most common objections systematically: taxes, costs, small caps, crashes and dependence on the short side. The authors are interested parties. It is still the best rebuttal available.
- **Subrahmanyam, A. (2018).** "Equity Market Momentum: A Synthesis of the Literature and Suggestions for Future Work." *Pacific-Basin Finance Journal* 51, 291–296.
- **Bouchaud, J.-P., Farmer, J. D. & Lillo, F. (2009).** "How Markets Slowly Digest Changes in Supply and Demand." In *Handbook of Financial Markets: Dynamics and Evolution*. — The microstructure survey most relevant to momentum.
- **Nagel, S. (2013).** "Empirical Cross-Sectional Asset Pricing." *Annual Review of Financial Economics* 5, 167–199.
- **Gu, S., Kelly, B. & Xiu, D. (2020).** "Empirical Asset Pricing via Machine Learning." *RFS* 33(5), 2223–2273. — The machine-learning survey for this domain.

3.5 Practitioner papers

These are white papers and practitioner-journal articles. They are less rigorously refereed than journal articles, and their authors have commercial interests. Several of them nonetheless contain the most directly usable results in the literature.

- ★ **Hurst, B., Ooi, Y. H. & Pedersen, L. H. (2017).** "A Century of Evidence on Trend-Following Investing." *Journal of Portfolio Management* 44(1), 15–29. (AQR.) — 1880–2016, 67 markets.
- ★ **Lempérière, Y., Deremble, C., Seager, P., Potters, M. & Bouchaud, J.-P. (2014).** "Two Centuries of Trend Following." *Journal of Investment Strategies* 3(3), 41–61. (CFM.) — arXiv:1404.3274.
- ★ **Dao, T.-L., Nguyen, T.-T., Deremble, C., Lempérière, Y., Bouchaud, J.-P. & Potters, M. (2017).** "Tail Protection for Long Investors: Trend Convexity at Work." *Journal of Investment Strategies*. — arXiv:1607.02410. The variance-difference decomposition of trend P&L.

- ★ **Fung, W. & Hsieh, D. (2001).** "The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers." *RFS* 14(2), 313–341. — Lookback straddles. The paper is academic, and practitioners use it as the model of the CTA payoff.
- **Geczy, C. & Samonov, M. (2016).** "Two Centuries of Price Return Momentum." *Financial Analysts Journal* 72(5), 32–56.
- **Levine, A. & Pedersen, L. H. (2016).** "Which Trend Is Your Friend?" *Financial Analysts Journal* 72(3), 51–66. — Shows that time-series regression, moving-average crossovers and other trend rules are nearly equivalent when their horizons match. The practical lesson is to stop tuning indicator forms and tune horizons instead.
- ★ **Baltas, N. & Kosowski, R. (2020).** "Demystifying Time-Series Momentum Strategies: Volatility Estimators, Trading Rules and Pairwise Correlations." In Satchell & Grant, eds., *Market Momentum.* (SSRN 2140091.) — The most useful implementation study. It covers which volatility estimator and which trading rule to use, and how correlation structure affects diversification and turnover.
- **Bruder, B., Dao, T.-L., Richard, J.-C. & Roncalli, T. (2013).** "Trend Filtering Methods for Momentum Strategies." SSRN 2289097. (Lyxor.) — Treats moving averages, L1 and L2 trend filtering, and Kalman filters explicitly as signal extraction.
- **Moreira, A. & Muir, T. (2017).** "Volatility-Managed Portfolios." *Journal of Finance* 72(4), 1611–1644.
- **Harvey, C., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M. & Van Hemert, O. (2018).** "The Impact of Volatility Targeting." *Journal of Portfolio Management* 45(1), 14–33. (Man Group.)
- **Frazzini, A., Israel, R. & Moskowitz, T. (2018).** "Trading Costs." SSRN 3229719. (AQR.) — Live-execution cost estimates from about $1.7 trillion of real trades. This is the best public evidence on momentum's true capacity. It is far more optimistic than academic estimates based on quoted spreads. **[Contested]** AQR runs momentum strategies, so its authors have an interest in the result.
- **Grinold, R. (1989).** "The Fundamental Law of Active Management." *Journal of Portfolio Management* 15(3), 30–37. — $\text{IR} \approx \text{IC}\sqrt{\text{breadth}}$.
- **Clarke, R., de Silva, H. & Thorley, S. (2002).** "Portfolio Constraints and the Fundamental Law of Active Management." *Financial Analysts Journal* 58(5), 48–66. — Adds the transfer coefficient, which explains why real portfolios capture far less than the theoretical IR.
- **Lim, B., Zohren, S. & Roberts, S. (2019).** "Enhancing Time-Series Momentum Strategies Using Deep Neural Networks." *Journal of Financial Data Science* 1(4), 19–38.

3.6 If you only read six things

1. Jegadeesh & Titman (1993) — the effect.
2. Lo & MacKinlay (1988, 1990) — what is actually being measured.
3. Moskowitz, Ooi & Pedersen (2012) **with** Huang, Li, Wang & Zhou (2020) — the effect and its most serious critique, together.
4. Daniel & Moskowitz (2016) — the tail, and why risk management is intrinsic.
5. Grinold & Kahn (1999) — how to turn a signal into a portfolio.
6. Sullivan, Timmermann & White (1999) — why most backtests are probably wrong.

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§3 Key takeaways

1. The literature has a small, stable core. Six papers and one book cover about 80% of what matters, and the rest is refinement.
2. **Read critiques alongside claims.** Every major momentum result has a serious published challenge. Reading each result together with its challenge is the fastest way to judge how reliable the result is.
3. Practitioner white papers from AQR, CFM, Man and Lyxor contain some of the most directly usable results: long-history validation, cost estimates and implementation studies. Their authors have commercial interests. Weight them accordingly, and prefer those with reproducible methodology.
4. Momentum practitioners under-read the microstructure literature. It explains the mechanism and the capacity constraint.

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4. Mathematical and statistical characterizations of momentum

4.0 Preliminaries

What all of these estimate

Every measure below estimates a functional of one object, the **conditional mean of future returns given the recent price path**. Write the target as

$$m_t(H) \;=\; \mathbb{E}\Big[\textstyle\sum_{h=1}^{H} r_{t+h} \;\Big|\; \mathcal{F}_t\Big]$$

Under the incomplete-adjustment model of §1.2, $m_t(H)$ depends on the past return path mostly through a few summaries. These are the *magnitude* of recent net movement, its *consistency*, and the *volatility* against which an analyst should judge both. These summaries define the whole design space. Almost every indicator ever invented trades off four quantities:

$$\underbrace{\text{signal strength}}_{\text{how far price moved}} \quad \text{vs.} \quad \underbrace{\text{signal reliability}}_{\text{how consistently}} \quad \text{vs.} \quad \underbrace{\text{scale}}_{\text{relative to what noise}} \quad \text{vs.} \quad \underbrace{\text{latency}}_{\text{how quickly detected}}$$

This view explains the taxonomy in §5. It also explains the empirical fact that most momentum measures correlate with each other at 0.8–0.95.

A warning about the estimand

Every measure also needs a second choice of target: **the conditional mean or the conditional Sharpe ratio.** These are different objects with different units. Confusing them produces position sizes that are wrong by a factor of volatility:

- $\mathbb{E}[r_{t+1} \mid \mathcal{F}_t]$ is an **expected return**. It is in return units.
- $\mathbb{E}[r_{t+1} \mid \mathcal{F}_t]/\sigma_t$ is an **expected Sharpe ratio**. It is dimensionless.

To keep these straight, *count the powers of σ in the final weight*. Do not memorize a rule about dividing once or twice. Under a mean–variance objective, the optimal position is

$$w_t \;=\; \frac{1}{\gamma}\cdot\frac{\mathbb{E}[r_{t+1} \mid \mathcal{F}_t]}{\sigma_t^2} \;=\; \frac{1}{\gamma}\cdot\frac{\text{expected Sharpe}}{\sigma_t}$$

with γ a risk-aversion constant. So the denominator of the weight should contain **two powers of σ in total if the signal estimates an expected return, and one if it estimates an expected Sharpe ratio.** The division can happen in the signal or in the sizing step. Only the total matters. The table shows three common designs that are internally consistent.

Signal s_t	Sizing rule	Total σ power	Implied belief
$r_{t-L:t}$ (raw past return)	$w \propto s_t/\sigma_t^2$	2	Past return predicts future *return*, with a coefficient that does not depend on volatility
$r_{t-L:t}/(\sigma_t\sqrt{L})$ (vol-normalized)	$w \propto s_t/\sigma_t$	2	Same belief, written in Sharpe units. It is algebraically identical to the row above
$\text{sign}(r_{t-L:t})$	$w \propto s_t/\sigma_t$	1	Only direction is informative. Expected Sharpe is constant given the direction (this is TSMOM, §4.6.1)

[Practice] Most professionals build the signal as an expected-Sharpe estimate and then size at $w \propto s_t/\sigma_t$, which is the second or third row. The genuine and very common error is losing count. A designer normalizes the signal by σ, then z-scores it against a window that *also* reflects volatility, then sizes by $1/\sigma$ again. The result is a power of σ that nobody

chose. Decide the intended power explicitly and write it in a comment. To check it, ask what the design implies about how expected return scales with volatility. §6.7 revisits this at the implementation level.

Notation for computational cost

This chapter gives costs in two forms. The **naive** cost is the per-bar cost of recomputing from scratch. The **streaming** cost is the per-bar cost with incremental state. For live systems, only the streaming cost matters. For research backtests over T bars and N assets, the naive cost applies unless the code uses vectorized operations.

The generic template used below

Each measure gets the same fields, in the same order: **Intuition → Definition → Assumptions → Strengths → Weaknesses → Cost → Robustness → Failure modes → When professionals prefer it.**

4.1 The price-difference family

These measures capture *how far price has moved*. They are the oldest and simplest measures. With normalization, they are still the most used.

4.1.1 Simple and cumulative return over a lookback

Intuition. This is the most direct measure of whether price went up.

Definition. Over a lookback of L bars ending at t, with an optional skip of S bars:

$$\text{MOM}_t^{\text{simple}}(L,S) = \frac{P_{t-S}}{P_{t-L}} - 1 \qquad \text{MOM}_t^{\text{cum}}(L,S) = \prod_{i=t-L+1}^{t-S}(1+R_i) - 1$$

The two are the same quantity when the returns are total returns of a single asset with no cash flows. They differ when the calculation compounds a *return series*, such as a strategy or a portfolio, instead of reading two prices. The canonical equity momentum signal uses $L = 12$ months and $S = 1$ month. This gives $P_{t-1\text{m}}/P_{t-12\text{m}} - 1$, an 11-month window ending one month before formation. In the standard "(2,12)" notation, this means using return lags 2 through 12 and skipping lag 1, the most recent month. Short-term *reversal* dominates that month (§1.5), so the signal skips it.

Assumptions. The measure assumes three things. The net displacement over the window is a sufficient statistic for the direction of future drift. The path taken is irrelevant. The window is the right horizon.

Strengths. It is transparent and non-parametric, needs no tuning beyond L and S, is directly comparable to the academic literature, and has minimal degrees of freedom.

Weaknesses.

- (a) **It is blind to the path.** A smooth 20% rise and a violent round trip that ends 20% up score identically, though their predictive content differs.
- (b) **It is sensitive to the endpoints.** The value depends entirely on two prices, and a single bad print at either end corrupts it.
- (c) **It depends on scale across assets.** A 20% move in a utility and a 20% move in a biotech are not comparable.
- (d) **It jumps at window edges.** The signal changes discretely when a large return leaves the window.

Computational cost. Naive $O(1)$, two price lookups. It is the cheapest measure.

Robustness. Moderate. **[Fact]** It is robust to specification, because any L in 6–12 months works in equities. It is fragile to bad data at the endpoints.

Failure modes.

- **Endpoint contamination:** a stale, erroneous or unadjusted price at $t - L$ silently sets the signal for the whole window.
- **Corporate-action leakage:** unadjusted splits and dividends produce enormous spurious momentum. In practice, this is the most common source of fake momentum alpha.
- **Window-edge discontinuity:** in the "drop-off effect", a single large old return leaves the window and flips the signal with no new information. This creates turnover uncorrelated with information.

- **Survivorship:** if the data drops delisted names, past losers disappear from the sample, and momentum looks better than it was.

When preferred. Use it as the *default* and as the *benchmark*. A fancier measure that cannot beat the 12-2 cumulative return after costs is not worth its complexity. It is also strongly preferred in academic replication, and wherever the team must defend the construction to a risk committee.

4.1.2 Log returns

Intuition. Work with log returns, which add across time instead of compounding.

Definition. $\text{MOM}_t^{\text{log}}(L, S) = p_{t-S} - p_{t-L} = \sum_{i=t-L+1}^{t-S} r_i$.

Why it matters. Four concrete reasons favor log returns, and none is aesthetic:

1. **Additivity across time.** The L-period log return is the sum of the one-period log returns. So windowed sums, regressions, EWMAs and filters are all linear operations. That is why every regression-based and filter-based method below operates on p_t, not P_t.
2. **Symmetry.** A round trip from A to B and back gives log returns of exactly equal size and opposite sign. Simple returns do not: the path out gives $+100\%$ and the path back gives -50%. Simple returns are bounded below by -100% and unbounded above, so their cross-sectional distribution is mechanically right-skewed. Logs remove that asymmetry.
3. **Better distributional behavior.** Log returns are closer to Gaussian, though still not Gaussian. That matters for every t-statistic computed on them.
4. **The Jensen gap is real.** $\mathbb{E}[\ln(1+R)] \approx \mu - \sigma^2/2$. A high-volatility asset with the same arithmetic mean has a lower geometric mean. Ranking on simple cumulative returns therefore carries a bias *toward high-volatility names*. This is one reason unadjusted cross-sectional momentum portfolios load on volatility.

Weaknesses. Log returns do not aggregate across the assets in a portfolio. The portfolio log return is not the weighted sum of the asset log returns. For portfolio accounting, use simple returns. For signal construction, use logs.

Cost. $O(1)$: one ln per bar, cached.

When preferred. Always, for signal construction. **[Practice]** Log returns are essentially universal in professional signal code. The sign is identical in both forms, and so is any sign-based rule. The choice therefore matters for magnitudes, rankings and regressions, not for a pure `sign()` trend rule.

4.1.3 Rate of change (ROC) and the "momentum indicator"

Intuition. ROC is exactly §4.1.1 expressed as a percentage. The classical "momentum indicator" is its unnormalized difference form.

Definition.

$$\text{ROC}_t(L) = 100 \cdot \left(\frac{P_t}{P_{t-L}} - 1 \right), \qquad \text{MOM}_t^{\text{raw}}(L) = P_t - P_{t-L}$$

Assessment. ROC is the same estimator as the simple lookback return and inherits all its properties. Essentially never use the *difference* form $P_t - P_{t-L}$. Its price units make it incomparable across assets, and it is not stationary within one asset over a long history. It exists only because analysts could compute it easily by hand in 1970.

When preferred. Use ROC for its familiar name, and otherwise use §4.1.1. The raw difference form is never the right choice.

4.1.4 Exponential (EWMA) momentum

Intuition. Replace the hard window with an exponentially decaying one. Recent returns get more weight, and the signal does not jump when old data leaves the window.

Definition. Practitioners parameterize the same filter three ways. The three are related by $\alpha = 1 - \lambda$:

$$\text{EWMA}_t = (1-\lambda)\, r_t + \lambda\, \text{EWMA}_{t-1} \quad \Longleftrightarrow \quad \text{EWMA}_t = (1-\lambda) \sum_{k \geq 0} \lambda^k r_{t-k}$$

- **Decay** $\lambda \in (0,1)$: the weight retained on the existing state each bar. Equivalently, the **smoothing constant** $\alpha = 1 - \lambda$ is the weight given to the new observation.
- **Half-life** h: the lag at which the weight has halved, $\lambda^h = \frac{1}{2}$, so $h = \ln 2 / \ln(1/\lambda)$.

- **Span** n: the convention used by most software, including pandas. The definition $\alpha = 2/(n+1)$ makes the EWMA's centre of mass match that of a simple n-bar window.

Equivalently, on prices, $\bar{P}_t = \alpha P_t + (1-\alpha)\bar{P}_{t-1}$, and the momentum signal is $p_t - \ln \bar{P}_t$ (see §4.1.5).

The **effective lookback** of an EWMA is its centre of mass, the average lag of the weight it applies:

$$\text{COM} \;=\; \sum_{k\geq 0} k\,\lambda^k(1-\lambda) \;=\; \frac{\lambda}{1-\lambda} \;=\; \frac{1-\alpha}{\alpha} \;=\; \frac{n-1}{2}$$

Use the centre of mass to match the horizon of an EWMA to a simple window. A simple window of length L has centre of mass $(L-1)/2$. So set $\lambda/(1-\lambda) = (L-1)/2$, which means span $n = L$. The last equality above explains the span convention. **Failing to match horizons is the most common reason two "equivalent" trend measures give different answers.**

Assumptions. The relevance of past information decays geometrically. This weighting is correct if the underlying state follows a random walk observed with noise, because the steady-state Kalman filter for a local-level model *is* an EWMA (see §4.7.1). This is a theoretical justification. It is not an argument from convenience.

Strengths. The filter has no window-edge discontinuity. The signal is smoother, so turnover is lower. The state is $O(1)$. Time-based decay, $\lambda = e^{-\Delta t/\tau}$, handles ragged or irregular sampling naturally.

Weaknesses. The memory is infinite, so old shocks decay but never leave the estimate completely. The effective window is harder to state. The first few $1/(1-\lambda)$ bars contain initialization bias.

Cost. Streaming $O(1)$ time and $O(1)$ memory. A naive backtest is a single $O(T)$ pass. It is the cheapest of all filters.

Robustness. High. The signal degrades smoothly when the parameter is wrong. **[Fact]** Levine & Pedersen (2016) show that trend signals based on EWMAs, MA crossovers and regressions produce very similar portfolios when horizons match. The functional form matters much less than the horizon.

Failure modes. Initialization bias: burn in for at least 5 half-lives before trusting the output. On data with gaps, a mismatch between calendar decay and bar decay quietly changes the horizon. A single outlier *never* leaves the estimate completely.

When preferred. Use it in live systems with latency or memory constraints, in high-frequency contexts, and whenever turnover matters. It is also the base primitive for MACD, ADX and volatility estimation. **[Practice]** It is by a wide margin the most-used smoother in production trading systems.

4.1.5 Moving-average displacement and crossover

Intuition. Compare the current price, or a fast average, with a slow average. If price is above its own recent average, it has been rising.

Definition. Let MA_n denote a simple or exponential moving average of length or span n. The measure has three related forms:

$$\text{Displacement:}\quad D_t(n) = \frac{P_t}{\text{MA}_n(P)_t} - 1 \;\approx\; p_t - \ln \text{MA}_n(P)_t$$

$$\text{Crossover:}\quad X_t(n_f, n_s) = \text{MA}_{n_f}(P)_t - \text{MA}_{n_s}(P)_t, \qquad n_f < n_s$$

$$\text{Normalized crossover:}\quad \tilde{X}_t = \frac{\text{MA}_{n_f} - \text{MA}_{n_s}}{\sigma_t \cdot P_t}$$

The key structural result. A crossover is a **band-pass filter** on the log price. MA_n is a low-pass filter. The difference of two low-pass filters passes the frequencies between them and attenuates both faster and slower components. The more useful view works on log prices and expands everything in terms of returns. Write $\text{MA}_n(p)_t = \frac{1}{n}\sum_{j=0}^{n-1} p_{t-j}$, and substitute $p_{t-j} = p_t - \sum_{i<j} r_{t-i}$. This gives the two identities that the rest of this section uses:

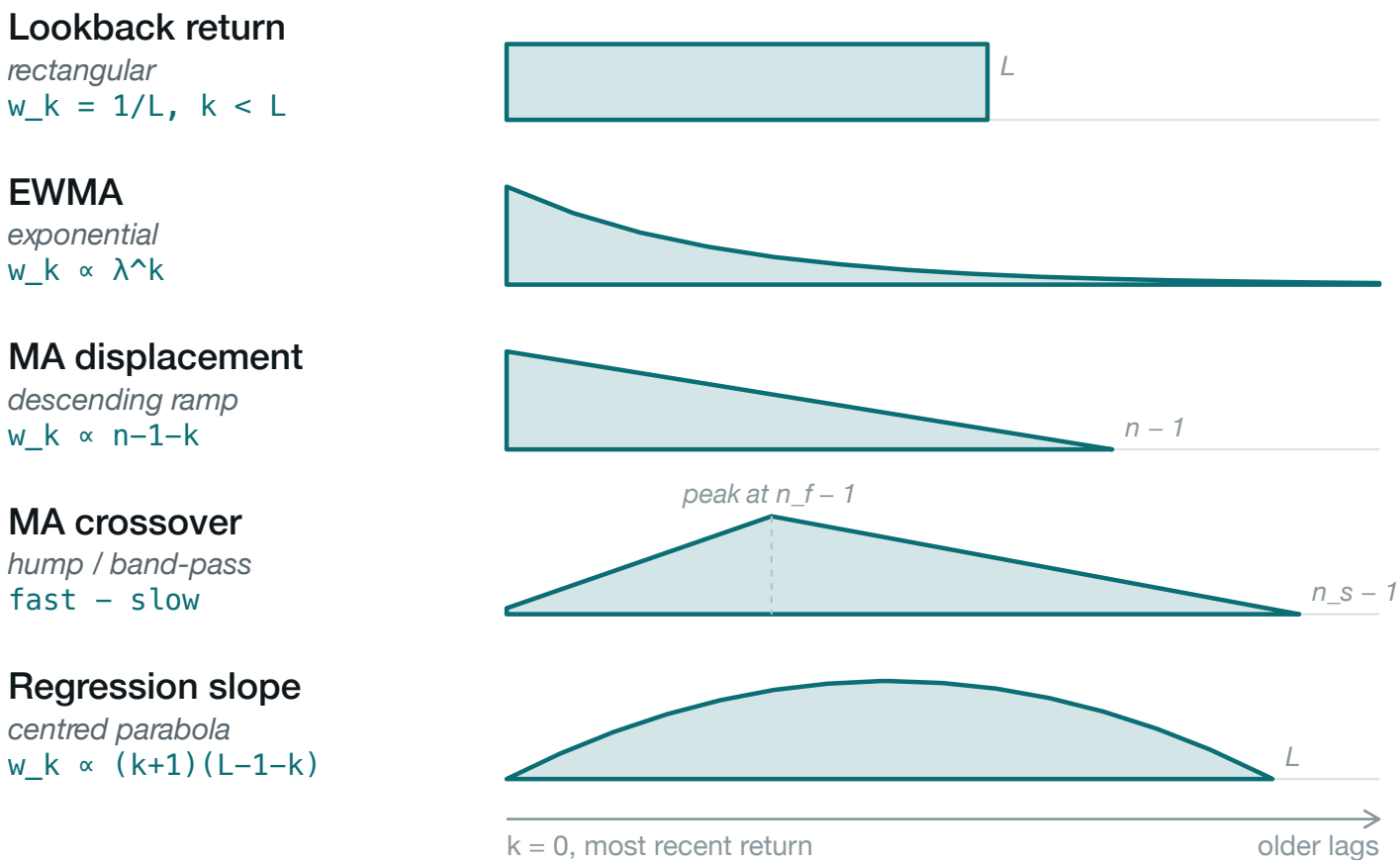
$$p_t - \text{MA}_n(p)_t \;=\; \sum_{k=0}^{n-2} \frac{n-1-k}{n}\, r_{t-k}$$

$$\text{MA}_{n_f}(p)_t - \text{MA}_{n_s}(p)_t \;=\; \sum_{k\geq 0} w_k\, r_{t-k}, \qquad w_k = \left(\frac{n_s - 1 - k}{n_s}\right)^+ - \left(\frac{n_f - 1 - k}{n_f}\right)^+$$

where $(x)^+ = \max(x, 0)$. Both identities apply to *simple* moving averages. The exponential case has the same structure, with geometric rather than linear decay: $p_t - \text{EMA}_\alpha(p)_t = \sum_k (1-\alpha)^{k+1} r_{t-k}$. The first kernel is a **descending ramp**. It puts the most weight on the most recent return and decays linearly to zero at lag $n-1$. The second kernel is **non-negative and hump-shaped**. It rises linearly to a peak at lag $n_f - 1$ and falls linearly to zero at lag $n_s - 1$. This shape is a band-pass, as claimed.

Every price-difference measure is a weighted sum of past returns, and the measures differ only in the kernel. The lookback return uses a rectangular kernel, the EWMA an exponential one, MA displacement a descending ramp, a crossover a hump, and the regression slope of §4.2.1 a centred parabola. *All of them are the same estimator with different weights.* This is the single most clarifying fact in Section 4. It explains why the measures correlate so highly. The figure plots the five kernels.

Every measure is a weighted sum of past returns. Only the kernel differs.



Kernel weights w_k applied to the return r_{t-k}, all normalized to unit sum.

Assumptions. A low-pass-filtered price is a usable estimate of an unobserved trend level. The deviation of price from that level is informative about direction, not about mean-reverting noise. The *opposite* assumption, that the deviation reverts to the mean, gives Bollinger-band contrarian trading from the identical statistic. **The same number is a momentum signal or a reversion signal, and the horizon determines which.**

Strengths. These measures are smooth and, unlike endpoint measures, use the whole path. They have low turnover, are extremely well understood, and are cheap. The displacement form measures the excess of price above a fair recent level.

Weaknesses. They lag. An MA_n lags price by roughly $(n-1)/2$ bars for an SMA, or $1/\alpha - 1$ bars for an EMA. Crossovers whipsaw in ranging markets. In that case the crossover passes a frequency band where the variance ratio is below 1. Crossovers also have two parameters instead of one.

Cost. SMA: $O(1)$ streaming with a ring buffer. EMA: $O(1)$ with a scalar. Naive: $O(n)$ per bar.

Robustness. High for the displacement form and moderate for crossovers. For crossovers, the ratio n_s/n_f matters more

than either level. **[Practice]** Ratios of 3:1 to 4:1 are conventional, and roughly optimal in the sense of maximizing band-pass gain per unit of turnover.

Failure modes. Choppy, ranging markets produce a rapid sequence of small losses, the typical loss pattern of a trend follower. Formally, crossover P&L is $\propto \mathrm{VR}(q) - 1$ at the band's centre frequency q. So if that band is mean-reverting, the losses are systematic, not random. Gaps make the MA and the price diverge discontinuously. On assets with strong seasonality, the band-pass may coincide with the seasonal frequency and pick up pure noise.

When preferred. Use these measures whenever turnover matters and the signal should use the whole path. **[Practice]** Ensembles of crossovers at several horizons are close to the industry standard for futures trend. An example is spans 8/24, 16/48 and 32/96, following the Baz et al. / Man AHL convention. Averaging across horizons is far more robust than choosing one.

4.1.6 MACD

Intuition. MACD is a crossover with a second smoothing applied to the crossover itself. It gives both the level of the trend and its rate of change.

Definition. With the conventional spans 12, 26 and 9:

$$\mathrm{MACD}_t = \mathrm{EMA}_{12}(P)_t - \mathrm{EMA}_{26}(P)_t$$
$$\mathrm{Signal}_t = \mathrm{EMA}_9(\mathrm{MACD})_t, \qquad \mathrm{Histogram}_t = \mathrm{MACD}_t - \mathrm{Signal}_t$$

A **volatility-normalized MACD** is far more useful, and it is what a professional would actually build. The construction follows Baz et al. (2015):

$$y_t = \frac{\mathrm{EMA}_{n_f}(P)_t - \mathrm{EMA}_{n_s}(P)_t}{\mathrm{std}_{63}(P)_t}, \qquad z_t = \frac{y_t}{\mathrm{std}_{252}(y)_t}, \qquad u_t = \frac{z_t \exp(-z_t^2/4)}{0.89}$$

The three steps are three separate normalizations.

1. The first divides a crossover in price units by a 63-day standard deviation *of prices*. This makes y_t dimensionless.
2. The second divides by the trailing 252-day standard deviation of y itself. This puts z_t on a roughly unit-variance scale, comparable across assets and eras.
3. The third applies a smooth **response function** $z \mapsto ze^{-z^2/4}$. The function is linear near zero, peaks at $z = \sqrt{2}$ with value $\sqrt{2}e^{-1/2} \approx 0.86$, and decays back toward zero for large $|z|$. The divisor 0.89 rescales it so that the maximum response is close to 1.

So the transform boosts moderate signals and *attenuates extreme ones*. It reflects the empirical view that very extreme trends are near exhaustion. This differs from clipping. Clipping holds an extreme signal at its cap, whereas this function reduces it toward zero. **[Practice]** This attenuation is common in production trend systems. It is one of the few areas where practitioner practice is clearly ahead of the academic literature.

Assumptions. The same as for the crossover, plus one more: the *acceleration* of the trend, shown by the histogram, carries information beyond its level.

Strengths. It combines level and change. The histogram is a genuine second-derivative signal, and everyone understands the indicator.

Weaknesses. It has three parameters, none justified beyond convention. In its raw form it is **not normalized**, so it is not comparable across assets, or across time within one asset. That alone disqualifies raw MACD from any cross-sectional application. Retail traders have heavily overfit it, so treat any published MACD backtest as data-snooped.

Cost. $O(1)$ streaming, with three EMA states.

Robustness. The *normalized* version is robust. The raw version is not, and the 12/26/9 parameters carry no special status.

Failure modes. Rules based on divergence ("price makes a new high, MACD does not") are the classic MACD application. They are notoriously prone to hindsight bias. Divergences are obvious after the fact. In real time they are frequent but unreliable. **[Practice] Recommendation: do not build a strategy on divergence signals without an unusually careful out-of-sample study.**

When preferred. Use the volatility-normalized form as one member of a multi-horizon trend ensemble. Raw MACD is a legacy artifact.

4.2 The regression family

These measures fit a model to the price path and use its parameters. They use the whole path by construction, and they provide a natural measure of *confidence*.

4.2.1 Rolling linear-regression slope

Intuition. Fit a straight line through the last L log prices. Its slope is the drift rate.

Definition. Regress $p_{t-L+1..t}$ on time $\tau = 1..L$:

$$p_\tau = a + b\tau + \varepsilon_\tau, \qquad \hat{b}_t = \frac{\sum_\tau (\tau - \bar{\tau})(p_\tau - \bar{p})}{\sum_\tau (\tau - \bar{\tau})^2} = \frac{\mathrm{Cov}(\tau, p)}{\mathrm{Var}(\tau)}$$

$\hat{b}_t$ is a log return per bar. Annualize it by multiplying by the number of bars per year. It is sometimes reported as the **annualized exponential regression slope**, $\left(e^{\hat{b}}\right)^{252} - 1$.

The kernel view. $\mathrm{Var}(\tau)$ is constant for a fixed window. So $\hat{b}_t$ is a linear combination of the *log prices*, with weights $\propto (\tau - \bar{\tau})$: negative on the first half of the window and positive on the second. Rewrite it in terms of *returns* by substituting $p_\tau = p_{t-L+1} + \sum_{u \le \tau} r_u$ and collecting terms. The result is a strictly positive, symmetric kernel weighted toward the centre:

$$\hat{b}_t \;=\; \frac{6}{L(L^2-1)} \sum_{k=0}^{L-2} (k+1)\,(L-1-k)\; r_{t-k}$$

The weights form a downward parabola, zero at both ends of the window and largest in the middle. This is why the slope is **much less sensitive to endpoints** than a lookback return. The advantage is genuine and underappreciated. It is the "regression slope" entry in the kernel chart of §4.1.5. The two views do not conflict. The weights have both signs and a ramp shape *on prices*, and they are positive and parabolic *on returns*.

Assumptions. The log price is locally linear in time, with constant drift over the window. The errors are additive, homoskedastic and uncorrelated. The last assumption is emphatically false for financial data. Financial errors show autocorrelation and heteroskedasticity. This corrupts the *standard error* but leaves $\hat{b}$ itself unbiased.

Strengths. The slope uses the whole path and is robust to endpoint noise. Its units are interpretable, as drift per unit of time. It yields R^2 and a t-statistic as byproducts at no extra cost, and it extends naturally to weighted or robust regression.

Weaknesses. It assumes a linear trend in log space, so it under-responds to accelerating trends and misspecifies curved paths. It is more expensive than a difference. It is still sensitive to the window edges, though less so than to endpoints.

Cost. Naive: $O(L)$ per bar per asset. **Streaming:** $O(1)$. Maintain running sums $\sum p$, $\sum \tau p$ and $\sum p^2$, with a ring buffer to subtract the departing observation. $\sum \tau$ and $\sum \tau^2$ are constants. The streaming version is worth implementing, because the naive version is a real bottleneck at $N \times L$ scale. Beware catastrophic cancellation in long-running sums. Use Welford-style updates, or recompute periodically in float64.

Robustness. Good. It is notably more robust than the lookback return to single-bar data errors, because a bad point in the middle of the window has bounded leverage.

Failure modes. A single extreme outlier at the *edge* of the window has maximum leverage on the slope, because regression leverage is highest at extreme x. Non-stationary volatility makes OLS over-weight the high-volatility part of the window. A log-linear fit badly misrepresents a parabolic blow-off.

When preferred. Use it when an interpretable drift estimate and a confidence measure are needed together, when data quality is imperfect, and when comparing trends across assets with different price levels. **[Practice]** It is widely used in equity momentum screens. The "annualized exponential regression slope × R^2" construction, popularized in Clenow's books, is a composite of slope and confidence.

4.2.2 Regression t-statistic

Intuition. A slope of 0.1% a day is strong evidence if the fit is tight and weak evidence if the fit is noisy. Divide the slope by its standard error.

Definition.

$$t_t = \frac{\hat{b}_t}{\mathrm{SE}(\hat{b}_t)}, \qquad \mathrm{SE}(\hat{b}_t) = \sqrt{\frac{\hat{\sigma}_\varepsilon^2}{\sum_\tau (\tau - \bar{\tau})^2}}, \qquad \hat{\sigma}_\varepsilon^2 = \frac{1}{L-2}\sum_\tau \hat{\varepsilon}_\tau^2$$

Since $\sum_\tau (\tau - \bar{\tau})^2 = L(L^2-1)/12$, this gives a useful scaling:

$$t_t \;=\; \hat{b}_t \,\frac{\sqrt{L(L^2-1)/12}}{\hat{\sigma}_\varepsilon} \;\sim\; \frac{\hat{b}_t}{\hat{\sigma}_\varepsilon}\cdot\frac{L^{3/2}}{\sqrt{12}}$$

At a fixed *per-bar* ratio of drift to noise, b/σ_ε, t grows like $L^{3/2}$. **Longer windows mechanically produce larger t-statistics for the same underlying trend strength.** Ranking signals by t across different lookbacks therefore systematically favors the longest one. Either fix L across the comparison, or rescale explicitly. When rescaling, choose the quantity carefully, because the two natural choices differ:

$$\frac{\hat{b}_t}{\hat{\sigma}_\varepsilon} = t_t \cdot \frac{\sqrt{12}}{L^{3/2}} \quad \text{(drift per bar, in per-bar noise units)}$$

$$\frac{\sqrt{L}\,\hat{b}_t}{\hat{\sigma}_\varepsilon} = t_t \cdot \frac{\sqrt{12}}{L} \quad \text{(drift over the whole window, in window-return units)}$$

The first is a per-bar Sharpe ratio. The second is the regression analogue of the volatility-normalized momentum of §4.3.1, $r_{t-L:t}/(\hat{\sigma}\sqrt{L})$. The second is usually the one to use when blending horizons in an ensemble. Analysts sometimes suggest rescaling by $t/\sqrt{L}$. This is neither of the two. It equals $\sqrt{12}\,L\,\hat{b}_t/\hat{\sigma}_\varepsilon$, which still grows linearly in L.

Assumptions. All the OLS assumptions apply, *plus* correct standard errors, which require iid homoskedastic residuals. **This assumption fails badly on financial data.** Inference requires HAC standard errors (Newey–West, 1987). For *signal construction*, the naive t is still a usable monotone transform, but do not read its size as a p-value.

Strengths. It is automatically volatility-normalized by the $\hat{\sigma}_\varepsilon$ in the denominator, so it is directly comparable across assets. That is its main advantage over the raw slope. It combines strength and consistency in one dimensionless number.

Weaknesses. It conflates the drift and its precision, two quantities an analyst may want separately. The $L^{3/2}$ scaling invalidates comparison across horizons. It is sensitive to residual autocorrelation, and it has no upper bound.

Cost. $O(1)$ streaming, with the same running sums as §4.2.1 plus $\sum p^2$.

Robustness. Good as a ranking statistic. Poor as an inferential statistic without HAC correction.

Failure modes. A collapse in volatility inflates t spuriously, through a small $\hat{\sigma}_\varepsilon$ in the denominator. A quiet, drifting market can produce enormous t-statistics that then reverse violently. Consider flooring $\hat{\sigma}_\varepsilon$ at a percentile of its own history. Residual autocorrelation can inflate t by a factor of 2–3.

When preferred. Use it for cross-sectional ranking across assets of different volatility, for screening universes, and whenever confidence in the trend should enter the signal. **[Practice]** It is a very common professional choice for equity momentum screens, precisely because of its built-in normalization.

4.2.3 R^2 and slope×R^2 composites

The regression R^2 measures how *linear*, and so how consistent, the trend has been. On its own it has no direction. It has two standard uses:

$$\text{Composite:}\quad \hat{b}_t \cdot R_t^2 \qquad\qquad \text{Filter:}\quad \hat{b}_t \cdot \mathbb{1}\{R_t^2 > c\}$$

The composite down-weights erratic trends smoothly, and the filter excludes them. **[Practice]** Both are common. The composite is preferable, because thresholds cause abrupt jumps in turnover.

Since $t^2 = \frac{R^2}{1-R^2}(L-2)$, slope, t and R^2 are three views of two underlying quantities, drift and noise. Using all three as separate ML features adds collinearity, not information.

4.2.4 The equivalence result to internalize

Levine & Pedersen (2016) show that time-series regression signals, moving-average crossovers and "past return" signals produce highly similar trend portfolios when their **horizons match**. The practical corollary follows:

> **Choosing the functional form of a trend measure is a low-value decision. Choosing the horizon (and the normalization) is a high-value decision.**

Spend the research budget accordingly. **[Fact, within the trend-following literature]** Researchers have robustly replicated the finding across futures trend research. The equivalence is looser in cross-sectional equity applications, where differences in normalization matter more.

4.3 The normalization family

Normalization adds most of the practical value in momentum signal construction. The raw price-difference measures of §4.1 are not comparable across assets, across time, or across horizons. Fixing that is worth more than any choice of indicator.

4.3.1 Volatility-normalized momentum

Intuition. A 10% move in an asset with 10% volatility is a three-sigma event. In an asset with 60% volatility, it is noise. Measure movement in units of its own noise.

Definition. The core construction appears in two forms:

$$s_t = \frac{r_{t-L:t}}{\hat{\sigma}_t \sqrt{L}} \qquad \text{(window displacement, in standard deviations of the window)}$$

$$s_t^{\text{bar}} = \frac{r_{t-L:t}/L}{\hat{\sigma}_t} \qquad \text{(drift per bar, in units of per-bar vol)}$$

where $\hat{\sigma}_t$ is an estimate of per-bar return volatility. The two are **not interchangeable**. Since $s_t^{\text{bar}} = s_t/\sqrt{L}$, they agree only up to a factor that depends on the horizon. For a single fixed L, either is fine, since a constant absorbs the difference. Across horizons, only the first is right, and confusing them is a common, silent bug.

The $\sqrt{L}$ explains this. Under a random walk, the L-period return has standard deviation $\sigma\sqrt{L}$. Dividing by $\hat{\sigma}_t\sqrt{L}$ therefore makes s_t approximately $N(0,1)$ *under the null of no momentum, at every* L. That common null scale lets an ensemble add signals from different lookbacks (§6.1). Dividing by L instead gives the intuitive "average return per bar" but shrinks long-horizon signals by $\sqrt{L}$ relative to short ones. A 252-day signal then enters an equal-weighted ensemble with roughly a third of the weight of a 21-day signal, which no designer intends.

The choice of $\hat{\sigma}$ matters more than people expect. The table lists the options in increasing order of statistical efficiency.

Estimator	Formula sketch	Notes
Rolling close-to-close SD	$\sqrt{\frac{1}{n-1}\sum(r_i - \bar{r})^2}$	Simplest and noisy. Equal weights are a poor model of vol dynamics
EWMA (RiskMetrics)	$\hat{\sigma}_t^2 = \lambda\hat{\sigma}_{t-1}^2 + (1-\lambda)r_t^2$	$O(1)$. $\lambda \approx 0.94$ daily is the classic value. Responsive
GARCH(1,1)	$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta\sigma_{t-1}^2$	Best pure-return model. Needs fitting. Mean-reverts to $\omega/(1-\alpha-\beta)$
Parkinson (1980)	$\frac{1}{4\ln 2}\left(\ln \text{Hi}_t/\text{Lo}_t\right)^2$	Uses the bar range. About 5× more efficient than close-to-close

Estimator	Formula sketch	Notes
Garman–Klass (1980)	uses open, high, low, close	More efficient still. Assumes no drift and no jumps
Yang–Zhang (2000)	combines overnight + open-to-close	Handles opening gaps and drift. **[Practice]** Often the best single choice for daily bars
Realized volatility	$\sum_{\text{intraday}} r_i^2$	Most efficient with intraday data. Needs microstructure-noise handling

[Fact] Baltas & Kosowski (2020) show that a more efficient volatility estimator in a TSMOM strategy materially reduces turnover with no statistically significant loss of performance. In their results the reduction exceeds a third. Volatility estimation matters.

Assumptions. Volatility is (a) persistent enough to be forecast from its recent history, and (b) the right scale for the signal. Both assumptions are well supported. A further implicit assumption is that expected return scales with volatility, so that dividing gives a stationary quantity.

Strengths. Normalization makes signals comparable across assets and across time. It removes the mechanical bias toward high-volatility names, and it greatly improves the quality of cross-sectional signals. It also uses the *most* predictable feature of returns, volatility, to scale the signal for the *least* predictable one, direction.

Weaknesses. It introduces a second estimation problem, with its own lookback parameter. In a volatility *collapse*, the denominator shrinks and the signal becomes very large, which is a real and dangerous failure mode. It also implicitly assumes the Sharpe-ratio target of §4.0. If the target is actually expected return, this is the wrong transform.

Cost. $O(1)$ streaming with EWMA volatility. $O(n)$ or more for GARCH refitting: fit infrequently and filter continuously.

Robustness. High. **[Practice] Recommendation: apply volatility normalization by default.** It is probably the single highest-value transformation among all signal transformations.

Failure modes.

- **Denominator collapse:** floor $\hat{\sigma}$ at, say, its 10th historical percentile, or blend it with a long-run estimate.
- **Mismatched volatility lookback:** if the volatility window is much shorter than the signal window, the signal carries the noise of the volatility window. **[Practice]** A volatility lookback of the same order as the signal lookback, or somewhat shorter, is conventional.
- **Regime shift in volatility:** after a structural change in volatility, the normalization is wrong for as long as the volatility window.
- **Stale volatility going into a crisis:** a backward-looking $\hat{\sigma}$ underestimates risk at the start of a shock. Signals *and* positions are then too large exactly when they should not be.

When preferred. Nearly always. In particular: whenever combining assets, ranking cross-sectionally, or building a multi-horizon ensemble.

4.3.2 Sharpe-like momentum

Intuition. Compute the *realized Sharpe ratio* over the lookback, and use it as the signal. A past with a high Sharpe ratio is a strong, consistent past.

Definition.

$$\text{SharpeMOM}_t(L) = \frac{\bar{r}_{t-L:t}}{\hat{\sigma}_{t-L:t}}\sqrt{A}$$

where $\bar{r}$ is the mean per-bar log return over the window and $\hat{\sigma}$ is its standard deviation over the *same* window. A is the number of bars per year, used for annualization.

Relationship to other measures. SharpeMOM is a rescaled t-statistic: $t = \bar{r}/(\hat{\sigma}/\sqrt{L}) = \text{SharpeMOM} \cdot \sqrt{L}/\sqrt{A}$. So Sharpe momentum, the t-statistic of a mean, and volatility-normalized momentum belong to one family. The regression t of §4.2.2 differs only in fitting a time trend rather than a mean. Its kernel on returns is parabolic and weighted toward the centre, where the mean's kernel is rectangular.

Assumptions. The Sharpe ratio inside the window predicts the Sharpe ratio after it. The *risk-adjusted* drift persists, not just the raw drift. A further assumption is that the in-window volatility is the right conditioning scale, rather than a forward-looking or longer-run estimate.

Strengths. It is dimensionless and directly comparable across assets. It ranks assets by exactly the quantity a mean–variance optimizer uses, and it penalizes erratic paths automatically.

Weaknesses. Using the *same window* for numerator and denominator induces a subtle negative dependence, because a large return in the window raises both. This attenuates extreme signals. The attenuation is sometimes desirable, but it shrinks the signal unintentionally. The measure is also very noisy for short L. The standard error of a Sharpe estimate on n observations is roughly $\sqrt{(1 + \text{SR}^2/2)/n}$ (Lo, 2002). A six-month window of daily data has $n \approx 126$, which gives a standard error of about $1/\sqrt{126} \approx 0.09$ *in per-day units*. Annualizing multiplies by $\sqrt{252} \approx 15.9$. So the annualized Sharpe ratio estimated over six months has a standard error above **1.4**, wider than the entire plausible range of true Sharpe ratios. **Sharpe estimates from short windows are almost pure noise.** Too few practitioners appreciate this.

Cost. $O(1)$ streaming, with running mean and variance via Welford.

Robustness. Moderate. Good for $L \geq$ 1 year of daily data, and poor below about 3 months.

Failure modes. Assets with low-volatility drift produce enormous Sharpe momentum right up until they break. Examples are a pegged currency, a short-volatility strategy, and an illiquid asset with stale prices. **This measure systematically over-ranks assets with understated volatility, caused by stale or smoothed pricing.** Guard against it with a liquidity filter and a stale-price detector.

When preferred. Use it for cross-sectional ranking over long lookbacks, in multi-asset-class contexts where volatilities differ by an order of magnitude, and wherever the downstream consumer is a mean–variance optimizer.

4.3.3 Z-scored momentum

Intuition. Express the signal in standard deviations relative to a reference distribution. Two very different reference distributions are in use, and conflating them is a common error.

Definition.

$$\text{Time-series z:} \quad z^{TS}_{i,t} = \frac{s_{i,t} - \mu_{i,t}(s)}{\sigma_{i,t}(s)} \qquad \text{(vs. this asset's own signal history)}$$

$$\text{Cross-sectional z:} \quad z^{CS}_{i,t} = \frac{s_{i,t} - \frac{1}{N}\sum_j s_{j,t}}{\text{sd}_j(s_{j,t})} \qquad \text{(vs. today's peers)}$$

The two answer different questions. The time-series z measures whether this asset is trending unusually strongly *for itself*. The cross-sectional z measures whether this asset is trending strongly *relative to its peers today*. The cross-sectional version demeans the signal, so it mechanically removes the market or common component. The resulting portfolio is dollar-neutral by construction. The time-series version does not remove the common component, and it leaves net market exposure.

Assumptions. For TS-z, the signal's own distribution is stationary over the standardizing window. For CS-z, the cross-section is comparable: same asset class, similar liquidity, similar volatility, and no strong industry clustering.

Strengths. Both put everything on a common scale for combination, which makes ensembles across measures and horizons meaningful. CS-z neutralizes common factors automatically. Both are easy to interpret.

Weaknesses. Standardization is sensitive to outliers *in the standardizing set*. A single extreme value inflates the denominator and shrinks everything else. **[Practice]** Winsorize at ±3σ first, or use a robust z based on the median absolute deviation, $(s - \text{median})/(1.4826 \cdot \text{MAD})$, instead. CS-z assumes a roughly symmetric cross-section. With skewed signals, ranks are safer.

Cost. TS-z: $O(1)$ streaming. CS-z: $O(N)$ per rebalance, or $O(N \log N)$ with ranks, which is negligible.

Robustness. High with robust statistics, and moderate without.

Failure modes.

- **Small N:** with $N < 30$, estimation noise in the cross-sectional moments dominates the cross-sectional z.
- **Non-stationary cross-section:** in a crisis, cross-sectional dispersion rises sharply, so today's z-scores are not comparable to last month's. Sizing on z then lets gross exposure change silently with dispersion.

- **Regime change in the signal's own distribution** invalidates TS-z.
- **Double normalization:** normalizing by volatility and *then* z-scoring over a window that also reflects volatility changes can over-shrink. Know what each step removes.

When preferred. Use them whenever combining several signals. CS-z is the standard preprocessing for cross-sectional equity factors. **[Practice]** Equity work often prefers rank-based normalization to z-scoring, precisely because ranks are fully immune to outliers. Analysts can map ranks to $[-1, 1]$ or convert them with a normal-score (Van der Waerden) transform. Ranking discards magnitude information, which matters if genuine signal strength varies.

4.3.4 Signal transforms: sign, rank, clip, and squash

The final step from a normalized signal to a position deserves explicit thought. The table lists the common choices, each of which assumes something different about the signal.

Transform	Form	Encodes
Sign	$\text{sign}(s_t)$	Only the direction is informative, and magnitude is noise. It is maximally robust, discards information and has minimal turnover
Linear	s_t	Signal magnitude is proportional to expected Sharpe. Highest turnover, most sensitive to outliers
Clipped linear	$\text{clip}(s_t, -c, c)$	Linear but bounded. The practical default
Rank	cross-sectional rank scaled to $[-1, 1]$	Only ordering is informative. Robust to outliers and to distributional change
tanh **/ squash**	$\tanh(ks_t)$	Smooth saturation. Differentiable (matters for ML)
Response function	$s\, e^{-s^2/4}$	Saturation *and decay*. The function reduces extreme trends toward zero

[Fact] Moskowitz, Ooi & Pedersen (2012) found that the *sign* of the past 12-month return performs comparably to versions scaled by magnitude in futures TSMOM. That is strong evidence that most of the information is in the direction. **[Contested]** Other studies find a modest benefit from scaling by magnitude. This chapter takes the view that the sign is a remarkably strong baseline, and any use of magnitude needs evidence that it adds value.

4.4 The oscillator family

Oscillators map the price path into a bounded range. They usually do so by comparing the current level with a recent range, or by comparing up-moves with down-moves. Their designers intended them for chart reading, and analysts *widely misuse* them. Most of them identify overbought and oversold conditions, which is a mean-reversion use, yet they are widely called "momentum indicators."

The essential clarification. Readers can interpret a bounded oscillator in two ways:

- **As a momentum signal:** a high value means a strong uptrend, so go long. This is trend continuation.
- **As a reversion signal:** a high value means overbought, so go short. This is mean reversion.

These are opposite trades from the same number. **The horizon, and the variance ratio at that horizon, entirely determine which reading is right. The indicator does not.** A source that says "RSI > 70 means sell" without specifying a horizon or providing evidence states an unsupported rule.

4.4.1 RSI (Relative Strength Index)

Intuition. RSI measures the fraction of total absolute movement over the last n bars that was upward.

Definition (Wilder, 1978). Let $U_i = \max(r_i, 0)$ and $D_i = \max(-r_i, 0)$, and use Wilder's smoothing, which is an EMA with $\alpha = 1/n$:

$$\overline{U}_t = \frac{(n-1)\overline{U}_{t-1} + U_t}{n}, \quad \overline{D}_t = \frac{(n-1)\overline{D}_{t-1} + D_t}{n}, \quad \mathsf{RS}_t = \frac{\overline{U}_t}{\overline{D}_t}$$

$$\mathsf{RSI}_t = 100 - \frac{100}{1 + \mathsf{RS}_t} = 100 \cdot \frac{\overline{U}_t}{\overline{U}_t + \overline{D}_t}$$

The last form is the most informative. **RSI is the smoothed share of up-moves in total absolute movement, rescaled to [0,100].** It is a normalized measure whose normalizer is total absolute movement, a robust proxy for volatility. A driftless symmetric series has an RSI of 50, and the signal is the deviation from 50.

Two practical notes apply. First, the name is a historical misnomer. RSI is a *single-asset* statistic, unrelated to cross-sectional relative strength (§4.6.3). Second, Wilder's smoothing is slower than its parameter suggests. With $\alpha = 1/n$, its centre of mass is $n - 1$ bars. By the identity of §4.1.4, that corresponds to a conventional EMA **span of** $2n - 1$. So the standard RSI(14) behaves like a 27-bar EMA, not a 14-bar one. Match horizons accordingly before comparing it with anything else.

Assumptions. The ratio of up-movement to total movement over n bars is informative, and the two are on the same scale, which holds by construction.

Strengths. RSI is bounded, so it is directly comparable across assets and time with no further normalization. That is a real advantage. It is robust to outliers compared with a raw return, because a single huge day contributes to both numerator and denominator. It is $O(1)$ streaming. It also saturates non-linearly, which is often desirable.

Weaknesses. The bounds are also the weakness. RSI **saturates** in strong trends. It stays at 80 or above for months and carries no further information exactly when the trend is strongest. The conventional 30/70 thresholds are convention only and have no derivation. Wilder's smoothing differs from the "standard" EMA, so implementations disagree, and the first n bars depend on initialization. Different data vendors report different RSI values for the same series, so verify any implementation against a known reference.

Cost. $O(1)$ streaming, with two EMA states.

Robustness. The structure is robust, because it is bounded and self-normalizing. The *thresholds* are not robust at all, and they are the most overfit parameters in technical analysis.

Failure modes.

- **Trend saturation:** shorting an "overbought" RSI in a strong trend is a classic way to lose money continuously. **[Fact]** In assets with positive drift, an unconditional "sell when RSI > 70" rule loses money.
- **Implementation divergence** across libraries: Wilder, simple and exponential smoothing differ.
- **Threshold overfitting:** the 30/70 levels are the single most data-snooped parameters in retail trading.

When preferred. Its most defensible modern use is as a *bounded, self-normalizing feature* in an ML model or an ensemble. It also works as a short-horizon **mean-reversion** signal in the band from 1 day to 2 weeks, where the variance ratio is genuinely below 1. Equity data support this use. **[Practice]** Professionals rarely use RSI as a standalone momentum signal. They commonly use it as one feature among many.

4.4.2 Stochastic oscillator

Intuition. The stochastic oscillator locates the current close within the recent high–low range. A close near the top of the range indicates strength.

Definition (Lane). Over n bars:

$$\%K_t = 100 \cdot \frac{P_t - \min_{i \in [t-n+1,\, t]} \mathsf{Lo}_i}{\max_{i \in [t-n+1,\, t]} \mathsf{Hi}_i - \min_{i \in [t-n+1,\, t]} \mathsf{Lo}_i}, \qquad \%D_t = \mathsf{SMA}_3(\%K)_t$$

Assessment. $\%K$ is the **normalized position within the recent range**. It is the same idea as the Donchian channel position and the 52-week-high measure (both in §4.4.3), with a shorter window and a percentage scale. It uses high and low data, which carry more information than closes alone.

Assumptions. Range position, not net displacement, is the informative summary. This statistic is *rank-like*, so it is robust to the distribution of returns. Analysts underrate this strength.

Strengths. It is bounded and robust, being essentially an order statistic. It uses information from within the bar, and it is cheap.

Weaknesses. Three numbers determine it entirely: the current close, the window maximum and the window minimum. So it discards the shape of the path. It is extremely sensitive to a single extreme high or low, which sets the denominator for n bars. It saturates in trends, like RSI, and it is very noisy at small n.

Cost. Naive $O(n)$. Streaming $O(1)$ amortized, using monotonic deques for the rolling minimum and maximum. That implementation is worthwhile for large $N \times T$.

Robustness. Moderate. The rank-like construction is robust, and the dependence on the minimum and maximum is not.

Failure modes. A single erroneous tick in the high or low corrupts the indicator for n bars. In most vendor feeds, bad high and low data are far more common than bad closes. The indicator saturates in trends. On gappy or thin instruments, the range is unrepresentative.

When preferred. Use it for short-horizon reversion, as a bounded ML feature, and when the analyst specifically wants information about the range within the bar. **[Practice]** It is more common in intraday work than in daily systematic work.

4.4.3 Channel position, Donchian breakout, and the 52-week high

Intuition. These measures test whether price is at a new extreme relative to its recent history.

Definition.

$$\text{Donchian breakout:} \quad \text{long if } P_t > \max_{i \in [t-n,\, t-1]} \text{Hi}_i, \quad \text{short if } P_t < \min_{i \in [t-n,\, t-1]} \text{Lo}_i$$

$$\text{Channel position:} \quad C_t(n) = \frac{P_t - \min_{i \in [t-n,\, t]} \text{Lo}_i}{\max_{i \in [t-n,\, t]} \text{Hi}_i - \min_{i \in [t-n,\, t]} \text{Lo}_i} \in [0, 1]$$

$$\text{52-week high proximity:} \quad \Phi_t = \frac{P_t}{\max_{i \in [t-252,\, t]} P_i} \in (0, 1]$$

The breakout rule excludes the current bar from its window. It compares today's price with the extremes of the previous n bars. Channel position includes the current bar. Using the same window for both would make a breakout impossible by construction, since P_t can never exceed a maximum it is part of.

Why this deserves separate treatment. [Fact, replicated] George & Hwang (2004) found that proximity to the 52-week high predicts returns *and largely subsumes* conventional cross-sectional momentum in their sample. The result is striking, because Φ_t uses only two prices and no return path at all. The interpretation is behavioral. The 52-week high is a psychologically salient **anchor**, and investors under-react to news that would push price through it. A behavioral mechanism makes a sharp prediction here, and the data confirm it. This is one of the cleanest such cases.

Assumptions. Extremes are salient reference points, and breaking one signals information not yet impounded. For the Donchian version, the distribution of future returns after a new n-bar extreme shifts favorably.

Strengths. These measures are extremely simple and non-parametric, using order statistics only. They are robust to the return distribution. They naturally produce the asymmetric, positively skewed payoff of trend following. The 52-week-high version has strong independent empirical support.

Weaknesses. The binary (breakout) versions depend severely on the path at the threshold, and repeated marginal breaks produce whipsaw. The maximum and minimum are maximally sensitive to outliers. The breakout form carries no information about magnitude. It is discontinuous, and so has high turnover near thresholds.

Cost. $O(1)$ amortized streaming with monotonic deques. $O(n)$ naive.

Robustness. The continuous forms, channel position and 52-week high, are robust. The binary breakout form is much less robust, and its performance is sensitive to n.

Failure modes. Whipsaw around the threshold is the dominant failure. The classic practitioner fix uses asymmetric entry and exit channels, for example a 55-bar entry and a 20-bar exit as in the Turtle system. That is a crude form of hysteresis. Bad high or low ticks also corrupt the signal. In markets that gap heavily, the "breakout price" is never available for execution.

When preferred. Use them in futures trend following, where breakouts remain a core primitive, and in equity momentum screens through 52-week-high proximity. They also suit any setting that needs a signal robust to assumptions about the return distribution. **[Practice]** Modern systematic work generally prefers continuous channel position to the binary breakout, because it carries the same information with far less turnover.

4.4.4 ADX and directional movement

Intuition. Separate two questions: *which direction* is price moving, and *how consistently*. ADX answers the second. It is the only classical indicator designed specifically to measure **trend quality rather than trend direction**.

Definition (Wilder, 1978). Directional movement per bar is

$$+\mathrm{DM}_t = \begin{cases} \mathrm{Hi}_t - \mathrm{Hi}_{t-1} & \text{if } (\mathrm{Hi}_t - \mathrm{Hi}_{t-1}) > (\mathrm{Lo}_{t-1} - \mathrm{Lo}_t) \text{ and } > 0 \\ 0 & \text{otherwise} \end{cases}$$

with $-\mathrm{DM}_t$ defined symmetrically by swapping the roles of the two comparisons. In words, a bar has up-directional movement only if it extended further above the previous bar's high than it extended below the previous bar's low. So at most one of $\pm$DM is nonzero on any bar. Let $\mathrm{TR}_t = \max(\mathrm{Hi}_t - \mathrm{Lo}_t,\ |\mathrm{Hi}_t - P_{t-1}|,\ |\mathrm{Lo}_t - P_{t-1}|)$ be the true range: the bar's range, widened to include any overnight gap. Let $\widetilde{\cdot}$ denote Wilder smoothing over n, with a default of 14. Then:

$$+\mathrm{DI}_t = 100\frac{\widetilde{+\mathrm{DM}}_t}{\widetilde{\mathrm{TR}}_t}, \qquad -\mathrm{DI}_t = 100\frac{\widetilde{-\mathrm{DM}}_t}{\widetilde{\mathrm{TR}}_t}$$

$$\mathrm{DX}_t = 100 \cdot \frac{|+\mathrm{DI}_t - -\mathrm{DI}_t|}{+\mathrm{DI}_t + -\mathrm{DI}_t}, \qquad \mathrm{ADX}_t = \widetilde{\mathrm{DX}}_t$$

How to read it. DX is the *normalized absolute imbalance* between up-directional and down-directional movement: a directional-consistency ratio in $[0, 100]$. ADX is its smoothed version. The absolute value makes it directionless by construction. **ADX is a conditioning variable, not a signal.**

Assumptions. The directional consistency of high and low extensions measures trend strength, and TR is the right normalizer. Both assumptions are heuristic but reasonable.

Strengths. It carries information genuinely orthogonal to direction, because it measures the *quality* of a trend. It is bounded, self-normalizing, and uses data from within the bar. Conceptually, it is a crude estimator of the same thing as the variance ratio and the efficiency ratio (§4.5): whether the market is trending or chopping.

Weaknesses. The definition has many parts and several arbitrary choices, such as the DM rule and the default of 14. The double smoothing makes it **very laggy**, so ADX typically confirms a trend well after the trend starts. Vendor implementations differ. It has no probabilistic interpretation.

Cost. $O(1)$ streaming, with several EMA states.

Robustness. Moderate. The concept is robust. The specific construction and the conventional threshold, ADX > 25 meaning "trending", are not.

Failure modes. Because of the lag, ADX often peaks near trend exhaustion. Using "rising ADX" as an entry filter therefore systematically enters late. A high ADX simply means *consistent movement*, which occurs both in healthy trends and in blow-off tops. It cannot distinguish the two.

When preferred. Use it as a **regime filter**, turning trend signals on when ADX is high and off when it is low, or as a continuous weight. **[Practice]** It is common as a filter. **Recommendation: for the same job, prefer the efficiency ratio or a variance-ratio estimate (§4.5).** Both are simpler and have a clearer statistical meaning. ADX's enduring popularity is partly historical.

4.5 Persistence and trend-quality metrics

These metrics say *whether the momentum hypothesis currently applies*. They do not say *which way* price will move. Given the regime dependence established in §1.4, this information is arguably more valuable.

4.5.1 Variance ratio (as a live statistic)

Definition. With T observations of r and aggregation q (Lo & MacKinlay, 1988):

$$\widehat{\mathrm{VR}}(q) = \frac{\hat{\sigma}_q^2}{q\,\hat{\sigma}_1^2}, \qquad \hat{\sigma}_q^2 = \frac{1}{m}\sum_{t=q}^{T}\left(\sum_{j=0}^{q-1} r_{t-j} - q\bar{r}\right)^2$$

Here $\hat{\sigma}_q^2$ is the variance of the *overlapping* q-period return, and $\hat{\sigma}_1^2$ is the variance of the one-period return. So the ratio is dimensionless and equals 1 under a random walk. The divisor m is $T-q+1$ for the simple estimator. Lo and MacKinlay's unbiased version uses $m = q(T-q+1)(1-q/T)$, which corrects the downward bias that comes from overlapping windows and from estimating $\bar{r}$. **Use the unbiased version for anything beyond a quick look.** The bias is largest exactly at the long horizons that matter most.

Under the random-walk null with heteroskedasticity, the standardized statistic $\sqrt{T}(\widehat{\mathrm{VR}}(q)-1)$ is asymptotically normal, with a computable variance. Lo and MacKinlay give both a homoskedastic version and a heteroskedasticity-robust version. **Use the robust version.** Financial returns are never homoskedastic, and under the naive version, volatility clustering alone produces apparent rejections of the random walk.

Strengths. The variance ratio is the *definition* of momentum, with a proper inferential theory. It finds the horizon at which momentum exists rather than assuming one.

Weaknesses. It needs a lot of data: estimating it precisely for q = 12 months takes decades. Overlapping windows induce autocorrelation in the estimator. It cannot serve as a fast-moving live signal.

Cost. $O(Tq)$ naive, and $O(T)$ with cumulative sums for each q.

When preferred. Use it in *research*, as the first diagnostic on any new market. Practitioners rarely use it as a live signal. [Practice] **Recommendation: before designing a momentum signal for an unfamiliar instrument, compute its variance-ratio profile.** Of all the habits in this chapter, this one is the most valuable.

4.5.2 Hurst exponent

Intuition. A single exponent describes how the range or variance of a series scales with the observation window. $H = 0.5$ is a random walk. $H > 0.5$ is persistent, or trending. $H < 0.5$ is anti-persistent.

Definition. For fractional Brownian motion, $\mathrm{Var}(p_{t+q} - p_t) \propto q^{2H}$. So regressing $\log \mathrm{Var}(p_{t+q} - p_t)$ on $\log q$ and halving the slope estimates H. The relation to the variance ratio is direct: $\mathrm{VR}(q) \propto q^{2H-1}$. **The Hurst exponent is a parametric summary of the variance-ratio profile.** It assumes the profile is a straight line in log-log space.

Strengths. It is one interpretable number, and the mental model ($H > 0.5$ means trend) is genuinely useful.

Weaknesses, which are severe. Estimates of H on financial data are notoriously unstable and depend on the estimator: rescaled range, detrended fluctuation analysis, wavelets and variance scaling all disagree. Short samples produce estimates of H biased away from 0.5. Volatility clustering alone can produce an apparent $H \neq 0.5$ with no return predictability whatsoever. And the fBm model that gives H its meaning describes returns poorly.

Cost. Typically $O(T \log T)$, and not suited to streaming.

Failure modes. The dominant failure is over-interpretation. An estimated $H = 0.58$ on 500 observations is very often statistically indistinguishable from 0.5. Always bootstrap a confidence interval. It will usually include 0.5.

When preferred. Use it for exploratory research on long histories. **[Practice] Recommendation: do not put a live Hurst estimate in a production signal without extraordinary evidence.** Treat published claims of regime switching based on the Hurst exponent skeptically.

4.5.3 Efficiency ratio (Kaufman)

Intuition. ER measures the net progress price makes per unit of total travel. A straight line scores 1, and a round trip scores 0.

Definition.

$$\mathrm{ER}_t(n) = \frac{|P_t - P_{t-n}|}{\sum_{i=t-n+1}^{t} |P_i - P_{i-1}|} \in [0,1]$$

Assessment. This is the cleanest trend-quality statistic available. It has a one-line formula, has no parameters beyond n, is bounded, and has an immediate geometric reading: net displacement divided by path length. A more irregular path has a longer path length for the same net displacement. So ER falls as the roughness of the path, its discrete fractal dimension, rises.

Calibrate it before using it. ER does *not* centre on a fixed value. Under a driftless random walk, the expected net displacement grows like $\sqrt{n}$, while the expected path length grows like n. So

$$\mathbb{E}[\mathrm{ER}_t(n)] \;\approx\; \frac{1}{\sqrt{n}} \qquad \text{(random-walk benchmark)}$$

That is 0.22 at $n = 20$ and 0.10 at $n = 100$. An ER of 0.3 is therefore evidence of trending at $n = 100$ and no evidence at all at $n = 10$. Any fixed threshold, such as "ER > 0.3 means trending", is silently a statement about one particular window length. The relation to the variance ratio is the same observation in a different form. The product $\mathrm{ER}_t(n)\sqrt{n}$ behaves like $\sqrt{\mathrm{VR}(n)}$, and ER itself does not. ER is an L^1 (absolute-deviation) analogue of the L^2 variance ratio. It is more robust to fat tails and has no inferential theory.

Kaufman's **Adaptive Moving Average (KAMA)** uses ER to interpolate the EMA smoothing constant between a fast and a slow value: $\alpha_t = [\mathrm{ER}_t(\alpha_{\text{fast}} - \alpha_{\text{slow}}) + \alpha_{\text{slow}}]^2$. The filter is then fast in trends and slow in choppy markets. The idea is elegant, and it is a good template for adaptive smoothing in general.

Strengths. ER is simple, bounded, cheap and interpretable. It is directionless, so it can serve as a pure conditioner. For most filtering purposes it is strictly preferable to ADX.

Weaknesses. It is sensitive to n. As shown above, its null level moves with n, so thresholds do not transfer across window lengths. A large gap inflates the numerator and denominator unequally. It has no inferential theory.

Cost. $O(1)$ streaming, with a rolling sum of $|\Delta P|$.

Failure modes. Under-sampling: with small n the null level is high (0.45 at $n = 5$) and the estimate is noisy, so short windows look "trending" almost all the time. Gappy markets distort it.

When preferred. Use it as a conditioner for regime or trend quality, and for adaptive smoothing. **[Practice]** Practitioners underuse it relative to its merit.

4.5.4 Autocorrelation, run statistics, and signal decay

Three direct persistence diagnostics are worth computing. The first is the rolling autocorrelation:

$$\text{Rolling autocorrelation:} \quad \hat{\rho}_k(t) = \frac{\sum (r_i - \bar{r})(r_{i-k} - \bar{r})}{\sum (r_i - \bar{r})^2}$$

It is very noisy. Under the null of independence, $\mathrm{SE}(\hat{\rho}_k) \approx 1/\sqrt{n}$. On 250 daily observations, the standard error is ≈ 0.063. An estimate must exceed roughly twice that, about ± 0.13, before it is distinguishable from zero at the 5% level. True daily ρ_1 is typically well under 0.05, so a one-year rolling window cannot detect it even in principle. **Rolling autocorrelation is essentially unusable as a live signal.** Reserve it for research on long samples. Run this calculation before writing any code.

The second is the decay of the signal's predictive power with horizon:

$$\text{Signal decay / IC term structure:} \quad \mathrm{IC}(h) = \mathrm{Corr}\big(s_t,\; r_{t+1:t+h}\big) \;\text{ as a function of } h$$

This is the most useful of the three persistence metrics, and it belongs in every research workflow. The IC term structure directly gives the optimal holding period: hold until the marginal IC no longer covers the marginal trading cost. It also reveals empirically where momentum turns to reversal, for the specific signal on the specific universe. That is far more actionable than any published horizon.

The third diagnostic uses **run statistics**, the distribution of the lengths of runs of same-sign returns, compared against a binomial null. This is a classic non-parametric test of independence. It is intuitive but weak, because it has low power against the small autocorrelations that actually generate momentum profits.

4.6 The relational family: the benchmark momentum is measured *against*

Every measure so far compared a single asset with its own past. This section changes the reference point. The choice of reference point is a **first-order decision**. It has larger consequences than any choice of indicator.

4.6.1 Time-series momentum (absolute momentum)

Intuition. Compare an asset with zero. Own it if it has been going up.

Definition. For asset i with lookback L and volatility target σ^*:

$$w_{i,t} = \frac{\sigma^*}{\hat{\sigma}_{i,t}} \cdot \text{sign}\big(r_{i,t-L:t}\big) \qquad \text{(Moskowitz, Ooi \& Pedersen, 2012)}$$

with the portfolio return $R^{\text{TSMOM}}_{t+1} = \frac{1}{N}\sum_i w_{i,t}\, r_{i,t+1}$.

The critical statistical caveat. Consider the expected profit of the unnormalized version:

$$\mathbb{E}\big[r_{i,t-L:t} \cdot r_{i,t+1}\big] \;=\; \underbrace{\sum_{k=1}^{L} \text{Cov}(r_{i,t+1-k},\, r_{i,t+1})}_{\text{genuine predictability}} \;+\; \underbrace{L\,\mu_i^2}_{\text{just a positive mean}}$$

The second term is positive whenever the asset has any nonzero unconditional drift, even with zero predictability. This is the core of the critique by Huang, Li, Wang & Zhou (2020). A TSMOM test that does not remove the unconditional mean partly tests whether the assets go up. That is already well known.

For a sign-based rule the algebra differs, but the same logic holds. $\mathbb{E}[\text{sign}(r_{t-L:t})r_{t+1}] > 0$ arises partly because $\Pr[\text{sign}(r_{t-L:t}) = +1] > 1/2$ when $\mu > 0$. **[Practice]** The fix is to demean, and there are three ways to do it. Test on returns in excess of the asset's own long-run mean. Alternatively, include a constant-long benchmark in the comparison. The best option is to evaluate the strategy against a passive long-only benchmark with matched volatility. A TSMOM strategy that does not beat a volatility-matched buy-and-hold has measured drift, not momentum.

Assumptions. The asset's own past predicts its own future. The sign is sufficient. Volatility is forecastable.

Strengths. It works on a single asset, with no cross-section needed. It can go short the whole market, so it naturally produces the convex, positively skewed "crisis alpha" profile. It is directly implementable in futures. It reduces risk explicitly through volatility targeting, and it is conceptually simple.

Weaknesses. It retains net market exposure. A TSMOM book is often net long, so part of its return is beta. It requires shorting, or at least the ability to go flat. It whipsaws in ranging markets. It is also vulnerable to the drift confound above.

Cost. $O(N)$ per rebalance, which is trivial.

Robustness. [Fact] Robustness is high across futures, and studies validate it over more than a century (Hurst, Ooi & Pedersen, 2017; Lempérière et al., 2014). **[Contested]** It is weaker in individual equities. The evidence is strongest for the diversified futures portfolio.

Failure modes. Sharp V-shaped reversals hurt most, because positions are maximally wrong at the turn. Sustained low-volatility ranges also hurt. During a macro shock, positions across a diversified book become correlated. The real diversification is then often much lower than the historical correlation matrix suggests, because trend books converge on the same trades. The dynamic-leverage adjustment of Baltas and Kosowski uses pairwise signed correlations to address exactly this.

When preferred. Use it in futures and macro trend programs, in overlay and tail-hedge applications, in any single-asset context, and whenever the strategy needs convexity.

4.6.2 Cross-sectional momentum (relative momentum)

Intuition. Compare each asset with its peers. Own the relative winners and short the relative losers.

Definition. Let the signal be $s_{i,t}$, typically $r_{i,t-12m:t-1m}$, and let $\bar{s}_t$ be its cross-sectional mean. Then

$$w_{i,t} = \frac{c}{N}\left(s_{i,t} - \bar{s}_t\right) \qquad \text{or, in the sorted-portfolio form,} \qquad w_{i,t} = \tfrac{1}{n}\mathbb{1}\{i \in \text{top decile}\} - \tfrac{1}{n}\mathbb{1}\{i \in \text{bottom decile}\}$$

By construction, $\sum_i w_{i,t} = 0$. The portfolio is dollar-neutral, but not beta-neutral.

The decomposition that matters. Follow Lo & MacKinlay (1990) and Jegadeesh & Titman (1995), and take the weighting scheme $w_{i,t} = \frac{1}{N}(R_{i,t-1} - \bar{R}_{t-1})$. Let Γ be the lag-1 cross-autocovariance matrix, with $\Gamma_{ij} = \mathsf{Cov}(R_{i,t-1}, R_{j,t})$, and let ι be a vector of ones. Then

$$\mathbb{E}[\pi_t] \;=\; \underbrace{\frac{N-1}{N^2}\,\mathsf{tr}(\Gamma)}_{\text{(A) own-autocovariance}} \;-\; \underbrace{\frac{1}{N^2}\sum_{i\neq j}\Gamma_{ij}}_{\text{(B) cross-serial (lead-lag)}} \;+\; \underbrace{\sigma_\mu^2}_{\text{(C) dispersion in means}}$$

where $\sigma_\mu^2 = \frac{1}{N}\sum_i(\mu_i - \bar{\mu})^2$.

The decomposition shows that cross-sectional momentum profits have **three completely different sources**:

- **(A)** Genuine persistence in individual returns. This is what everyone assumes momentum is.
- **(B)** Negative *lead-lag* structure. Suppose stock i's move today predicts stock j's move tomorrow *positively*, as when large caps lead small caps. Then term (B) *reduces* momentum profits. Contrarian profits come from that positive lead-lag. Lewellen (2002) argues that momentum in size and book-to-market portfolios comes largely from this channel and not from (A). The result is genuinely surprising.
- **(C)** Pure cross-sectional dispersion in *unconditional* expected returns. **This source requires no predictability whatsoever.** If some stocks simply have permanently higher expected returns, a strategy that buys past winners will overweight them and earn $\sigma_\mu^2 > 0$. This profit comes from risk-premium exposure and resembles a timing signal.

A profitable cross-sectional momentum backtest is not evidence of return persistence. A claim of persistence must show that the profit survives the removal of (C). One method demeans each asset's returns by its own full-sample mean. That is an in-sample adjustment. It is usable for attribution and not for trading. Another method tests within groups of stocks with similar expected returns ex ante.

Assumptions. The cross-section is comparable: same asset class, similar liquidity and risk. Relative ranking is the informative quantity. Shorting is feasible.

Strengths. It is market-neutral by construction, so it hedges the common factor and isolates the relative signal. It has large breadth. By the Fundamental Law (IR $\approx$ IC$\sqrt{\text{breadth}}$), that supports a much higher information ratio than TSMOM for the same IC. Because it is rank-based, it normalizes itself across time. It also has a huge academic literature and standard benchmarks.

Weaknesses. It requires a cross-section, so it is useless for a single asset. It requires shorting, and the costs, borrow constraints and crash risk concentrate on the short side. It is **prone to crashes**. The conditional-beta problem of §1.6 belongs specifically to cross-sectional momentum. Unless neutralized, it also takes unhedged industry and factor bets implicitly.

Cost. $O(N \log N)$ per rebalance for sorting, which is negligible.

Robustness. [Fact] Very high in equities across countries and centuries, and weaker but present in other asset classes.

Failure modes.

- **Momentum crashes** after bear markets (Daniel & Moskowitz, 2016).
- **Unintended factor bets:** winners and losers differ systematically in beta, industry, size and volatility. Without neutralization, the strategy is an uncontrolled factor portfolio.
- **An infeasible short leg:** historically, the profits concentrate in the short leg, where borrowing is hardest and costs are highest (Novy-Marx & Velikov, 2016).
- **Universe contamination:** including microcaps inflates paper returns dramatically. Traders cannot realize that inflation.

When preferred. Use it for equity long/short, for any large homogeneous cross-section, and whenever market neutrality is required.

4.6.3 Relative strength

"Relative strength" names two different things. Distinguish them before using the term.

(a) Ratio-based relative strength is a *pairwise* or *benchmark-relative* momentum:

$$\mathsf{RS}_{i,t} = \frac{P_{i,t}/P_{i,t-L}}{P_{B,t}/P_{B,t-L}} \qquad \text{or, in log form,} \qquad r_{i,t-L:t} - r_{B,t-L:t}$$

This is momentum of the *ratio series* P_i/P_B, the relative price. It is the natural signal for sector, country and style rotation, and the "relative rotation graph" family visualizes it.

(b) Rank-based relative strength is the cross-sectional percentile of the momentum signal, such as the IBD "RS rating", a percentile from 1 to 99. It is simply the signal of §4.6.2 expressed as a rank.

Assessment. Form (a) is genuinely distinct and useful. It is a two-asset time-series momentum on a spread, and it inherits the properties of §4.6.1 applied to the ratio. Its main hazard is that the ratio's volatility differs from the asset's volatility. So re-estimate σ on the ratio series, not on the legs. Form (b) is cross-sectional momentum. Treat it as such.

When preferred. Use it for sector, country and style rotation. Use it for pairs and spread trading. Use it for benchmark-relative mandates whose objective genuinely is relative return.

4.6.4 Residual (idiosyncratic) momentum

Intuition. Remove the part of an asset's past return that common factors explain, and rank on the remainder. If Nvidia rose 40% because semiconductors rose 40%, the move is a sector exposure and carries no stock-specific signal.

Definition (Blitz, Huij & Martens, 2011). For each asset, estimate a factor model over a rolling window, typically 36 months:

$$r_{i,\tau} = \alpha_i + \beta_i^{\mathsf{MKT}}\mathsf{MKT}_\tau + \beta_i^{\mathsf{SMB}}\mathsf{SMB}_\tau + \beta_i^{\mathsf{HML}}\mathsf{HML}_\tau + \varepsilon_{i,\tau}$$

Then form the momentum signal on the standardized residuals over the ranking window:

$$s_{i,t}^{\mathsf{resid}} = \frac{\sum_{\tau=t-12}^{t-2} \hat{\varepsilon}_{i,\tau}}{\hat{\sigma}(\hat{\varepsilon}_i)\sqrt{11}}$$

Assumptions. The factor model is correctly specified, and its betas are stable over the estimation window. The residual is the "true" idiosyncratic signal.

Strengths. [Fact] Blitz, Huij and Martens report that residual momentum achieves risk-adjusted returns roughly comparable to or better than conventional momentum, with substantially lower volatility. It largely avoids the dynamic factor exposures that cause momentum crashes. This is the most important practical claim. Residual momentum has a much better crash profile because it does not accumulate the conditional beta that causes the crash losses.

Residual momentum has further strengths. Its factor and industry concentration is much lower. The residual standardization is a natural volatility adjustment. It also makes the strategy's claim of alpha honest by construction.

Weaknesses. It adds a factor-model estimation step. This step has its own specification risk, because the user must choose the factors and the window. Error in the estimated betas propagates into the signal, and turnover is higher. It also *removes* the industry-momentum component, which Moskowitz & Grinblatt (1999) showed is a real source of returns. The strategy deliberately gives up a profitable component to gain lower crash risk. Whether the exchange is worth it depends on the risk budget.

Cost. $O(N \cdot K^2 \cdot W)$ for rolling regressions with K factors over a window W. It is the most expensive signal in §4.6, though still trivial at $N \sim$ thousands with vectorized normal equations and incremental updates.

Robustness. Good, but sensitive to the choice of factor model. **[Practice]** Many firms residualize with a commercial risk model, such as Barra or Axioma, rather than the Fama–French factors. That is generally an improvement.

Failure modes. Betas estimated over one window can be unstable, especially through a regime change. Betas estimated over 2019–2021 were badly wrong for 2022. If the factor model omits a true common factor, that factor's return enters the "residual" momentum signal. The strategy then carries an unintended factor bet again. Careless implementation of overlapping ranking and estimation windows can introduce look-ahead.

When preferred. Use it for equity long/short where crash risk is the binding constraint, and for portfolios that must be factor-neutral under their mandate. It also combines well with conventional momentum, because the two are imperfectly correlated.

4.6.5 Dual momentum: combining absolute and relative

[Practice] A widely used composition comes from Antonacci (2014). First, select the cross-sectional winner among a set of assets (relative momentum). Then require the winner to have positive absolute momentum against cash or T-bills as well (time-series momentum). Otherwise, hold cash.

The two reference points fail in different ways. Relative momentum keeps the portfolio in the *best* asset, but not necessarily in a *good* one. In 2008 it held the least-bad equity market. Absolute momentum exits falling markets but gives no guidance on which asset to hold. Combining them addresses both failures. The empirical support comes mostly from backtests, and the specific published parameterizations are certainly data-snooped. The *structural* argument is nonetheless sound, and the principle of combining generalizes well.

4.7 Probabilistic and state-space approaches

The measures above are estimators without explicit models. State-space methods make the model explicit. This gives three benefits. They handle noise in a principled way. They produce a *distribution* over the trend rather than a point estimate. They also handle missing data and irregular sampling naturally.

4.7.1 Kalman filter / local linear trend

Intuition. Posit an unobserved trend that evolves smoothly and is observed through a noisy price. Infer the trend and its slope optimally, updating recursively. The output is a trend estimate and a *posterior variance*. The posterior variance measures how much confidence the estimate deserves.

Definition. The **local linear trend** model has state $x_t = (\mu_t, \beta_t)'$, the level and the slope:

$$\begin{aligned} \mu_t &= \mu_{t-1} + \beta_{t-1} + \eta_t, & \eta_t &\sim N(0, \sigma_\eta^2) \\ \beta_t &= \beta_{t-1} + \zeta_t, & \zeta_t &\sim N(0, \sigma_\zeta^2) \\ p_t &= \mu_t + \varepsilon_t, & \varepsilon_t &\sim N(0, \sigma_\varepsilon^2) \end{aligned}$$

In matrix form, $x_t = Fx_{t-1} + w_t$ and $p_t = Hx_t + \varepsilon_t$, with $F = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $H = (1 \quad 0)$. The Kalman recursion gives $\hat{x}_{t|t}$ and $P_{t|t}$. The **momentum signal is the filtered slope** $\hat{\beta}_{t|t}$, with a natural t-statistic $\hat{\beta}_{t|t} / \sqrt{[P_{t|t}]_{22}}$.

The result that connects this to everything else. Take the simpler **local level** model, with $\beta \equiv 0$. Its steady-state Kalman filter is *exactly* an EWMA, with gain

$$K = \frac{\sqrt{q^2 + 4q} - q}{2}, \qquad q = \frac{\sigma_\eta^2}{\sigma_\varepsilon^2} \quad \text{(the signal-to-noise ratio)}$$

The EWMA is the optimal filter for a random-walk signal in white noise, and its decay parameter states a signal-to-noise ratio. It is not an ad hoc smoother. This justifies a century of practitioner smoothing after the fact. It also shows how to set λ. Estimate q rather than guess it. In the same way, the local linear trend filter is a second-order generalization that corresponds to a double EMA. Bruder, Dao, Richard & Roncalli (2013) work through these correspondences explicitly.

Assumptions. The dynamics are linear and Gaussian, the variances are known or estimated, and the state dimension is correct. Gaussianity is false for financial data. Even so, the Kalman filter remains the *minimum mean-squared-error linear* estimator whatever the distribution, which is a useful robustness property.

Strengths. It derives from an explicit model and is optimal within its class. It delivers uncertainty quantification as part of the filter output, which is rare and valuable. It handles missing observations and irregular sampling natively. It costs $O(1)$ per step in the state dimension. It also extends naturally to multivariate settings (common trends across assets), to time-varying parameters, and to non-Gaussian and nonlinear variants (EKF and UKF, particle filters).

Weaknesses. The user must specify or estimate the variances $\sigma_\eta^2, \sigma_\zeta^2, \sigma_\varepsilon^2$. These hyperparameters largely determine the effective lookback. So the model reparameterizes parameter selection and does not remove it. The new parameters are arguably more interpretable, which is a genuine gain. Maximum-likelihood estimation of the variances on financial data is unstable. It often reaches boundary solutions, such as $\sigma_\zeta^2 \to 0$, which implies a constant slope. The model assumes Gaussian noise, so it handles jumps badly. It interprets a single jump as a large change in trend.

Cost. $O(d^3)$ per step for state dimension d. Here $d = 2$, so the cost is trivial. Maximum-likelihood fitting costs $O(T)$ per likelihood evaluation, times the optimizer iterations. Fit offline and filter online.

Robustness. Moderate. The filter is robust to mild misspecification and sensitive to jumps and to misestimated variances. **[Practice]** Robustify it with a Student-t observation density, through a particle filter or a variational approximation. A simpler option is to winsorize the innovation $p_t - H\hat{x}_{t|t-1}$ at a few posterior standard deviations. The simpler option is cheap and effective.

Failure modes. Misspecified variances produce either a signal that never moves or one that follows noise. The filter reads jumps as trend changes. The posterior variance overstates confidence, because it is correct only if the model is correct, which it usually is not. Do not treat $\hat{\beta}/\text{SE}$ as a calibrated t-statistic.

When preferred. Use it when sizing needs uncertainty estimates, and with irregular or missing data, as in intraday data, illiquid assets, or several markets with different holidays. It also suits multivariate settings that need a common trend across related assets, and any case that calls for interpretable hyperparameters. **[Practice]** Practitioners use it more in fixed income and FX than in equities, and more in research than in production trend systems. Empirically, it does not beat a well-tuned EWMA ensemble by enough to justify the complexity.

4.7.2 Regime-switching models

Intuition. Instead of one trend, posit K latent states with different means and volatilities, and infer the probability of each state.

Definition. A Hamilton (1989) Markov-switching model has a latent state $S_t \in \{1..K\}$ with transition matrix Π, and

$$r_t \mid S_t = k \;\sim\; N(\mu_k, \sigma_k^2)$$

The forward filter gives $\Pr[S_t = k \mid \mathcal{F}_t]$. A natural signal is $\sum_k \Pr[S_t = k \mid \mathcal{F}_t]\, \mu_k/\sigma_k$, an expected Sharpe ratio weighted by probability. Extensions include switching AR coefficients, so that momentum itself depends on the regime. They also include regimes in a factor model and hidden semi-Markov models with explicit duration distributions.

Strengths. It directly models the regime dependence that §1.4 identified as first-order. It produces calibrated *probabilities* rather than scores. It can encode the empirical fact that high-volatility states differ in both their means and their persistence.

Weaknesses. It is notoriously prone to overfitting in sample. With enough states it can fit anything, and the fitted states often lack meaning out of sample. EM suffers from label switching and local optima. **The most common practical failure is that the fitted regimes are only volatility regimes.** A rolling standard deviation would identify those at a fraction of the complexity. Regime identification is also inherently lagged. A new regime becomes visible only after enough evidence accumulates.

Cost. EM (Baum–Welch) costs $O(TK^2)$ per iteration, and filtering costs $O(K^2)$ per step. Refit infrequently.

Failure modes. Look-ahead bias is frequent in this model. Fitting the model on the full sample and then using smoothed state probabilities $\Pr[S_t \mid \mathcal{F}_T]$ is a severe and common error. Only *filtered* probabilities $\Pr[S_t \mid \mathcal{F}_t]$, from a model fit on data through t, are legitimate. Many published regime-switching results do not meet this standard.

When preferred. Use it as a *conditioning* layer over simpler momentum signals, scaling exposure by the probability of a trending regime, rather than as the signal itself. **[Practice]** It is widely used for risk overlays and drawdown control, and rarely for generating the primary signal.

4.7.3 Bayesian online change-point detection

Intuition. Maintain a posterior over how long the current regime has lasted, and update it with each observation. A change point resets the run length.

Reference. Adams & MacKay (2007), “Bayesian Online Changepoint Detection” (arXiv:0710.3742).

Assessment. Conceptually, this is the right tool for trend initiation and exhaustion (§1.6), because it represents the *detection* problem explicitly. It gives a posterior over run length, from which the expected age of the trend follows directly. Pruning low-probability run lengths reduces the naive cost of $O(t)$ per step. Performance depends heavily on the prior for the hazard rate and on the observation model.

[Practice] The method is elegant and occasionally used, but not mainstream. Its main practical value is as a *risk* signal rather than a directional one. A typical rule is “the probability that the regime just changed is high, so reduce size.”

4.8 Spectral and frequency-domain approaches

Intuition. A price series is a superposition of components at different frequencies. Momentum is *excess power at low frequencies* relative to a random walk. Every trend filter is a low-pass or band-pass operation. Viewing filters this way makes the design trade-offs explicit.

The core objects.

For a linear filter with impulse response $\{h_k\}$, the output is $y_t = \sum_k h_k p_{t-k}$, with **transfer function**

$$H(\omega) = \sum_k h_k e^{-i\omega k}, \qquad |H(\omega)| = \text{gain}, \quad -\frac{\arg H(\omega)}{\omega} = \text{phase lag at frequency } \omega$$

The transfer function is genuinely useful. For any smoother, it gives exactly how much of each frequency the smoother passes. It also gives how many bars of lag the smoother introduces at that frequency. An SMA of length n has $H(\omega) = \frac{1}{n}\frac{\sin(n\omega/2)}{\sin(\omega/2)}e^{-i\omega(n-1)/2}$. The lag is a constant $(n-1)/2$ bars. The gain has **sidelobes**, the sinc pattern. The sidelobes pass some high-frequency energy and can even invert its sign. An EMA has smoothly decaying gain and a lag that depends on frequency. The sidelobes explain why an SMA sometimes behaves strangely.

Spectral density and the variance ratio. Let $f_r(\omega)$ be the spectral density of returns. Under a random walk, returns are white noise, and $f_r(\omega)$ is flat. Momentum means $f_r(0) > \sigma^2/2\pi$, which is excess power at zero frequency. The long-horizon variance ratio satisfies

$$\lim_{q\to\infty} \text{VR}(q) = \frac{2\pi f_r(0)}{\sigma_r^2}$$

The long-horizon variance ratio equals the normalized spectral density of returns at zero frequency. This connects to §1.2 and gives the whole chapter one unifying quantity.

Methods in use. The table summarizes them.

Method	What it does	Assessment
Fourier / periodogram	Decompose into fixed sinusoids	Assumes stationarity, which is false. Useful for filter design and diagnosis. Poor for prediction. Genuine periodicity in returns is rare outside known seasonals
Wavelet MRA	Time-*and*-frequency localized decomposition	Suited to non-stationary series. Gives multi-horizon momentum components that are orthogonal by construction. Boundary effects at the right edge (the current bar) are the main practical problem. Use only causal or undecimated variants with proper boundary handling
Hodrick–Prescott filter	Penalized-smoothness trend extraction	**Do not use for signals.** It is two-sided, so it uses future data, which is an immediate look-ahead bug. Even its one-sided variant has documented artifacts. Hamilton (2018), "Why You Should Never Use the Hodrick-Prescott Filter," is the definitive critique
L^1 trend filtering	Piecewise-linear trend via ℓ_1 penalty on second differences	Kim, Koh, Boyd & Gorinevsky (2009). Produces exactly the "trend with kinks" structure practitioners draw by hand. It has the same two-sided problem unless run causally

Method	What it does	Assessment
Empirical mode decomposition	Adaptive data-driven decomposition	Attractive in principle. It has no theory, has mode-mixing problems and is not causal. **[Practice]** Treat with skepticism

Assumptions. Linearity, and stationarity for Fourier methods. Financial series satisfy neither.

Strengths. Spectral methods make the trade-off between lag and smoothness *explicit and computable*, rather than a matter of taste. They provide a principled decomposition across horizons. They show that most indicators are the same filter under different names.

Weaknesses. Non-stationarity violates the assumptions of the framework. Edge effects are worst at the most recent observation. Traders can act only on that observation, and it is where the estimate is needed most. Two-sided filters make look-ahead very easy to introduce.

Cost. FFT costs $O(T \log T)$. Wavelet MRA costs $O(T)$. ℓ_1 trend filtering is a convex program that costs $O(T)$ with specialized solvers.

Failure modes. Look-ahead through two-sided filtering is the dominant failure, and it is very common. Any filter that is symmetric in time uses future data. Examples include HP, a Butterworth filter applied with `filtfilt`, centered moving averages, and most `scipy.signal` defaults. Check whether any attractive "trend extraction" chart is causal. The second failure is spurious cycles. Finite samples of a random walk produce apparent periodicities. This is the Slutsky–Yule effect. Differencing and averaging *create* cycles from noise.

When preferred. Use these methods for *diagnosis* and *filter design*, not for generating signals directly. **[Practice]** Understanding a smoother's transfer function is a useful and mostly neglected competence. It reveals the smoother's true lookback and lag, which are the quantities that matter. Direct spectral trading signals are a niche, with limited public evidence of success.

4.9 Machine-learning feature representations

Intuition. Rather than choosing one momentum measure, supply many measures and let a learned model find the mapping from past returns to expected future returns.

Feature designs, roughly in order of increasing structure

The table lists the common representations.

Representation	Form	Notes
Multi-horizon normalized returns	$\{r_{t-L:t}/(\hat{\sigma}_t\sqrt{L})\}$ for $L \in \{5, 21, 63, 126, 252\}$	The standard choice. Compact and interpretable, and it spans the horizon space. **[Practice]** Start here. It is hard to beat
Normalized MACD panel	$u_t(n_f, n_s)$ for several span pairs	Baz et al. (2015) construction. A smoothed, saturating basis over horizons
Indicator set	RSI, ADX, stochastic, ER, channel position, …	Highly collinear. Adds multiple-testing risk more than information. Use only with strong regularization
Raw return sequence	$(r_{t-k})_{k=0}^{K}$ into a CNN/LSTM/Transformer	Lets the model learn the kernel. Needs a lot of data and is prone to overfitting
Path signatures	Iterated integrals of the price path (rough-path theory)	Lyons; Levin, Lyons & Ni (2013). A principled, order-truncated basis for *path-dependent* functionals. It genuinely captures order-of-events information that all §4.1 measures discard. Underused. Dimension grows fast with truncation order

Representation	Form	Notes
Wavelet coefficients	Causal MRA detail coefficients	Orthogonal multi-horizon basis, with boundary effects
Order-book tensors	LOB levels and flows	For intraday only (Zhang, Zohren & Roberts, 2019; Sirignano & Cont, 2019)
Learned latents	Autoencoder / self-supervised embeddings	Attractive but hard to validate. Opacity is a real operational cost

Labels and objectives

The choice of *target* matters at least as much as the features. The four main choices are:

- **Fixed-horizon return** $r_{t+1:t+h}$. It is simple, but it ignores the path and the risk.
- **Volatility-scaled return** $r_{t+1:t+h}/\hat{\sigma}_t$. It targets the Sharpe ratio rather than the return, and it usually behaves better.
- **Triple-barrier labeling** (López de Prado, 2018). Label each observation by the first of three barriers that the price reaches: a profit target, a stop loss or a time limit. This encodes the actual trading decision, including its dependence on the path. **[Practice]** It is valuable. *Meta-labeling* is a related idea. A primary model gives the direction, and a secondary model predicts whether to act. It is genuinely useful for separating the signal from the sizing.
- **Direct Sharpe optimization** (Lim, Zohren & Roberts, 2019). Make the loss function the negative Sharpe ratio of the resulting position series. This is elegant, because it optimizes the actual objective and skips the mapping from prediction to position entirely.

Assumptions, strengths, weaknesses

Assumptions. The mapping from past to future is stable enough to learn, and the sample has enough effectively independent observations. The second assumption is the binding constraint, and data routinely violate it. Thirty years of daily data give about 7,500 observations. With overlapping labels and cross-sectional correlation, the *effective* sample size is one to two orders of magnitude smaller. **This is the central difficulty of machine learning in finance, and no amount of model sophistication fixes it.**

Strengths. Machine learning captures interactions that hand-built signals miss, such as momentum conditional on volatility conditional on dispersion. It handles many horizons coherently. It can learn the nonlinear saturation that practitioners impose by hand. **[Fact]** Gu, Kelly & Xiu (2020) show real out-of-sample gains from nonlinear methods, with momentum features consistently among the most important.

Weaknesses. The risk of overfitting is severe, and the usual defense, i.i.d. cross-validation, is invalid. With a low signal-to-noise ratio, high-capacity models mostly fit noise. Non-stationarity makes the learned mapping less accurate over time. Opacity impedes risk management and post-mortems. When an opaque model loses money, it is impossible to tell whether the edge weakened or the pipeline failed.

Cost. Feature construction costs $O(NT \cdot F)$. Training takes from minutes, for gradient-boosted models, to hours or days, for deep networks. Inference is trivial. The real cost is the time spent on *research iteration*, and the compute needed for honest nested cross-validation.

Robustness. [Contested] Tree ensembles with heavy regularization and few features are reasonably robust. The view taken here, from the public evidence, is that deep sequence models on raw returns are not robust. Kelly, Malamud & Zhou (2024) argue that heavily over-parameterized, ridge-regularized models can be robust ("the virtue of complexity"). That argument contradicts the conventional preference for parsimony. The evidence does not settle the question.

Failure modes that matter in practice:

1. **Normalization leakage.** Computing z-scores, scalers or feature medians over the full sample before splitting it. This is the most common leak in practice, and it is silent.
2. **Overlapping labels.** With h-day forward returns, adjacent samples share $h-1$ days of outcome. Naive k-fold cross-validation puts near-identical samples in training and test sets. The fix has two parts. **Purging** drops training samples whose label window overlaps the test set. An **embargo** drops a further buffer after the test set (López de Prado, 2018).
3. **Survivorship and point-in-time errors.** Using today's index membership, restated fundamentals, or a universe free of delistings.
4. **Cross-sectional correlation.** N assets on the same day are not N independent samples. Weighting samples by uniqueness helps, and a block bootstrap by date is better.

5. **Hyperparameter search on the test set.** An out-of-sample period used for tuning is no longer out of sample. Nested cross-validation or a locked holdout is the only defense.
6. **Regime shift.** A model trained on 2010–2019 learned a regime of falling rates and suppressed volatility.

When preferred. [Practice] Use machine learning for *combining* signals, *conditioning* them on state, and *sizing*, not for discovering raw directional alpha from scratch. The professional consensus, as far as published work and hiring patterns reveal it, is that machine learning contributes to momentum in the ensembling and risk layers. A gradient-boosted model over 10–20 well-motivated, orthogonalized features, with purged cross-validation, is a defensible design. A Transformer on raw returns generally is not.

4.10 Comparison table

The table compares the measures on the attributes that discriminate between them. The robustness ratings are this chapter's assessment, on a 1–5 scale. They synthesize the evidence discussed above. Treat them as an opinionated summary and not as a measurement. "Cost" is the streaming cost per bar per asset.

Measure	Family	Path-aware	Normalized	Bounded	Cost	Robustness	Typical use
Lookback / cumulative return	Price-diff	✗	✗	✗	$O(1)$	★★★★☆	Benchmark, XS ranking
Log return sum	Price-diff	✗	✗	✗	$O(1)$	★★★★☆	Default building block
ROC	Price-diff	✗	✗	✗	$O(1)$	★★★★☆	Same as lookback return
EWMA momentum	Price-diff	✓	✗	✗	$O(1)$	★★★★★	Live systems, low turnover
MA displacement	Price-diff	✓	partial	✗	$O(1)$	★★★★☆	Trend state
MA crossover	Price-diff	✓	✗	✗	$O(1)$	★★★★☆	Futures trend ensembles
MACD (normalized)	Price-diff	✓	✓	~	$O(1)$	★★★☆☆	Ensemble member
Regression slope	Regression	✓	✗	✗	$O(1)$†	★★★★☆	Interpretable drift
Regression t-stat	Regression	✓	✓	✗	$O(1)$†	★★★★☆	XS screens
Slope × R^2	Regression	✓	partial	✗	$O(1)$†	★★★★☆	Quality-weighted trend
Vol-normalized momentum	Normalization	✗	✓	✗	$O(1)$	★★★★★	**Default transform**
Sharpe momentum	Normalization	✓	✓	✗	$O(1)$	★★★☆☆	Long-lookback XS ranking
Z-score (TS / XS)	Normalization	—	✓	✗	$O(1)$/$O(N)$	★★★★☆	Signal combination

Measure	Family	Path-aware	Normalized	Bounded	Cost	Robustness	Typical use
RSI	Oscillator	✓	✓	✓	$O(1)$	★★★☆☆	ML feature; ST reversion
Stochastic %K	Oscillator	✗	✓	✓	$O(1)$‡	★★★☆☆	ML feature; intraday
Channel position	Oscillator	✗	✓	✓	$O(1)$‡	★★★★☆	Trend state, robust
Donchian breakout	Oscillator	✗	✓	✓	$O(1)$‡	★★★☆☆	Futures trend entry
52-week high proximity	Oscillator	✗	✓	✓	$O(1)$‡	★★★★☆	Equity XS momen-tum
ADX / DMI	Persistence	✓	✓	✓	$O(1)$	★★★☆☆	Regime filter
Efficiency ratio	Persistence	✓	✓	✓	$O(1)$	★★★★☆	Regime filter, adaptive α
Variance ratio	Persistence	✓	✓	✗	$O(q)$	★★★★★§	Research diagnostic
Hurst exponent	Persistence	✓	✓	✓	$O(T \log T)$	★★☆☆☆	Research only
IC term structure	Persistence	—	✓	✓	$O(T)$	★★★★★	Holding-period choice
Time-series momen-tum	Relational	✗	✓	~	$O(1)$	★★★★☆	Futures/macro trend
Cross-sectional momen-tum	Relational	✗	✓	~	$O(N \log N)$	★★★★☆	Equity long/short
Relative strength (ratio)	Relational	✗	✗	✗	$O(1)$	★★★☆☆	Sector/country rotation
Residual momen-tum	Relational	✗	✓	✗	$O(NK^2W)$	★★★★☆	Factor-neutral equity L/S
Kalman slope	State-space	✓	✓¶	✗	$O(d^3)$	★★★☆☆	Uncertainty-aware sizing
Regime probability	State-space	✓	✓	✓	$O(K^2)$	★★☆☆☆	Risk overlay
Wavelet compo-nents	Spectral	✓	✗	✗	$O(T)$	★★☆☆☆	Multi-horizon research
ML ensemble	ML	✓	✓	~	varies	★★★☆☆	Combination & condi-tioning

† $O(L)$ naive, $O(1)$ with running sums. ‡ $O(1)$ amortized with monotonic deques, $O(n)$ naive. § As a *diagnostic*, and not usable as a live signal. ¶ Via the posterior variance.

[Fact] At a matched horizon, most of these measures correlate at 0.80–0.95 with one another on the same data. The *large*

differences in realized performance come from four sources. They are (i) horizon, (ii) normalization, (iii) the reference point (own past, peers or factors), and (iv) position sizing and cost control. Differences *between indicator formulas at a fixed horizon* are small. Budget research time in that order.

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§4 Key takeaways

1. Every momentum measure estimates $\mathbb{E}[r_{t+1:t+H} \mid \mathcal{F}_t]$. The measures differ in **kernel shape, normalization, reference point and latency**.
2. **The kernel view unifies §4.1–4.2.** Applied to past returns, the lookback return is a rectangular kernel. The EWMA is an exponential kernel. MA displacement is a descending ramp, a crossover is a hump (band-pass), and the regression slope is a centred parabola. This is why the measures correlate so highly. It is also why Levine and Pedersen find them nearly equivalent once horizons match.
3. **Normalize by volatility.** It is the highest-value transformation available. It uses the most predictable feature of returns, and it makes signals comparable across everything. Decide explicitly whether the estimand is expected return or expected Sharpe ratio.
4. **The reference point is a first-order choice.** The asset's own past (TSMOM) gives convexity, net market exposure and viability on a single asset. Peers (XSMOM) give market neutrality, high breadth and crash exposure. Factor residuals give lower crash risk, at the cost of discarding industry momentum.
5. **Know where the strategy's profit comes from.** The Lo–MacKinlay decomposition shows that cross-sectional momentum profits can come from own-autocovariance, from lead-lag structure, *or* from pure dispersion in unconditional means. The last requires no predictability at all. The analogous $L\mu^2$ term confounds naive TSMOM tests.
6. **Oscillators are conditioners, not signals.** RSI, the stochastic oscillator and ADX are bounded and self-normalizing, which makes them good ML features and good regime filters. Their conventional thresholds are the most overfit numbers in finance.
7. **Persistence metrics deserve more attention than they get.** The variance-ratio profile and the IC term structure answer whether momentum applies here, and at what horizon. That question is more valuable than which indicator to use.
8. **The EWMA is the steady-state Kalman filter** for a random walk in noise. Each smoothing constant implies a signal-to-noise ratio. Estimate the ratio instead of guessing it.
9. **Spectral thinking is for diagnosis and filter design.** Its main practical use is to prevent look-ahead bias from two-sided filters, a very common bug.
10. **The realistic contribution of machine learning is combination, conditioning and sizing**, not raw directional discovery. Its dominant risk is overfitting, because the effective sample size is much smaller than the nominal one.

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5. Taxonomy

5.1 The problem with one-dimensional taxonomies

Textbooks usually classify momentum measures in a flat list, such as "price-based, oscillator-based, regression-based". The list is misleading, because a given measure belongs to several categories at once. Consider a volatility-normalized regression t-statistic computed cross-sectionally on factor residuals. It is regression-based, normalization-based, cross-sectional and residual all at once. A flat taxonomy cannot represent it.

The useful structure is **a set of orthogonal design axes**. Any momentum measure is a point in this space. Building a new measure means choosing a coordinate on each axis. The axes also make the design space enumerable.

5.2 The master form

Nearly every measure in §4 has the form

$$s_{i,t} \;=\; g\left(\frac{\sum_{k\geq 0} h_k\left(r_{i,t-k} - b_{i,t-k}\right)}{\mathcal{N}_{i,t}}\right)$$

The form has four independently chosen components, listed in the table.

Component	Role	Choices
h_k	**Kernel** — how past returns are weighted	rectangular (lookback return), exponential (EWMA), descending ramp (MA displacement), hump/band-pass (crossover), centred parabola (regression slope), order-statistic (channel position), learned (ML)
$b_{i,t}$	**Benchmark** — what the return is measured against	0 (TSMOM), risk-free rate, an index or peer asset (relative strength), the cross-sectional mean (XSMOM), a fitted factor model (residual momentum)
$\mathcal{N}_{i,t}$	**Normalizer** — the scale	1 (raw), $\widehat{\sigma}_{i,t}\sqrt{L}$ (vol-normalized), in-window SD (Sharpe), total absolute movement (RSI, ER), window range (stochastic, channel), cross-sectional SD (XS z-score), residual SE (regression t)
$g(\cdot)$	**Transform** — signal to position	identity, sign, clip, rank, tanh, $ze^{-z^2/4}$, threshold

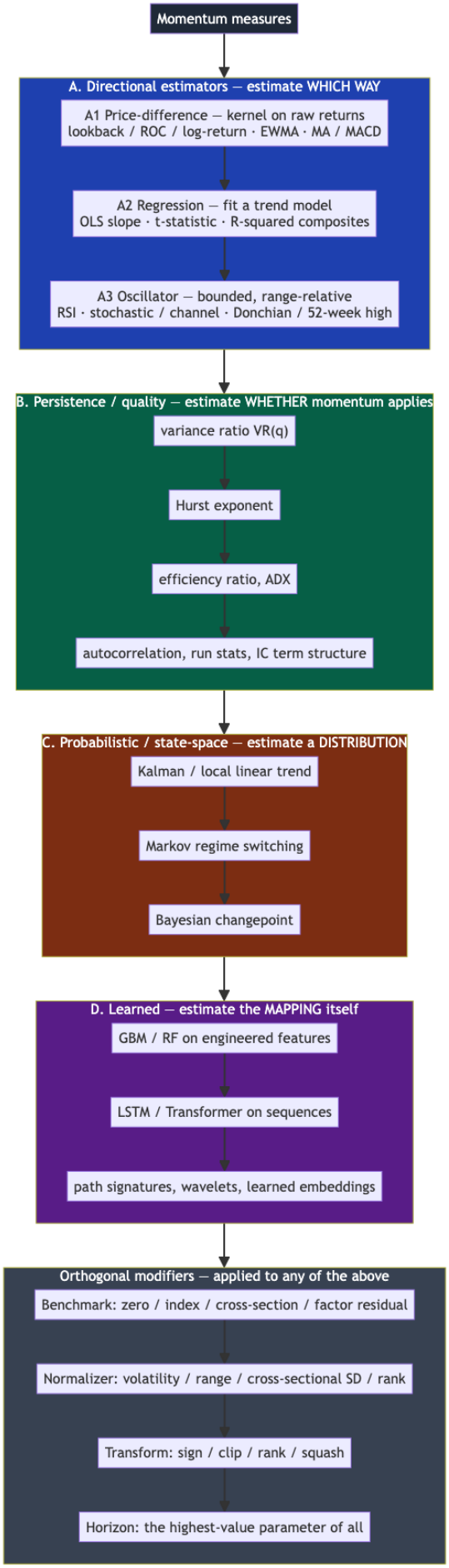
Two things follow immediately. First, **the design space is a product of choices and not a list.** The literature has hundreds of "indicators", but only a few dozen genuinely distinct points in this space. Second, **the axes are not equally important.** Empirically, b and $\mathcal{N}$ drive results, and so does the kernel's *horizon*, though not its *shape*. g matters for turnover and tails. The kernel's shape barely matters at all.

Three families fall partly outside the master form and need separate treatment:

- **persistence metrics**, which estimate *whether* momentum applies rather than its direction.
- **state-space methods**, which produce a posterior rather than a point estimate.
- **learned representations**, where the model estimates h and g and the designer does not choose them.

5.3 The taxonomy

The diagram arranges the estimator families and the modifiers that apply to all of them.


Momentum measures
A. Directional estimators — estimate WHICH WAY
A1 Price-difference — kernel on raw returns
lookback / ROC / log-return · EWMA · MA / MACD
A2 Regression — fit a trend model
OLS slope · t-statistic · R-squared composites
A3 Oscillator — bounded, range-relative
RSI · stochastic / channel · Donchian / 52-week high
B. Persistence / quality — estimate WHETHER momentum applies
variance ratio VR(q)
Hurst exponent
efficiency ratio, ADX
autocorrelation, run stats, IC term structure
C. Probabilistic / state-space — estimate a DISTRIBUTION
Kalman / local linear trend
Markov regime switching
Bayesian changepoint
D. Learned — estimate the MAPPING itself
GBM / RF on engineered features
LSTM / Transformer on sequences
path signatures, wavelets, learned embeddings
Orthogonal modifiers — applied to any of the above
Benchmark: zero / index / cross-section / factor residual
Normalizer: volatility / range / cross-sectional SD / rank
Transform: sign / clip / rank / squash
Horizon: the highest-value parameter of all

The critical structural point is the separation between the four *estimator families* (A–D) and the four *modifiers* (N1–N4). The modifiers apply to essentially any estimator, in any combination. "Cross-sectional momentum" is a **choice of benchmark**. A trader can apply it to a regression slope, an EWMA, an RSI or an ML output. It is not a category alongside "regression momentum". Missing this leads to comparisons between combinations that are not comparable, in the belief that they are alternatives.

5.4 The benchmark axis, made explicit

This axis needs its own treatment, because analysts most often leave it implicit. Every measure subtracts *something* before measuring movement, even when that something is zero:

$$\tilde{r}_{i,t} = r_{i,t} - b_{i,t}$$

The table lists the common benchmarks and what each one isolates.

Benchmark $b_{i,t}$	Resulting measure	Portfolio property	What it isolates
0	Time-series / absolute momentum	Net long or short; directional	Total return direction
r_t^f (risk-free)	Excess-return momentum	Same, cash-aware	Risk premium direction
$r_{B,t}$ (an index)	Relative strength	Long/short vs. benchmark	Benchmark-relative performance
$\bar{r}_t$ (cross-sectional mean)	Cross-sectional momentum	Dollar-neutral	Idiosyncratic + factor tilts
$\widehat{\beta}_i' f_t$ (factor model)	Residual momentum	Dollar- and factor-neutral	Pure idiosyncratic
$\widehat{\beta}_i' f_t$ with industry dummies	Industry-neutral momentum	Also industry-neutral	Within-industry idiosyncratic

Reading down the table, each row is a more aggressive projection. Each step removes a component of return, and with it both that component's risk and its expected return. **The choice of benchmark determines which risks earn the strategy a return.** Cross-sectional momentum earns a return for taking industry and factor bets. Residual momentum forgoes that return in exchange for a much better crash profile. Neither is right. They are different products.

An identity ties the first and fourth rows together. **Cross-sectional momentum is time-series momentum applied to market-relative returns.** The two are the same operator with different benchmarks, not different phenomena. Goyal & Jegadeesh (2018) make essentially this point formally. They show that the difference between the two is largely a time-varying net long position in the market.

5.5 Relationships and equivalences

The table lists exact and approximate identities between the measures. Knowing them prevents "diversifying" across measures that are the same measure.

Relationship	Status
ROC $\equiv$ simple lookback return	Exact (rescaling)
Sharpe momentum $\equiv$ t-stat of the in-window mean $\times\sqrt{A/L}$	Exact
Regression $t \equiv$ slope / residual SE, with $t \propto L^{3/2} \cdot (b/\sigma_\varepsilon)$	Exact
$t^2 = \frac{R^2}{1-R^2}(L-2)$	Exact. Slope, t and R^2 are three expressions of two quantities
MA crossover $\equiv$ band-pass filter on log price $\equiv$ hump-kernel weighted sum of returns	Exact
MA crossover $\approx$ difference of two lookback returns of different lengths	Approximate
EWMA $\equiv$ steady-state Kalman filter for a local-level model	Exact, at steady state
Double EMA $\approx$ local-linear-trend Kalman filter	Approximate
VR$(q) = 1 + 2\sum_{k<q}(1-k/q)\rho_k$	Exact

Relationship	Status
$\lim_{q\to\infty} \mathrm{VR}(q) = 2\pi f_r(0)/\sigma_r^2$	Exact. The long-horizon variance ratio is the normalized spectral density of returns at zero frequency
Hurst exponent $\equiv$ log-log slope of the VR profile ($\mathrm{VR}(q) \propto q^{2H-1}$)	Exact under fBm; approximate otherwise
$\mathrm{ER}(n)\sqrt{n} \approx$ a robust, L^1 analogue of $\sqrt{\mathrm{VR}(n)}$ (note the $\sqrt{n}$: ER itself is $\approx n^{-1/2}$ under a random walk)	Heuristic
ADX $\approx$ a smoothed, range-based directional-consistency ratio	Heuristic; same purpose as ER
XSMOM $\equiv$ TSMOM on market-relative returns	Exact given equal weights
Residual momentum $\equiv$ XSMOM with a multi-factor rather than single-mean benchmark	Structural
Trend-following P&L $\propto \sigma^2_{\text{long}} - \sigma^2_{\text{short}} \propto \mathrm{VR}(q) - 1$	Leading order (Dao et al., 2017)
Trend-following payoff $\approx$ long lookback straddle	Empirical (Fung & Hsieh, 2001)

The last three rows say that a trend follower's P&L, the variance ratio and an option payoff are three descriptions of one thing. Of all the connections in this chapter, this one is the most important to understand.

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§5 Key takeaways

1. Momentum measures form a **product space** and not a list. The factors are kernel × benchmark × normalizer × transform, with horizon as a further dimension of the kernel.
2. Four estimator families answer different questions: **directional** (which way), **persistence** (whether momentum applies), **probabilistic** (how confident) and **learned** (the mapping itself). A complete system uses at least the first two.
3. **The choice of benchmark is orthogonal to the choice of estimator**, and it has larger consequences. Zero gives convexity and net exposure. The cross-sectional mean gives market neutrality. Factor residuals give factor neutrality and a much better crash profile.
4. Many "different" measures are provably the same. Check the equivalence table before assuming that an ensemble adds diversification.
5. Horizon is the highest-value parameter, and kernel shape the lowest. Allocate research effort accordingly.

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6. Practical implementation

This section covers the decisions that determine whether a correct signal becomes a profitable strategy. [Practice] The ordering of impact is roughly **look-ahead bias > costs and capacity > normalization and vol scaling > horizon > everything else.** The first two can turn a real edge into a fictitious one or an unimplementable one. Most people spend their time on the last item.

6.1 Lookback selection

Do not optimize a single lookback. The parameter surface for momentum is noisy. The lookback that maximizes the in-sample Sharpe ratio is a heavily biased estimate of the best out-of-sample lookback. Instead, follow five steps:

1. **Derive a prior from the structure of the data.** Compute the variance-ratio profile (§4.5.1) and the IC term structure (§4.5.4) on a *research* sample. These show which horizons carry momentum in this market, before any model fit.
2. **Check that the parameter surface is a plateau, not a peak.** Plot the Sharpe ratio (or the IC) against the lookback. A broad region of similar performance is evidence of a real effect. An isolated spike is evidence of overfitting. **[Practice]** The diagnostic is cheap. **Recommendation: treat a spiky surface as disqualifying.**

3. **Ensemble across horizons rather than choosing one.** Average the normalized signals from several lookbacks:

$$s_t = \frac{1}{M}\sum_{m=1}^{M}\frac{r_{t-L_m:t}}{\widehat{\sigma}_t\sqrt{L_m}}, \qquad L_m \in \{21, 63, 126, 252\}$$

   [Fact] Ensembling across horizons reliably improves out-of-sample robustness, at a small cost in in-sample Sharpe ratio. The mechanism is simple. Ensembling averages over parameter uncertainty, and the horizon at which momentum is strongest varies over time.
4. **Space horizons geometrically**, not arithmetically. Lookbacks of 21, 63, 126 and 252 days span the space efficiently. Lookbacks of 60, 70, 80 and 90 do not span it efficiently, because those four signals are nearly the same signal.
5. **Match horizons across measure types**, using the effective lookback (the centre of mass). Otherwise the "ensemble" will accidentally overweight one horizon.

A caution about the canonical 12-2. It is a good default for US equity cross-sectional momentum, and it is the standard for comparability. But it is also the most-searched parameter pair in the history of finance, and its apparent optimality partly reflects that search. In a new market, derive the lookback afresh.

6.2 Sampling frequency

A precise and underappreciated result applies here (Merton, 1980):

> **Increasing sampling frequency improves the estimate of a volatility. It does not improve the estimate of a drift.**

Take a diffusion with drift μ and volatility σ, observed over a calendar span T with n samples. Then $\mathrm{SE}(\widehat{\mu}) = \sigma/\sqrt{T}$, *independent of* n. Meanwhile $\mathrm{SE}(\widehat{\sigma}) \approx \sigma/\sqrt{2n}$, which shrinks with n. The consequences are direct:

- **For the momentum signal, a drift-like quantity, the calendar span of the history matters.** Ten years of daily data contain roughly the same information about drift as 10 years of 5-minute data. This is why data fundamentally limit momentum research, and why histories spanning decades and broad cross-sections are so valuable.
- **For volatility, the normalizer, use the highest frequency that the pipeline can handle cleanly.** Intraday realized volatility is far more precise than daily close-to-close volatility. It needs careful handling of microstructure noise. Use 5-minute sampling or a noise-robust estimator rather than tick-by-tick data.

Three other considerations bear on frequency. First, match the bar frequency to the horizon. Daily bars suit multi-month momentum, and tick data for a 6-month signal adds considerable noise and no benefit. Second, intraday data carry intraday seasonality, the U-shaped pattern in volume and volatility. Remove it before any normalization. Third, for multi-market portfolios, decide explicitly what "daily" means. Each market's own close is non-synchronous and creates spurious lead-lag. A common snapshot time avoids that.

6.3 Overlapping windows

Monthly signals formed from 12-month lookbacks share 11 months of data between consecutive observations. This has three consequences:

1. **Overlap inflates test statistics.** For h-period overlapping observations, the effective sample size is roughly T/h, not T. A naive t-statistic can exceed the correct value by a factor of $\sqrt{h}$ in the worst case.
2. **The fix for inference** is HAC standard errors, using Newey & West (1987) with a lag truncation of at least $h-1$ (Hansen & Hodrick, 1980, for the case of overlapping forecasts). Researchers know that even these are undersized in small samples. The honest approach reports both HAC-corrected statistics and a block-bootstrap distribution (§7.7.1).
3. **A cleaner alternative for strategy evaluation** forms overlapping portfolios properly. Jegadeesh and Titman's original approach holds H sub-portfolios at once, each formed a month apart and each held for H months. The aggregate portfolio then rebalances monthly, with $1/H$ turnover. This uses all the data without pretending the observations are independent. It also produces a return series that standard tools can evaluate.

[Practice] Report the equivalent non-overlapping sample size prominently. "20 years of monthly data with a 12-month look-back" sounds like 240 observations and behaves more like 20.

6.4 Normalization and smoothing

§4.3 covered normalization statistically. The implementation notes follow.

- **Always normalize before combining.** Simple averaging of signals on different scales weights them in effect by their variances and not by intent.

- **Separate signal smoothing from position smoothing.** Smoothing the signal changes the prediction and adds lag. Smoothing the position changes only turnover. To reduce costs, smooth the *position*: $w_t = \theta w_t^{\text{target}} + (1-\theta) w_{t-1}$. This is more than a heuristic. It approximates the optimal policy under quadratic transaction costs (Gârleanu & Pedersen, 2013). In that policy the optimal trade is a partial step toward an "aim" portfolio.
- **No-trade bands** are the other standard cost control. Rebalance only when $\left|w^{\text{target}} - w^{\text{current}}\right|$ exceeds a threshold. Under proportional costs, the optimal policy genuinely has this form. Bands introduce path dependence into the backtest, so implement them in the simulator, not as a filter applied afterwards.
- **Avoid double smoothing.** Almost certainly nobody has computed the effective lag of an EWMA of a moving average of a smoothed price. Compute the transfer function of the composition, or at least its centre of mass.

6.5 Handling gaps, holidays and non-synchronous data

These problems are practical and frequent:

- **Overnight gaps.** Close-to-close returns include the overnight move, and open-to-close returns do not. Signals built on one and executed against the other are inconsistent. Decide once.
- **Trading halts and limit moves.** A halted or limit-locked instrument has no tradable price. Carrying the last print forward creates artificial zero-return bars, which deflate volatility estimates and inflate the Sharpe ratio.
- **Cross-market holidays.** In a global futures portfolio, markets close on different days. Forward-filling introduces artificial zero returns and spurious cross-market lead-lag. **[Practice]** Prefer computing returns only on days a market actually traded, and handle the resulting ragged panel explicitly. State-space methods have a genuine advantage here, because they handle missing observations natively.
- **The futures roll.** This is the largest source of silent error in futures momentum research. Build a continuous series by one of two methods. Back-adjustment subtracts the roll gap from all history. It can produce negative prices in long histories, and it makes percentage returns meaningless before the adjustment point. Ratio adjustment multiplies instead. It preserves positive prices and the meaning of returns, and it is generally preferable. **Never compute returns across an unadjusted roll.** The resulting spurious jump is often larger than any real signal. Back-adjusted price *levels* are not real prices, so any measure that uses price levels, such as channel position or the 52-week high, needs careful computation.
- **Currency.** For international portfolios, decide whether signals use local-currency or base-currency returns. The two differ. Using base-currency returns unintentionally mixes FX momentum into equity momentum.

6.6 Outliers

Momentum signals are sums of returns, and sums are not robust. A single erroneous print can dominate a lookback window.

Two thresholds serve two different jobs. A loose threshold bounds values automatically. A tight threshold flags values for human attention.

- **Winsorize, don't drop.** Clip every return at, say, ±5 standard deviations based on the median absolute deviation (§4.3.3). This preserves the observation count and the direction, while bounding the influence of any single bar on the signal. Dropping creates its own biases, because the dropped bars are not missing at random.
- **Flag and inspect separately.** Raise an alert on the genuinely implausible. Examples are returns beyond about 10 robust standard deviations, price moves inconsistent with the bar's own high and low, and zero-volume bars with nonzero returns. These cases need a human, not a rule. Log and inspect them, and do not repair them silently.
- **Distinguish errors from events.** A −40% day may be a bad tick or a real crash, and the two need opposite handling. Cross-check four things. Check whether the move appears in the high and low. Check whether there is corresponding volume. Check whether related instruments move. Check whether there was a corporate action. **[Practice]** In equity data, the most common cause of a "−40% day" is an unadjusted corporate action. The second most common is a real event.
- **Use robust estimators where they are cheap.** Examples are median-based volatility (MAD), trimmed means, and the Theil–Sen regression slope instead of OLS. The loss of efficiency is small, and the gain in robustness is large.
- **Always winsorize cross-sectionally before z-scoring** (§4.3.3).

6.7 Volatility scaling

Volatility scaling applies in three distinct places, and each does something different:

$$\underbrace{s_{i,t} = \frac{r_{i,t-L:t}}{\hat{\sigma}_{i,t}\sqrt{L}}}_{\text{(1) signal normalization}} \qquad \underbrace{w_{i,t} = s_{i,t} \cdot \frac{\sigma^*}{\hat{\sigma}_{i,t}}}_{\text{(2) position sizing}} \qquad \underbrace{W_t = w_t \cdot \frac{\sigma_p^*}{\hat{\sigma}_{p,t}}}_{\text{(3) portfolio vol targeting}}$$

Step (1) makes signals comparable across assets and horizons. Step (2) equalizes each asset's contribution to risk. Step (3) stabilizes risk at the portfolio level over time. They answer three different questions, and all three are standard.

Applying (1) and (2) together divides by $\hat{\sigma}$ twice. By the accounting rule of §4.0, that is exactly right when the raw signal predicts on the *return* scale. The mean–variance weight is $\mathbb{E}[r]/\sigma^2$, and the two divisions supply its two powers of σ, one from each step. It is one division too many if the signal is already a direction or a bounded score. In TSMOM, for example, $w \propto \text{sign}(\cdot)\,\sigma^*/\hat{\sigma}$ has a single power. Step (3) differs in kind. It divides by an estimate of *portfolio* volatility, not asset volatility, and it does not enter the per-asset count at all. **[Practice]** Write down the intended total power of $\hat{\sigma}$ before writing the code. The failure mode is ending up with a power that nobody chose.

Consider (3), portfolio volatility targeting. **[Fact]** Harvey et al. (2018) find that it improves risk-adjusted returns for risk assets and for trend strategies. The main reason is that, for equities, volatility is persistent and negatively related to subsequent returns. **[Contested]** The benefit is smaller once the analysis includes realistic costs and the higher turnover. It is weakest for assets without a strong relation between volatility and return, such as bonds and commodities.

Four practical cautions apply:

- Use a *forecast* of volatility, not a trailing realization. The two differ most at the times when the choice matters most.
- The volatility estimate lags when a shock begins. For risk purposes, consider blending a fast and a slow estimator and taking the larger.
- Cap leverage independently of the volatility target. Otherwise a regime of collapsing volatility makes the target require extreme gross exposure.
- A volatility target is a *feedback loop*. Many funds targeting volatility at once produce correlated deleveraging.

6.8 Transaction costs, liquidity, and capacity

Momentum is a *high-turnover* strategy, so costs are large. They decide whether the anomaly makes money in practice.

Cost model. A usable decomposition is

$$\text{cost per trade} \;=\; \underbrace{\tfrac{1}{2}\,\text{spread}}_{\text{immediate}} \;+\; \underbrace{Y\,\sigma\sqrt{Q/V}}_{\text{impact (square-root law)}} \;+\; \underbrace{\text{fees, borrow, financing}}_{\text{explicit}}$$

Every term is a **fraction of the notional traded**. So the terms add, and they compare directly with an alpha also expressed in return units. The spread is the relative bid-ask spread, σ is the daily return volatility, and Q/V is the order as a fraction of daily volume (§1.3). The square-root term binds at scale, and its concavity is the key economic fact. **Doubling the size of a trade raises its cost per share by only $\sqrt{2}$.** Capacity therefore degrades gradually rather than catastrophically. It does degrade, however, and total cost still rises by $2\sqrt{2}$.

Capacity. A rough but genuinely useful calculation follows. Suppose a strategy has expected gross alpha α per unit of turnover, and trades Q against volume V. Net alpha goes to zero when $Y\sigma\sqrt{Q/V} = \alpha$, which gives

$$Q^* \;\approx\; V\left(\frac{\alpha}{Y\sigma}\right)^2$$

Run this calculation before building anything. It answers "how much money can this hold" in one line. Because capacity depends on the square of α/σ, small differences in edge produce large differences in capacity.

The evidence conflicts. [Contested] Academic estimates of costs use quoted spreads and effective-spread proxies (Lesmond, Schill & Zhou, 2004; Novy-Marx & Velikov, 2016). These estimates find that costs consume momentum's profits substantially or entirely, especially in the short leg and in small caps. Frazzini, Israel & Moskowitz (2018) used about $1.7 trillion of AQR's own live executions. They find real-world costs roughly an order of magnitude lower than those proxies, and conclude that momentum remains implementable at very large scale. Both sides have a point. Quoted-spread proxies genuinely overstate the costs a patient, opportunistic trader pays. AQR's estimates come from a firm with best-in-class execution and an interest in the answer. **[Practice]** The view taken here is that costs are much lower than naive proxies suggest *if* the trader trades patiently and can shape participation. Otherwise they are much higher than AQR's numbers.

Turnover control. The main methods, in order of effectiveness, are:

1. longer holding periods, guided by the IC term structure. Hold until the marginal IC no longer covers the marginal cost.
2. position smoothing, or partial adjustment toward the aim portfolio.
3. no-trade bands.
4. cost-aware portfolio optimization, with an explicit penalty on transaction costs.
5. trading the *change* in the signal rather than rebalancing to target.
6. crossing internally against other strategies.

Liquidity screening. Exclude names below a minimum dollar volume. Cap position size at a fraction of average daily volume (ADV). **[Practice]** Common limits are 5–10% of ADV for a *position* and 5–20% of ADV for a *participation rate*. Liquidity is itself correlated with momentum. The names most attractive to trade after a big move are often the ones whose liquidity has just deteriorated.

6.9 Regime dependence and parameter sensitivity

§1.4 established **regime conditioning** as first-order. The main practical approaches are listed below in increasing order of complexity. In the view taken here, the confidence they deserve decreases in the same order.

1. **Volatility scaling** (§6.7). It conditions on regime implicitly and robustly, by reducing exposure in exactly the high-volatility states where momentum crashes occur. **[Fact]** It captures most of the available benefit, which is why the result of Barroso and Santa-Clara is so valuable.
2. **Conditioning on the market state.** Reduce momentum exposure after market declines (Cooper, Gutierrez & Hameed, 2004; Daniel & Moskowitz, 2016). This is simple and well documented, and it has few parameters.
3. **Conditioning on trend quality.** Scale by the efficiency ratio, ADX, or an estimated variance ratio.
4. **Explicit regime models** (§4.7.2). These are powerful and dangerous, and the risk of look-ahead is severe.

Parameter sensitivity. Test it deliberately:

- **Plateau test.** Vary each parameter over a wide range and plot the performance surface. The aim is plateaus.
- **Perturbation test.** Jitter all parameters at random by ±20% simultaneously, 1,000 times, and examine the *distribution* of outcomes. If the median is far below the tuned value, the strategy is overfit.
- **Count the degrees of freedom.** Every threshold, window and filter is a parameter, including the ones chosen "obviously". Feed the honest count into the multiple-testing correction (§7.6).
- **Prefer forms with few or no parameters** where performance is comparable. A signal with two parameters that performs 90% as well as one with eight is the better signal.

6.10 Avoiding look-ahead bias

Look-ahead bias invalidates many backtests. It is a family of bugs and not a single bug. Most instances are silent. They produce excellent results with no error message.

A checklist, ordered roughly by how often each item causes real damage in practice:

1. **Retroactively adjusted prices.** Adjustment factors for splits and dividends apply to the *entire history* when the event occurs. If a database stores prices adjusted as of today, a backtest "on 2015-06-01" sees prices adjusted for a 2018 split. For total-return momentum, this is usually acceptable, because the adjusted series is the correct total-return series. For anything that uses price *levels*, such as a $5 minimum-price filter, channel position or the 52-week high, it is a genuine leak. Use point-in-time unadjusted prices with as-of adjustment factors.
2. **Survivorship bias.** A universe built from securities listed today excludes every company whose price fell to zero. **[Fact]** This inflates momentum backtests substantially, because momentum's short leg consists precisely of the names that later fail. Use a point-in-time universe that includes delisted securities with proper delisting returns. CRSP provides these, and many vendors do not.
3. **Look-ahead in index membership.** Using today's S&P 500 constituents for a 2005 backtest is a severe and very common leak. It selects companies that *became* large.
4. **Timing from signal to execution.** If the signal uses the close of day t, the trade cannot happen at the close of day t. Trade at the open of $t+1$ or the close of $t+1$, or model a realistic execution window. **[Practice]** The "signal at close, fill at the same close" bug is the most common single cause of an implausibly good backtest. Its effect is largest for short-horizon signals.
5. **Two-sided filters.** Anything centred or applied with `filtfilt`, the HP filter, and smoothed (as opposed to filtered) state-space estimates. See §4.8.
6. **Full-sample normalization.** Computing z-scores, winsorization thresholds, scalers or PCA loadings over the whole sample. Use expanding or rolling windows.

7. **Restated fundamentals and macro data.** If the strategy conditions momentum on any fundamental or macro variable, use point-in-time vintages. GDP, earnings and index levels are all revised.
8. **Mapping of corporate actions and identifiers.** Ticker reuse, where a new company takes over a delisted ticker, silently splices two unrelated price series. Map by permanent identifiers.
9. **Parameter selection on the full sample.** This is the most consequential and least visible form. A lookback chosen by looking at full-sample results turns the "out-of-sample" test into an in-sample one. This is the subject of §7.

[Practice] **Recommendation: shift every signal back by one extra bar and rerun the backtest.** If performance falls sharply, the edge comes from the last bar, and the timing assumptions need very close scrutiny. Legitimate multi-month momentum should be almost unaffected by a one-day delay.

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§6 Key takeaways

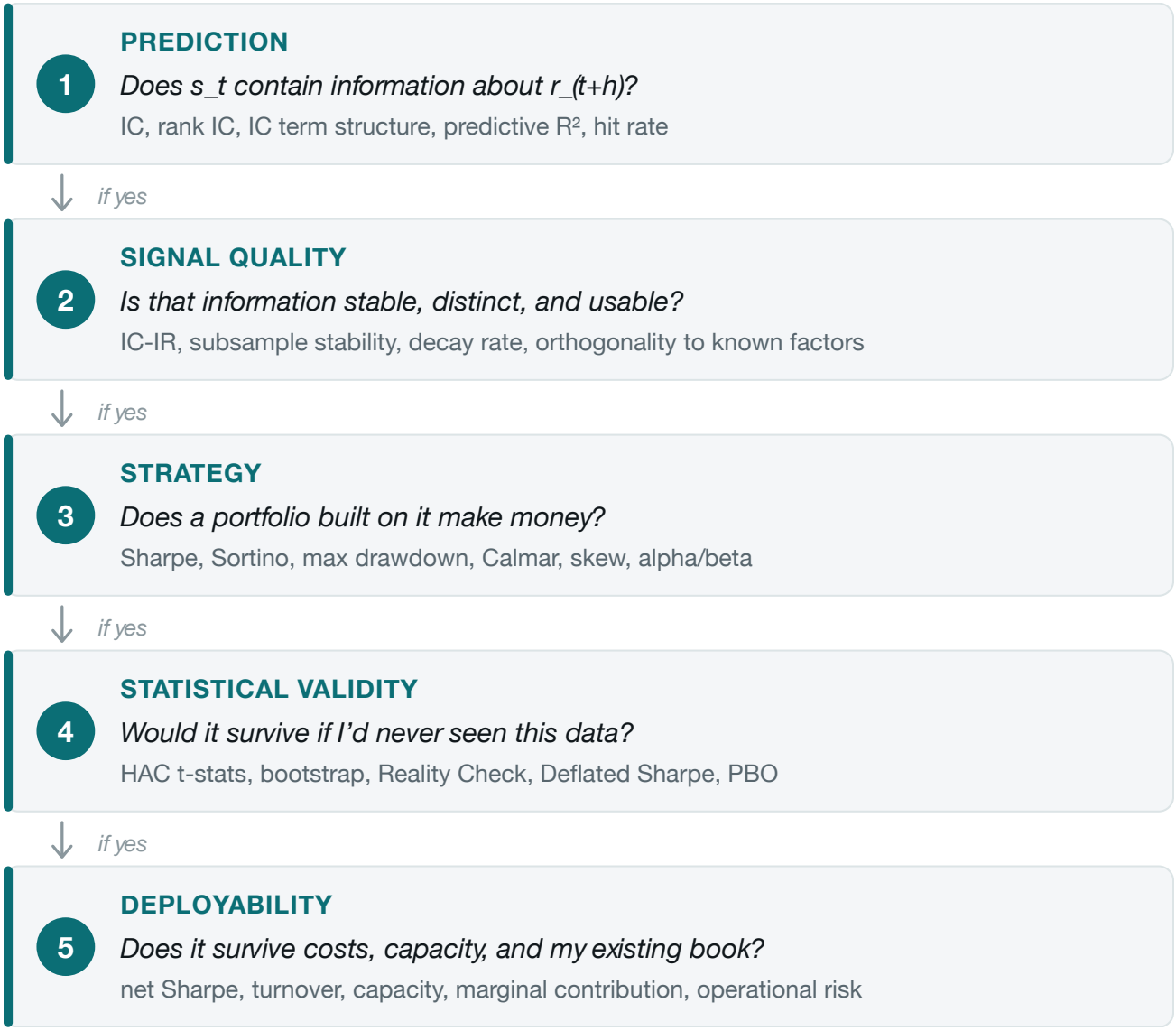
1. The ordering of impact is **look-ahead bias > costs and capacity > normalization and vol scaling > horizon > indicator choice.** Spend time in that order.
2. **Don't optimize a lookback. Ensemble lookbacks spaced geometrically**, and demand a plateau in the parameter surface, not a peak.
3. **Higher sampling frequency helps volatility estimation, not drift estimation** (Merton, 1980). Calendar span limits momentum research. Use high-frequency data for the denominator only.
4. Overlapping windows inflate t-statistics by up to $\sqrt{h}$. Use HAC errors, and report the effective sample size.
5. **Futures roll adjustment and corporate-action adjustment are the two main silent sources of error** in price-based research. Ratio-adjust futures, and use point-in-time adjustment factors for equities.
6. Volatility scaling applies in three places: signal, position and portfolio. Applying it more than once is fine, but it must be deliberate.
7. **Compute capacity before building:** $Q^* \approx V(\alpha/Y\sigma)^2$. The square-root impact law makes this a one-line calculation.
8. The evidence on costs conflicts. Quoted-spread proxies overstate them for patient traders. Live-execution studies from interested parties understate them for everyone else. Model them for the strategy at hand.
9. **Volatility scaling** achieves most of the benefit of regime conditioning. The simplest method captures most of the benefit and carries the least risk of overfitting.
10. Run the **one-extra-bar delay test**. It catches a large share of timing leaks in one line.

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7. Testing momentum-based trading signals

7.1 The evaluation ladder

The most common evaluation mistake is applying strategy-level metrics to a signal, or signal-level metrics to a deployment decision. Evaluation proceeds in stages. Each stage has its own question, its own metrics and its own failure modes, as the figure shows.



Fail fast at the top. Stage 1 is cheap and rejects most ideas. Stage 5 is expensive. Running a full portfolio backtest on a signal whose IC has not been measured is the wrong order. Most people nonetheless follow that order, because backtests are more fun than correlations.

7.2 Validation protocols

7.2.1 In-sample vs. out-of-sample — and the meta-problem

The standard prescription is to develop on a training period and validate on a held-out period. This is necessary but *insufficient*, for the reason below:

> **A held-out sample can be used only once.** As soon as a researcher looks at the out-of-sample results and goes back to modify the strategy, that sample becomes in-sample. Every later "out-of-sample" test on it is contaminated, and the contamination compounds silently.

In practice, researchers iterate dozens of times against the same "out-of-sample" period. The result is a strategy fitted to the full history through a slow, undocumented search. Because the search is undocumented, no one can correct for it. This is

why the multiple-testing methods of §7.6 exist. It is also why the only genuinely clean out-of-sample test is **live paper trading on data that did not exist when the researcher built the model**.

[Practice] A workable discipline splits the data three ways. A *train* period allows free development. A *validation* period allows iteration, and the researcher counts every iteration. A **locked holdout** allows exactly one test, at the end, with a decision rule registered in advance. Record the number of configurations tried, because §7.6 needs it.

7.2.2 Rolling vs. expanding windows

The two schemes differ in what they treat as the training set.

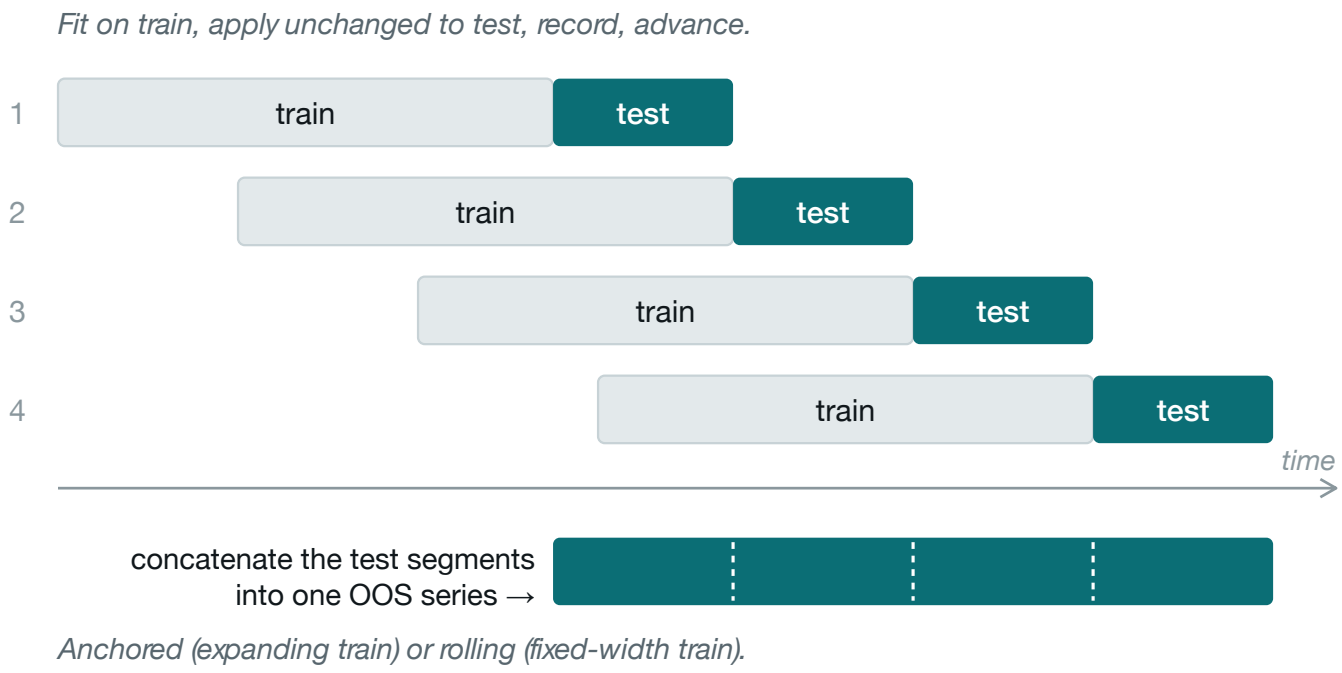
Scheme	Training set	Rationale	Best when
Expanding	All data from start to t	Uses maximum data, and parameter estimates stabilize	The DGP is stable and parameters are constants
Rolling	Fixed window ending at t	Adapts to regime change and discards stale data	The DGP drifts and recent data is more relevant

[Practice] For momentum, use expanding windows for *structural* estimates, such as factor loadings, long-run volatility and the shape of the IC term structure. Use rolling windows for *state* estimates, such as current volatility and the current regime. A common hybrid weights observations with exponential decay. It interpolates continuously between the two schemes, and it is usually better than either.

Both schemes raise a subtle issue. With an expanding window, later periods use more data in estimation than earlier ones. So measured performance improves over time for purely statistical reasons. Do not read that improvement as the strategy getting better.

7.2.3 Walk-forward validation

Walk-forward validation is the core protocol for developing time-series strategies. The figure shows its structure.



At each step, fit or select the parameters on the training window, apply them unchanged to the test window, record the results, and advance. Concatenating the test segments gives a single out-of-sample track record. The model predicted every observation genuinely from prior information alone.

Strengths. It respects causality, and it simulates the actual process of research and redeployment. It produces a usable out-of-sample series. It also reveals parameter *instability* over time. Plot the selected parameters. If they change erratically, the optimization is fitting noise.

Weaknesses. It is expensive, with one fit per step. It consumes data, because early observations only ever serve as training. It can still overfit. If a researcher tunes the walk-forward *design* (window lengths, refit frequency, parameter grid) by looking at the concatenated out-of-sample result, the overfitting moves to the meta level. **[Practice]** This meta-overfitting is extremely common and rarely acknowledged.

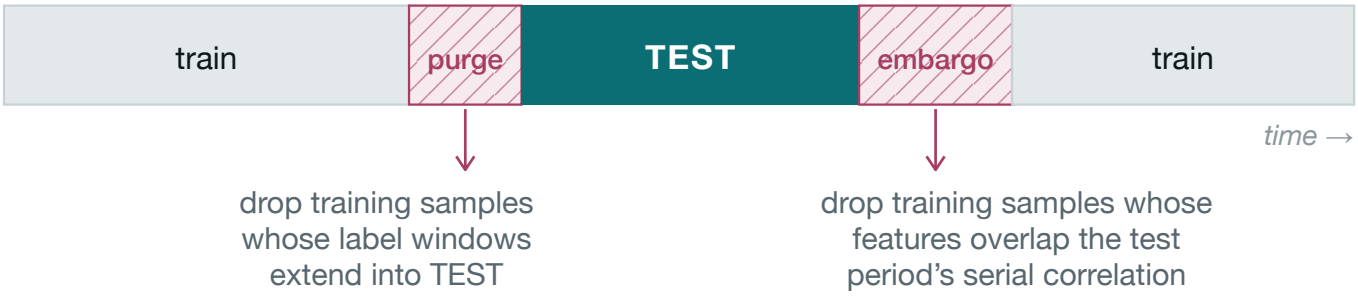
7.2.4 Cross-validation for time series

Standard k-fold cross-validation is invalid on time series, for two reasons. First, it trains on the future to predict the past. Second, and more insidiously, **overlapping labels leak information across folds**. With an h-day forward-return label, the observations at t and at $t+1$ share $h-1$ days of outcome. If one is in the training set and the other in the test set, the test uses data the model trained on.

López de Prado (2018) gives three fixes:

- **Purging.** Remove from the training set any observation whose label window overlaps the time span of the test set.
- **Embargo.** Also remove training observations for a buffer period *after* the test set. This handles serial correlation in features that would otherwise leak backward.
- **Combinatorial Purged Cross-Validation (CPCV).** Instead of one train/test split per fold, form all $\binom{K}{k}$ combinations of k test groups out of K. This generates many distinct backtest paths. The result is a *distribution* of out-of-sample performance rather than a single number. That is the right output, because a single out-of-sample Sharpe ratio carries almost no information about its own uncertainty.

The figure shows a purged, embargoed split.



[Practice] For cross-sectional equity work, also **group by date**. All assets on the same day must go to the same fold, because they are not independent observations.

7.2.5 What good practice actually looks like

A defensible protocol has eight steps, in order:

1. Fix the universe, the data pipeline and the cost model *before* looking at any performance.
2. Explore on a training period, form hypotheses, and count the configurations tried.
3. Validate walk-forward, with purging where labels overlap.
4. Compute the full suite of metrics (§7.3–7.5) on the concatenated out-of-sample series.
5. Apply a multiple-testing correction, using the honest trial count (§7.6).
6. Bootstrap for confidence intervals (§7.7).
7. Touch the locked holdout once.
8. Paper trade before allocating capital. **[Practice]** A paper-trading period of 6–12 months catches implementation bugs, differences between data vendors, and misestimated costs that no backtest will.

7.3 Stage 1–2 metrics: forecasting and signal quality

7.3.1 Information coefficient

$$\mathrm{IC}_t = \mathrm{Corr}_i\big(s_{i,t},\ r_{i,t+1:t+h}\big) \quad \text{(cross-sectional, computed each period)}$$

The IC is the main signal-quality metric. Report the full time series of IC_t, not just its mean.

Calibration for what "good" means. [Fact] A monthly cross-sectional IC of 0.02–0.05 is a useful equity signal, and 0.05–0.10 is excellent. Search for a bug or a leak when the IC stays above 0.15 over a large liquid universe. **Useful financial signals explain a tiny fraction of variance.**

The IC-IR, the information ratio of the IC, is

$$\text{IC-IR} = \frac{\overline{\text{IC}}}{\text{sd}(\text{IC}_t)}, \qquad t\text{-stat} = \text{IC-IR} \cdot \sqrt{T}$$

The IC-IR is more informative than the mean IC alone. Consider two signals. The first has a mean IC of 0.03 and an IC standard deviation of 0.05. The second has a mean IC of 0.05 and a standard deviation of 0.20. The first signal is far better.

The Fundamental Law (Grinold, 1989) relates the IC to the achievable information ratio:

$$\text{IR} \;\approx\; \text{IC} \cdot \sqrt{\text{breadth}}$$

Clarke, de Silva & Thorley (2002) add the **transfer coefficient** TC, the correlation between the ideal portfolio and the actual one, after constraints and costs:

$$\text{IR} \;\approx\; \text{TC} \cdot \text{IC} \cdot \sqrt{\text{breadth}}$$

This is the most useful formula for deciding what to work on. Raising the IC from 0.03 to 0.035 is hard. Raising the transfer coefficient from 0.5 to 0.58 has the same effect and is often much easier. To do it, relax a constraint, reduce a cost, or widen a band. **[Practice]** Most practitioners' realized TC is 0.3–0.6. So implementation, not prediction, loses most of the theoretical alpha. Edge as Information develops this framework in full.

7.3.2 Rank correlation (Spearman IC)

$$\text{IC}_t^{\text{rank}} = \text{Corr}\big(\text{rank}(s_{\cdot,t}),\ \text{rank}(r_{\cdot,t+1})\big)$$

For cross-sectional work, strongly prefer the rank IC to the Pearson IC. Financial returns have fat tails, and a single extreme return can dominate a Pearson correlation across hundreds of names. Consider a Pearson IC on a 500-stock universe in a month when one biotech stock rises 300%. It essentially measures that one stock. The Spearman correlation is immune to this.

[Practice] Report both. A large gap between the Pearson and Spearman ICs means the signal's apparent value is concentrated in the extremes. The cause may be real, because momentum has fat-tailed payoffs. It may also be one data error.

7.3.3 Predictive R^2 and why it looks so bad

For a univariate predictor, $R^2 \approx \text{IC}^2$. An IC of 0.05 gives $R^2 = 0.25\%$.

This small number is no reason for discouragement and no argument against the signal. Campbell & Thompson (2008) show that a monthly out-of-sample R^2 of 0.5% is economically large for a mean-variance investor. Returns are almost all noise. Explaining a small fraction of a large variance is worth a great deal to an investor who can lever and diversify.

Use the **out-of-sample** R^2 against a benchmark forecast, usually the historical mean:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t (r_t - \bar{r}_{t-1})^2}$$

It can be negative, and for published predictors it frequently is. Goyal & Welch (2008) found negative values for most predictors of the equity premium. A negative R^2_{OOS} strongly disqualifies a predictor.

7.3.4 Hit rate, confusion matrices, precision and recall

The hit rate is the fraction of directional calls that are correct. Two cautions apply.

(a) The right benchmark is the hit rate that the IC implies, not 50%. For a jointly normal signal and return, the orthant-probability identity gives

$$\Pr[\text{sign}(s) = \text{sign}(r)] = \frac{1}{2} + \frac{\arcsin(\text{IC})}{\pi}$$

An IC of 0.05 corresponds to a hit rate of **51.6%**. A hit rate of 55% implies an IC near 0.16. For a liquid universe, that IC should prompt suspicion, not pleasure. A reported hit rate of 70% on liquid instruments indicates a bug, a tiny sample, or a strategy with catastrophic tail losses.

(b) The hit rate is nearly irrelevant to profitability. Expected value determines profitability:

$$\mathbb{E}[\text{P\&L}] = p \cdot \mathbb{E}[\text{win}] - (1-p) \cdot \mathbb{E}[|\text{loss}|]$$

[Fact] Trend-following strategies typically have hit rates of 30–45%. They are profitable because the ratio of win size to loss size exceeds 2:1. Optimizing for hit rate destroys trend strategies. It pushes the strategy to take profits early and let losses run, which is the opposite of what makes trend strategies work. **A high hit rate combined with a low payoff ratio is the pattern of a short-volatility strategy.** Such a strategy looks excellent until it fails.

Confusion matrices, precision and recall fit when the signal is a classifier. Examples are a meta-labeling model that decides whether to *act* on a primary signal, and a regime classifier. Precision is the fraction of trades taken that were profitable. Recall is the fraction of profitable opportunities that the strategy captured. This framing is natural and useful for meta-labeling specifically (López de Prado, 2018), because the primary model sets the direction and the secondary model makes a binary decision.

For a directional forecast of continuous returns, confusion matrices **discard magnitude information**, and the P&L depends on magnitude. Use them as diagnostics, not objectives. They can show whether the signal is asymmetric between longs and shorts, or fails in one direction only.

7.3.5 IC term structure and decay

Plot IC(h) against the forecast horizon h. This one chart shows three things:

- The optimal holding period is the horizon where the marginal IC stops covering the marginal cost.
- The chart shows how fast the edge decays. Steep decay means high turnover, so costs dominate.
- The chart shows where momentum turns into reversal *for this signal on this universe*. This is more reliable than any published horizon.

[Practice] Recommendation: make this chart mandatory in every review of a momentum signal. It is more informative than the equity curve.

7.4 Stage 3 metrics: strategy performance

7.4.1 Sharpe ratio

$$\text{SR} = \frac{\mathbb{E}[R_p] - R_f}{\text{sd}(R_p)}, \qquad \text{SR}_{\text{ann}} = \text{SR}_{\text{period}} \cdot \sqrt{A}$$

Annualizing by $\sqrt{A}$ assumes iid returns, and it is wrong when returns are autocorrelated. Lo (2002) gives the correction. For returns with autocorrelations ρ_k, the q-period Sharpe ratio is

$$\text{SR}(q) = \frac{q}{\sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\rho_k}} \cdot \text{SR}$$

Positive autocorrelation in *strategy* returns is common in illiquid strategies and in strategies with smoothed prices. With such autocorrelation, naive annualization **overstates** the Sharpe ratio, sometimes by 50% or more. Check the strategy's own return autocorrelation before annualizing.

Standard error. For iid returns (Lo, 2002),

$$\mathrm{SE}(\widehat{\mathrm{SR}}) \approx \sqrt{\frac{1 + \mathrm{SR}^2/2}{T}}$$

The standard example shows how large the standard error is. Ten years of monthly data give $T = 120$ observations. An annual Sharpe ratio of 1.0 is a monthly Sharpe ratio of $1/\sqrt{12} \approx 0.29$. Then $\mathrm{SE}(\widehat{\mathrm{SR}}_{\text{monthly}}) = \sqrt{(1 + 0.29^2/2)/120} \approx 0.093$. Annualizing multiplies by $\sqrt{12}$. This gives a standard error for the *annualized* Sharpe ratio of roughly **0.33**.

A backtest that reports an annualized Sharpe ratio of 1.0 over 10 years has a 95% confidence interval of about $1.0 \pm 1.96(0.33) = [0.35,\ 1.65]$. **A decade of data is consistent with a true Sharpe ratio anywhere from "barely worth trading" to "excellent."** Give any single backtest number weight in proportion to that width. The interval is this wide before accounting for the *selection* of this strategy from many. This is the strongest argument for the multiple-testing corrections of §7.6.

Weaknesses. The Sharpe ratio ignores skew and kurtosis. So it flatters short-volatility strategies and penalizes the positive skew of trend following. Sharpe ratios do not combine across strategies in an intuitive way. A strategy can also inflate its Sharpe ratio by smoothing returns.

7.4.2 Sortino ratio

$$\text{Sortino} = \frac{\mathbb{E}[R_p] - \tau}{\sqrt{\mathbb{E}[\min(R_p - \tau, 0)^2]}}$$

The Sortino ratio penalizes only downside deviation relative to a target τ. **It suits momentum specifically.** Trend-following returns are positively skewed, and the Sharpe ratio penalizes their upside volatility as if it were risk. **[Practice]** Report both. A Sortino ratio much higher than the Sharpe ratio is evidence of the desired positive skew. The reverse is a warning.

One caveat applies. The downside deviation uses fewer observations, only the negative ones, so it is noisier than the full standard deviation. On the same data, the Sortino ratio has a wider confidence interval than the Sharpe ratio.

7.4.3 Maximum drawdown and Calmar

Let W_t be the equity curve. The drawdown at t is the shortfall from the running peak. The maximum drawdown (MDD) is its worst value over the sample:

$$\mathrm{DD}_t = 1 - \frac{W_t}{\max_{s \le t} W_s}, \qquad \mathrm{MDD} = \max_{t \le T} \mathrm{DD}_t, \qquad \text{Calmar} = \frac{\text{annualized return}}{\mathrm{MDD}}$$

The normalization goes *inside* the maximum. Each drawdown is relative to the peak that preceded it, not to a single global peak.

The statistical properties of MDD are poor, and too few practitioners know it:

- It is a **single-realization extreme-value statistic**: one number from one path. Its sampling distribution is very wide.
- It **increases mechanically with sample length.** A 30-year backtest shows a larger MDD than a 10-year one for the same process. Comparing MDDs across strategies with different histories is meaningless.
- It is **highly sensitive to the start date**.

[Practice] Use MDD for *operational* checks: whether the business can survive the drawdown, and whether it breaches a risk limit. Do not use it to compare strategies. For comparison, prefer the *distribution* of drawdowns from a block bootstrap (§7.7). That distribution shows what drawdowns the process generates, not which one happened to occur. Report the expected maximum drawdown and its 95th percentile, not the realized one.

The Calmar ratio inherits all of MDD's problems and adds the noise of the return estimate. It is popular with allocators. Treat it as a communication tool.

7.4.4 Alpha, beta, and factor attribution

Regress the strategy's returns on a factor model:

$$R_{p,t} - R_{f,t} = \alpha + \beta_{\text{MKT}}\text{MKT}_t + \beta_{\text{SMB}}\text{SMB}_t + \beta_{\text{HML}}\text{HML}_t + \beta_{\text{UMD}}\text{UMD}_t + \varepsilon_t$$

For a momentum strategy, **including UMD is the essential test**. If a novel signal's alpha vanishes against UMD, the signal has rebuilt the momentum factor. That result still has value, because the signal may be a cheaper or higher-capacity implementation. It is a different claim, and the author should state it as such.

Two additions go beyond the standard regression:

1. **Conditional beta.** Given §1.6, estimate beta separately in up and down markets, and in high- and low-volatility states. Alternatively, use an interaction term $\beta_{\text{down}} \cdot \text{MKT}_t \cdot \mathbb{1}\{\text{bear}\}$. **[Fact]** An unconditional beta near zero together with a strongly time-varying conditional beta is the characteristic pattern of momentum. Reporting only the unconditional number omits the crash risk.
2. **Option-like exposure.** Regress on $\max(\text{MKT}, 0)$ and $\min(\text{MKT}, 0)$ separately, or include MKT^2. Trend strategies show positive convexity, the payoff shape of a long straddle. Cross-sectional momentum shows negative convexity in panics.

7.5 Statistical significance done properly

The t-statistic has four problems specific to this domain:

1. **Autocorrelation and overlapping data** inflate naive t-statistics. Use Newey–West with a lag of at least $h - 1$.
2. **Fat tails** make the normal approximation poor in small samples. Bootstrap.
3. **The threshold should be above 2.** Harvey, Liu & Zhu (2016) argue for $t > 3.0$ for a *newly proposed* factor, because of the intensity of collective search. For a strategy found after trying 50 configurations, even 3.0 is too lenient.
4. **Cross-sectional dependence.** N stocks on the same day carry one observation's worth of macro information, not N. Cluster standard errors by date, or use Fama–MacBeth with appropriate corrections.

7.6 Multiple testing and data snooping

This is the most important part of Section 7. The core problem is:

> Test M independent strategies with no true edge. The expected maximum Sharpe ratio among them is roughly $\text{SE}(\text{SR}) \cdot \sqrt{2 \ln M}$. With $M = 1{,}000$ and a 10-year monthly backtest (SE ≈ 0.33), the expected best Sharpe ratio from *pure noise* is about **1.2**.

A backtested Sharpe ratio of 1.2 after a thousand trials is the result that noise alone produces. The effect is large.

7.6.1 White's Reality Check (White, 2000)

The Reality Check tests the null that the best of M candidate strategies has no superior predictive ability over a benchmark. It accounts for the selection of the best strategy.

Procedure.

1. Let $f_{m,t}$ be model m's performance relative to the benchmark at t, and $\bar{f}_m$ its mean. The test statistic is $V = \max_m \sqrt{T}\bar{f}_m$.
2. Generate bootstrap resamples (the stationary bootstrap, §7.7) of the *joint* performance series across all M models.
3. For each resample b, compute $V_b^* = \max_m \sqrt{T}(\bar{f}_{m,b}^* - \bar{f}_m)$.
4. The p-value is the fraction of the V_b^* that exceed V.

Key property. It resamples all models *jointly*, so it handles the correlation among them. This matters because 1,000 momentum rules are not 1,000 independent tests.

Limitation. The test can be conservative when the set includes many poor models, because its "worst-case" null is unrealistically pessimistic. **Hansen's (2005) SPA test** fixes this by studentizing and down-weighting clearly inferior models. **Prefer SPA in practice.** To identify *which* models are significant, not just whether any is, use the stepwise multiple testing of **Romano & Wolf (2005)**.

7.6.2 Deflated Sharpe Ratio (Bailey & López de Prado, 2014)

The deflated Sharpe ratio adjusts an observed Sharpe ratio for four things. They are the number of trials, the variance of the Sharpe ratios across trials, the non-normality of returns and the sample length.

First, compute the expected maximum Sharpe ratio under the null of no skill across M independent trials:

$$\mathbb{E}[\max_m \mathrm{SR}_m] \approx \sqrt{\mathrm{Var}(\mathrm{SR}_m)}\left[(1-\gamma)\,Z^{-1}\big(1-\tfrac{1}{M}\big)+\gamma\,Z^{-1}\big(1-\tfrac{1}{Me}\big)\right]$$

with $\gamma \approx 0.5772$ (Euler–Mascheroni) and Z^{-1} the inverse normal CDF. This becomes the benchmark SR_0. The DSR is then the Probabilistic Sharpe Ratio evaluated against that benchmark:

$$\mathrm{DSR} = Z\left(\frac{(\widehat{\mathrm{SR}}-\mathrm{SR}_0)\sqrt{T-1}}{\sqrt{1-\hat{\gamma}_3\widehat{\mathrm{SR}}+\frac{\hat{\gamma}_4-1}{4}\widehat{\mathrm{SR}}^2}}\right)$$

where $\hat{\gamma}_3$ is the skewness and $\hat{\gamma}_4$ the kurtosis of the strategy's returns. The skewness and kurtosis terms make the DSR robust to non-normality. Negative skew and fat tails widen the effective denominator and lower the DSR.

Interpret the DSR precisely. The DSR is the probability that the strategy's *true* Sharpe ratio exceeds SR_0. The search process is expected to produce this level from strategies with **no skill at all**. The DSR is not the probability that the true Sharpe ratio exceeds zero. A DSR of 0.6 does not mean "60% likely to be profitable". It means the result is barely distinguishable from the best of M coin flips. Conventional practice sets the bar at DSR > 0.95.

Strengths. It is directly usable and handles non-normality. It also forces the researcher to state M, which is the main discipline it imposes. **Weaknesses.** It requires an honest M, which nobody has. It also assumes a specific distribution of trial Sharpe ratios.

A related and useful tool is the **Probability of Backtest Overfitting (PBO)**. Researchers compute it through combinatorially symmetric cross-validation (Bailey, Borwein, López de Prado & Zhu, 2017). Split the sample into S blocks and form every split into train and test halves. In each split, select the best configuration in sample, and measure how often it ranks below the median out of sample. A PBO above about 0.5 means the selection procedure is worse than random.

7.6.3 False discovery rate

When testing many signals, the aim is sometimes to control the *proportion* of false positives among the discoveries, not the probability of any false positive. In that case use Benjamini–Hochberg (1995). Sort the p-values in ascending order, find the largest k with $p_{(k)} \le \frac{k}{M}q$, and reject all hypotheses up to k. The procedure is far less conservative than Bonferroni. It is generally the right choice when screening a library of signals.

[Practice] The honest, simple version of all this is to **log every configuration tested, including the ones abandoned after five minutes.** The number in that log is M. Most researchers underestimate that number by an order of magnitude.

7.7 Bootstrapping and Monte Carlo

7.7.1 Which bootstrap

The plain iid bootstrap is **invalid** on financial time series. It destroys serial dependence, including volatility clustering and the autocorrelation that momentum depends on. Use one of these instead:

- **Block bootstrap.** Resample contiguous blocks of length b. This preserves dependence within blocks. The choice of b matters, and b should exceed the signal's memory.
- **Stationary bootstrap** (Politis & Romano, 1994). Use random block lengths drawn from a geometric distribution with mean $1/p$. This produces a stationary resampled series. The result is less sensitive to the choice of block length. **[Practice]** It is the default, and White's Reality Check assumes it.
- **Circular block bootstrap.** Wrap around the end of the series, so that every observation is equally likely to be resampled.

Uses. The bootstrap gives confidence intervals for the Sharpe ratio, the IC and the drawdown, and the null distribution for the Reality Check and SPA. Most valuably, it gives the *distribution of the maximum drawdown*. That distribution is the only sound way to interpret the statistic.

7.7.2 Permutation and randomization tests

These tests are useful and underused:

- **Shuffle the signal, keep the returns.** This destroys any relation between signal and return, while preserving both marginal distributions and the dependence in the return series. Repeat 1,000 times to get the null distribution of the performance metric. If the real strategy falls inside that distribution, stop.
- **Shuffle returns across assets within a date.** This preserves the cross-sectional distribution and the market factor, while destroying asset-specific predictability. It isolates whether the edge comes from asset selection or from market timing.
- **Randomize entry timing.** Keep position sizes and holding periods, and randomize when trades start. This tests whether the *timing* carries information, or only the exposure profile.

7.7.3 Monte Carlo

- **Synthetic price paths.** Simulate from a fitted model, such as GARCH, a jump diffusion or regime switching, *with and without* a momentum component. Then verify that the pipeline detects the effect when it is present and does not detect it when it is absent. **This is the best way to validate a research pipeline, and almost nobody does it.** A pipeline that reports a Sharpe ratio of 1.5 on data generated as a pure random walk has a bug. This method finds such a bug much faster than reasoning about the code.
- **Randomizing trade order.** Reshuffle the sequence of realized trade P&Ls to get a distribution of equity curves and drawdowns. This destroys serial dependence, so it *understates* drawdown risk. Treat the result as a lower bound.
- **Parameter Monte Carlo.** Sample parameters from plausible priors instead of optimizing them, and report the resulting distribution of performance. This shows what the strategy would have earned without hindsight.

7.8 Which metrics matter for which purpose

The table matches metrics to the four purposes of evaluation.

Purpose	Primary metrics	Secondary	Actively misleading here
Forecasting — does the signal predict?	Rank IC, IC-IR, IC term structure, R^2_{OOS}	Pearson IC, hit rate	Sharpe (conflates prediction with sizing); MDD
Signal quality — is it stable and distinct?	IC-IR, subsample IC stability, decay rate, correlation to existing signals, factor-neutralized IC	Turnover-adjusted IC, IC by sector/size/regime	Total return; single-period IC
Portfolio construction — how is it traded?	Transfer coefficient, marginal contribution to portfolio Sharpe, factor exposures, conditional beta, correlation to existing book	Turnover, breadth	Standalone Sharpe (ignores diversification); hit rate
Deployability — should this get capital?	Net-of-cost Sharpe, capacity Q^*, turnover, bootstrapped drawdown distribution, DSR/PBO, Sortino	Calmar, realized MDD, operational complexity	Gross Sharpe; in-sample anything; realized MDD alone

Two points apply across the table:

- **Use only the metrics appropriate to the stage.** A signal with an excellent IC and a terrible standalone Sharpe ratio may be a superb *addition* to a portfolio. Judging it on its standalone Sharpe ratio rejects it wrongly. Conversely, a strategy with a great Sharpe ratio and 800% annual turnover is not deployable, however good its signal is.
- **Always report net of realistic costs.** A momentum backtest gross of costs measures a different and much less interesting quantity than the net result. **[Practice]** Report gross and net side by side. The *gap* is itself diagnostic. A strategy whose gap exceeds its net Sharpe ratio depends on the cost model, not on alpha.

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§7 Key takeaways

1. **Evaluate in stages**, prediction → signal quality → strategy → statistical validity → deployability, and stop at the first stage that fails.
2. **The IC is the core signal metric.** A monthly cross-sectional IC of 0.02–0.05 is genuinely good. Use the Spearman version for fat tails, and report the IC *series* and its IR, not just the mean.
3. $\text{IR} \approx \text{TC} \cdot \text{IC} \cdot \sqrt{\textbf{breadth}}$ shows where to spend effort. Implementation (TC), not prediction (IC), loses most alpha.
4. **A hit rate of 51.6% corresponds to an IC of 0.05.** Anything much above that on liquid instruments is a bug. The hit rate is nearly irrelevant to profitability. Expected value determines profit, and trend strategies are profitable at hit rates of 30–45%.
5. **A 10-year backtest showing a Sharpe ratio of 1.0 has a 95% interval of roughly [0.35, 1.65].** $\text{SE}(\widehat{\text{SR}}) \approx \sqrt{(1 + \text{SR}^2/2)/T}$, about 0.33 annualized on monthly data. Account for this interval before trusting any single number.
6. **The maximum drawdown is a one-observation extreme-value statistic** that grows mechanically with sample length. Use the bootstrapped *distribution* of drawdowns for comparison, and the realized one for operations.
7. **Multiple testing is the largest threat.** The expected best Sharpe ratio from M noise strategies is $\approx \text{SE}(\text{SR})\sqrt{2 \ln M}$, about 1.2 for 1,000 trials on 10 years of monthly data. Use SPA (preferred over the plain Reality Check), the Deflated Sharpe Ratio and PBO, and keep an honest log of trials.
8. **Use purged, embargoed, date-grouped cross-validation.** Standard k-fold leaks through overlapping labels and cross-sectional correlation.
9. **Bootstrap with blocks, not iid.** And run the pipeline on synthetic random-walk data. If it finds alpha there, it has a bug.
10. **Only one clean out-of-sample test exists.** Every test after the first look is in-sample. Paper trading on genuinely new data is the only fully honest validation.

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8. Current best practices

A caveat comes first. No one outside a firm knows exactly how it trades. The inferences in this section come from four sources. They are published research by authors affiliated with practitioners, methodologies disclosed in fund documents and index construction rules, conference material, and the observable characteristics of returns. Treat the whole section as **[Practice]**, with an added step of inference.

8.1 The central shift: momentum is a building block, not a strategy

Essentially no sophisticated firm runs a standalone momentum strategy. These firms treat momentum as:

- **a risk premium**, to be harvested cheaply and at scale alongside value, carry, quality and defensive;
- **a source of convexity**, whose main role in a portfolio is to diversify and hedge the tail of a long-only book;
- **one input among many** to a combined signal, rather than a decision rule.

This view of momentum changes every design decision. If momentum is a portfolio component, the objective is its *marginal contribution* to the combined portfolio, not its *standalone* Sharpe ratio. If it is a source of convexity, the design must protect its positive skew, and optimizing its hit rate would destroy that skew. If it is a risk premium, fees, capacity and cost efficiency matter more than small improvements to the signal.

8.2 What has stood the test of time

[Fact] 1. The effect itself. Momentum survived the replication crisis better than almost any other anomaly. It works across 200 years, dozens of countries and every major asset class, and researchers documented it out of sample after publication. It is the most robust finding in the field.

[Fact] 2. Diversification is the largest single improvement to the Sharpe ratio. Diversified futures trend programs have historically achieved Sharpe ratios far above those of any individual market, because trend signals are only weakly correlated across markets most of the time. Nothing else in this chapter improves risk-adjusted return as much as adding uncorrelated markets. The century-long evidence of Hurst, Ooi and Pedersen depends on this effect.

[Fact] 3. Volatility scaling. It applies at the asset level (equal risk contribution), at the strategy level (the risk-managed momentum of Barroso and Santa-Clara), and at the portfolio level (volatility targeting). Practitioners now use it universally. It is the most valuable practical refinement of the last two decades.

[Fact] 4. Ensembling across horizons. Averaging signals across geometrically spaced lookbacks is more robust out of sample than any single horizon. It is standard in production trend systems.

[Fact] 5. Skipping the most recent period in cross-sectional equity momentum, to avoid short-horizon reversal.

[Fact] 6. Combining momentum with value. The two are negatively correlated (Asness, Moskowitz & Pedersen, 2013), so the combination is materially better than either alone. The value/momentum pair is the basis of modern multi-factor investing.

[Fact] 7. Costs and turnover treated as first-class design constraints, not as later adjustments. The tools are cost-aware portfolio optimization, no-trade bands, and partial adjustment toward an aim portfolio.

[Practice] 8. Robust signal transforms. Sign, rank, clipping and saturating response functions replace raw magnitudes. The evidence that magnitude adds much beyond direction is weak.

8.3 What has been superseded or demoted

Single-indicator systems. The evidence that the *form* of an indicator matters much at a fixed horizon is weak (Levine & Pedersen, 2016). Describing a system by a specific indicator, such as "a MACD strategy", is a retail way to describe a strategy.

Fixed oscillator thresholds. RSI 30/70, ADX 25 and similar thresholds have no derivation and are heavily data-snooped. Where professionals use oscillators, they use them as continuous conditioners or ML features.

Unmanaged momentum. Running cross-sectional momentum without volatility scaling and without factor neutralization is now considered a strictly dominated design, because cheap methods now reduce the crash risk.

Single-market trend following. Diversified implementations dominate it, for the reason given in §8.2.

"Momentum is purely behavioral." The conditional-risk literature (Daniel & Moskowitz, 2016; Kelly, Moskowitz & Pruitt, 2021) is serious enough that treating all momentum returns as riskless alpha is naïve. Some fraction is compensation for a conditional beta that appears at exactly the worst moment.

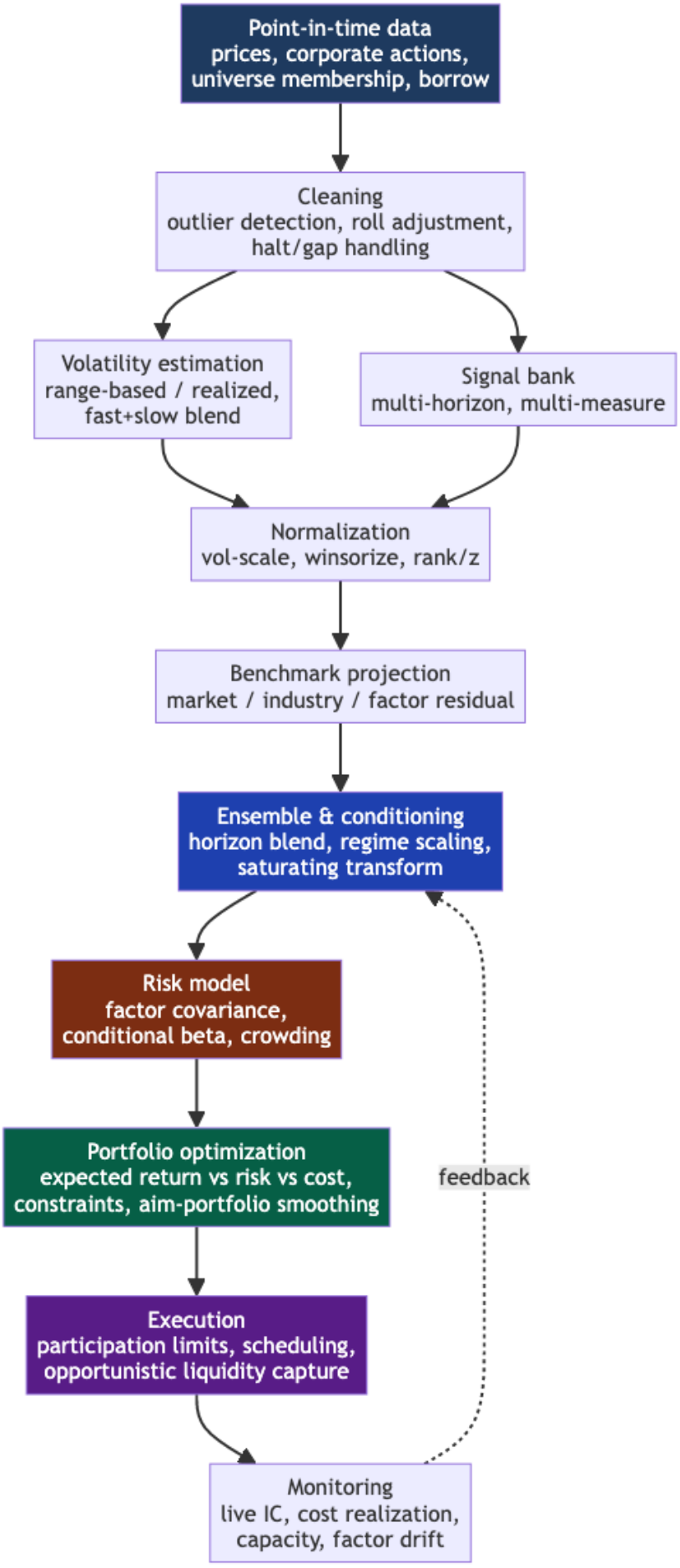
Grid-search parameter optimization. Ensembling and regularization have replaced it. Where practitioners still optimize, they use walk-forward selection with explicit accounting for multiple testing.

Naive TSMOM tests. Since Huang et al. (2020), researchers do not take seriously a claim of time-series momentum that does not control for the unconditional mean.

Chart-pattern recognition as a primary signal. Lo, Mamaysky & Wang (2000) tested it most rigorously. Their verdict was "some information, unclear profitability."

8.4 How a sophisticated implementation looks today

The diagram combines the methodology disclosed across the industry into the rough structure of a modern momentum implementation.


Point-in-time data
prices, corporate actions,
universe membership, borrow
Cleaning
outlier detection, roll adjustment,
halt/gap handling
Volatility estimation
range-based / realized,
fast+slow blend
Signal bank
multi-horizon, multi-measure
Normalization
vol-scale, winsorize, rank/z
Benchmark projection
market / industry / factor residual
Ensemble & conditioning
horizon blend, regime scaling,
saturating transform
Risk model
factor covariance,
conditional beta, crowding
Portfolio optimization
expected return vs risk vs cost,
constraints, aim-portfolio smoothing
Execution
participation limits, scheduling,
opportunistic liquidity capture
Monitoring
live IC, cost realization,
capacity, factor drift
feedback

Three observations follow from the diagram:

1. **The signal bank is a small part of the system.** Most of the diagram, and most of the headcount at a serious firm, deals with data, risk, portfolio construction and execution. This is the clearest indication of where the competitive edge now lies.
2. **Monitoring feeds back into conditioning.** Firms track realized IC and realized costs against expectations continuously. A material divergence triggers an investigation.
3. **The risk model is separate from the signal.** Firms measure and control factor exposures explicitly, rather than accepting whatever exposures arise.

The competitive edge has moved from the signal to its implementation. Everyone knows about 12-2 momentum, and nobody's edge comes from discovering it. The edge comes from trading it more cheaply, at greater capacity and with better crash management, and from combining it with signals that offset its weaknesses.

8.5 An honest note on expected performance

[Fact] Both diversified trend following and equity cross-sectional momentum were materially weaker in the 2010s than in the three decades before. Trend following faced a low-volatility, mean-reverting environment dominated by central banks. Equity momentum suffered severe reversals, among them March–May 2009 and November 2020, the rotation after the vaccine announcement.

[Contested] Two explanations compete. One is alpha decay from crowding and lower fees. The other is an ordinary run of bad luck. A strategy with a Sharpe ratio around 0.5–1.0 has long drawdown-prone periods as a matter of arithmetic. Both explanations have reasonable arguments. For example, a strategy with a true Sharpe ratio of 0.7 has roughly a 1-in-4 chance of a negative 3-year period, and a meaningful chance of a flat decade. **A weak decade alone is not evidence of decay.**

[Practice] Recommendation: plan for a true Sharpe ratio of 0.4–0.8 for a well-built, realistically costed, diversified momentum program at meaningful scale. Backtests showing 2.0 or more almost always measure overfitting, unrealistic costs, or capacity that does not exist.

8.6 Where active research is focused

1. **Stock momentum may be derivative. [Contested]** The factor-momentum literature (Gupta & Kelly, 2019; Ehsani & Linnainmaa, 2022; Arnott et al., 2023) argues that momentum in individual stocks may be a *consequence* of autocorrelation in factor returns. If true, that changes both the explanation of the mechanism and the optimal implementation. Researchers are actively working on this question.
2. **The share of momentum returns that is conditional risk.** Kelly, Moskowitz & Pruitt (2021) and the IPCA program address this question. If a large fraction of momentum returns compensates for time-varying beta, the "alpha" framing is wrong, and the implications for hedging follow directly.
3. **Crash prediction and dynamic hedging.** This work goes beyond volatility scaling. It predicts the conditional beta, hedges the option-like exposure explicitly, and uses options rather than dynamic replication.
4. **Machine learning: where, and how much.** The claim of a "virtue of complexity" (Kelly, Malamud & Zhou, 2024) contradicts decades of orthodoxy that favored parsimony. Whether heavily over-parameterized models help in low-signal financial settings is one of the most consequential open questions in the field.
5. **Measuring crowding.** Positioning data, factor-return correlation, dealer gamma and short-interest metrics can serve as inputs to a momentum strategy aware of capacity and crowding. Momentum is unusual in that its own popularity plausibly changes its behavior. The feedback loop is real, and Vayanos and Woolley (2013) formalize one version of it.
6. **Alternative data and momentum in information space.** News sentiment, revision momentum, flow and positioning data, and options-implied signals are candidates. The question is whether they are new sources of momentum or faster versions of the same one.
7. **Intraday and microstructure momentum.** This is prediction from order flow with deep learning (Sirignano & Cont, 2019; Zhang, Zohren & Roberts, 2019). Its capacity is tightly constrained, but the effect is real and growing.
8. **Cross-asset and macro momentum.** Trends in macro variables, such as inflation, growth and policy, can condition asset momentum. Their interaction with carry is also under study.
9. **New asset classes.** Crypto shows strong momentum characteristics, in a short, non-stationary and heavily retail-driven history. **[Contested]** The sample is too short to support confident claims, and the microstructure differs materially.

10. **Non-linear response functions.** The empirical shape of the optimal map from signal strength to position, including where saturation and attenuation should begin, is under-studied relative to its practical importance.

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§8 Key takeaways

1. **Momentum is a component, not a product.** Firms harvest it as a risk premium, use it as a source of convexity, and combine it with offsetting signals, especially value.
2. **What survived:** the effect itself, diversification, volatility scaling at three levels, ensembling across horizons, skipping a month, the value/momentum combination, cost-aware construction and robust transforms.
3. **What was superseded:** single-indicator systems, fixed oscillator thresholds, unmanaged momentum, single-market trend, grid-search optimization, naive TSMOM tests and the purely behavioral framing.
4. **The edge has moved from the signal to its implementation.** Firms now differ in data quality, risk modeling, portfolio construction, execution and capacity management.
5. **Calibrate expectations honestly:** a realistic Sharpe ratio at scale is 0.4–0.8, with decade-long weak periods that are statistically ordinary. A weak decade is not proof of decay.
6. **Open research questions:** factor momentum versus stock momentum; how much is conditional beta; the complexity of ML models; crowding; microstructure momentum; and the shape of the response function.

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9. Synthesis

9.1 A unifying conceptual framework

The whole chapter reduces to three linked ideas.

(I) The generating mechanism: incomplete adjustment

Price impounds information *gradually*, for four reasons. Information diffuses through a heterogeneous population. Agents are biased. Traders must split large orders. Markets reprice risk with a lag. Formally (§1.2), the log price is a distributed lag on value innovations, $p_t = c + \sum_j \Psi_j \delta_{t-j}$. Here Ψ_j is the fraction of an innovation impounded within j periods, with $\Psi_j \to 1$ and $\Psi_0 < 1$. Returns then take the per-period response $\psi_j = \Psi_j - \Psi_{j-1}$ as their moving-average weights. Their autocorrelation is $\rho_k = \sum_j \psi_j \psi_{j+k} / \sum_j \psi_j^2$. This one assumption produces momentum ($\psi_j \geq 0$), later reversal (overshoot, $\Psi_K > 1$, then $\psi_j < 0$), and horizon dependence together. It also explains why momentum's size varies with the *frictions* in a market, not with anything about the assets themselves.

(II) The observable signature: the variance ratio

The impulse response is unobservable, but its integral is observable. The variance-ratio profile

$$\mathrm{VR}(q) = \frac{\mathrm{Var}(p_t - p_{t-q})}{q\,\mathrm{Var}(p_t - p_{t-1})} = 1 + 2\sum_{k<q}\left(1 - \tfrac{k}{q}\right)\rho_k \xrightarrow[q\to\infty]{} \frac{2\pi f_r(0)}{\sigma_r^2}$$

shows, for each horizon q, whether prices diffuse faster (trend) or slower (reversion) than a random walk. The P&L of a trend follower is also, to leading order, proportional to $\mathrm{VR}(q) - 1$ at the strategy's horizon (Dao et al., 2017). **The quantity measured, the quantity traded and the quantity that pays are the same.**

The same quantity has three views:

- **Time domain:** positive autocorrelation of returns at lags $\leq q$.
- **Frequency domain:** excess spectral power at low frequencies.
- **P&L domain:** long-horizon variance exceeding short-horizon variance, which is equivalent to a long straddle position (Fung & Hsieh, 2001).

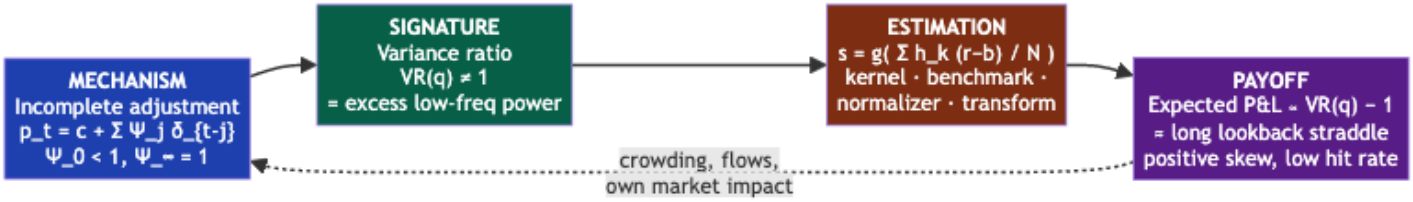
(III) The estimation problem: one master form

Every measure is a choice of four things:

$$s_{i,t} \;=\; g\left(\frac{\sum_{k\geq 0} h_k\,(r_{i,t-k} - b_{i,t-k})}{\mathcal{N}_{i,t}}\right)$$

the **kernel** h (which horizon, which weighting), the **benchmark** b (measured against what), the **normalizer** $\mathcal{N}$ (in what units) and the **transform** g (how aggressively to act).

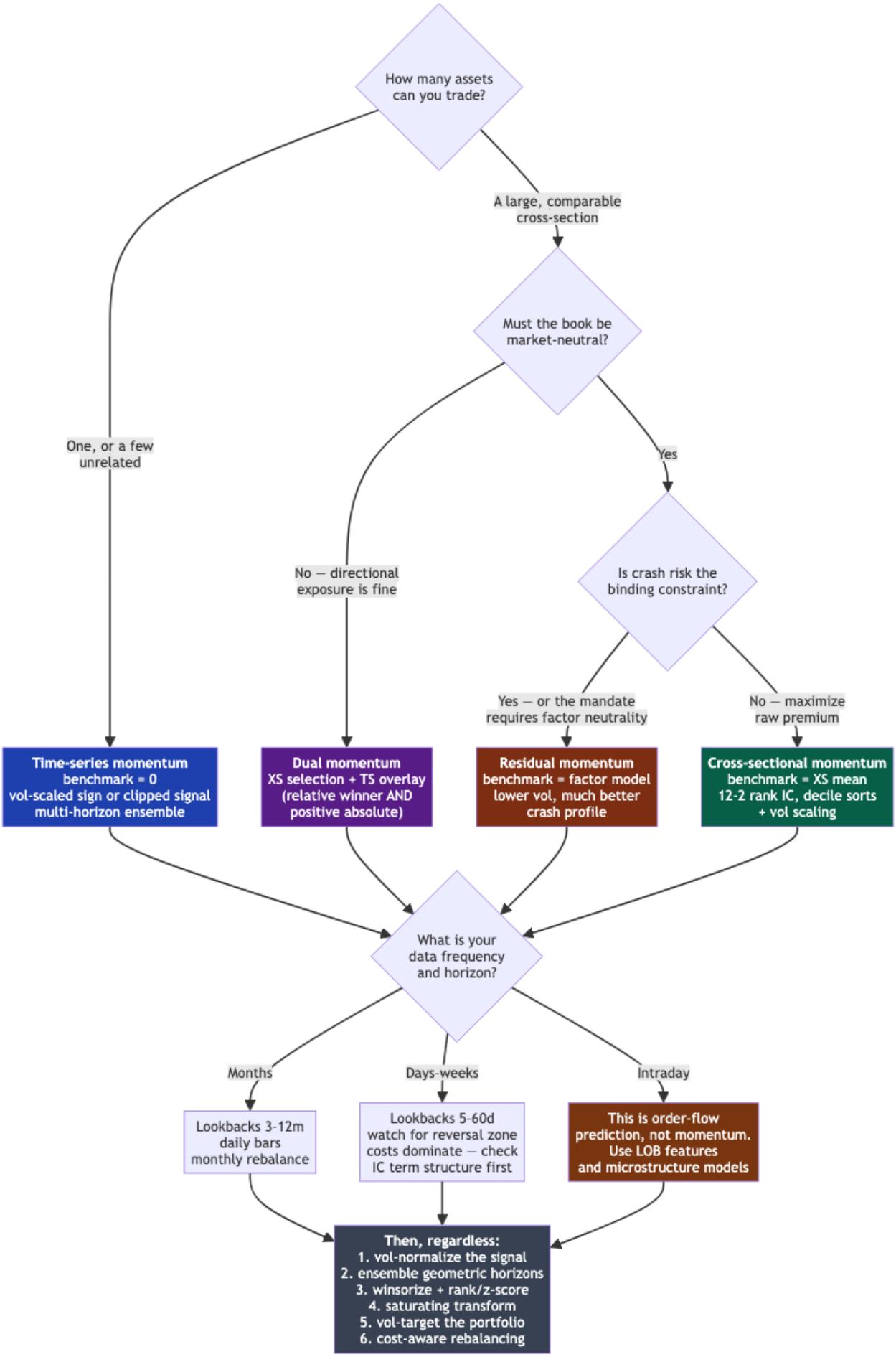
The empirical ordering of importance is: **horizon > benchmark ≈ normalizer > transform > kernel shape.** The diagram links the three ideas to the payoff.



The dashed feedback arrow shows that trend followers' own flows contribute to the impulse response they try to detect. This is why crowding is a continuing concern. It is also why the effect is self-limiting, not self-eliminating.

9.2 A decision tree for choosing a momentum measure

The tree maps the trader's constraints to a choice of momentum measure.


How many assets can you trade?
One, or a few unrelated
A large, comparable cross-section
Must the book be market-neutral?
No — directional exposure is fine
Yes
Is crash risk the binding constraint?
Yes — or the mandate requires factor neutrality
No — maximize raw premium
Time-series momentum
benchmark = 0
vol-scaled sign or clipped signal
multi-horizon ensemble
Dual momentum
XS selection + TS overlay
(relative winner AND positive absolute)
Residual momentum
benchmark = factor model
lower vol, much better crash profile
Cross-sectional momentum
benchmark = XS mean
12-2 rank IC, decile sorts
+ vol scaling
What is your data frequency and horizon?
Months
Days-weeks
Intraday
Lookbacks 3-12m
daily bars
monthly rebalance
Lookbacks 5-60d
watch for reversal zone
costs dominate — check IC term structure first
This is order-flow prediction, not momentum.
Use LOB features
and microstructure models
Then, regardless:
1. vol-normalize the signal
2. ensemble geometric horizons
3. winsorize + rank/z-score
4. saturating transform
5. vol-target the portfolio
6. cost-aware rebalancing

The tree assumes an answer to a prior question: whether momentum exists in this market at this horizon at all. Compute a variance-ratio profile and an IC term structure before anything else. If $\mathrm{VR}(q) \leq 1$ across the candidate horizons, and the IC term structure is flat or negative, no choice of indicator will help.

9.3 Recommendations for building a systematic momentum strategy from scratch

The roadmap below has stages. **Do not skip stages. Do not proceed to the next stage until the current stage is clean.** Roughly half of the stages have nothing to do with momentum.

Stage 0 — Infrastructure (takes weeks, and is worth the time)

- Point-in-time data: prices, corporate actions as *as-of* adjustment factors, universe membership, delisting returns, and borrow availability.
- A backtester with explicit, auditable timing: the signal is computed on bar t and executed on $t + 1$. Make the lag a parameter, so the delay test of §6.10 can be run.
- A realistic cost model: half-spread + $Y\sigma\sqrt{Q/V}$ + fees + borrow.
- **Validate the pipeline on synthetic data** (§7.7.3). A pure random walk must produce a Sharpe ratio near zero. A synthetic series with a known injected AR(1) must produce a detectable signal of the right sign and roughly the right size. This is the most valuable hour of the whole project.

Stage 1 — Characterize the market before modeling it

- The variance-ratio profile with heteroskedasticity-robust confidence bands, per market and per era.
- The IC term structure for a simple lookback-return signal across horizons.
- Volatility dynamics: persistence, clustering, and the relation between volatility and subsequent returns.
- The liquidity profile: the distribution of ADV, spreads, and how they behave under stress.

Decision point. If momentum is not visible here, stop. Stopping now saves months.

Stage 2 — Build the simplest thing that could work

$$s_{i,t} = \frac{r_{i,t-252:t-21}}{\hat{\sigma}_{i,t}\sqrt{231}}, \qquad w_{i,t} = \mathrm{clip}\big(\text{rank-}z(s_{i,t}),\, -2,\, 2\big) \cdot \frac{\sigma^*}{\hat{\sigma}_{i,t}}$$

Rebalance monthly, with no optimization, no filters and no regime conditioning. **This is the benchmark, and it is hard to beat.** Every later addition must beat it, net of costs and out of sample.

Stage 3 — Add the refinements that are known to work, in order of evidence

1. **A multi-horizon ensemble**, with geometric spacing: 21, 63, 126 and 252 days.
2. **Portfolio volatility targeting.**
3. **Risk-managed scaling** by the strategy's own trailing volatility (Barroso & Santa-Clara).
4. **Factor and industry neutralization**, or residual momentum, if crash risk binds.
5. **Cost-aware rebalancing:** no-trade bands, and partial adjustment toward the aim portfolio.
6. **Diversification across markets and asset classes.** This is the largest single improvement available, and it belongs earlier if the markets are accessible.

Measure the marginal contribution of each addition separately, net of costs and out of sample. **[Practice]** Expect roughly half of them to add less than hoped, and the diversification step to add more.

Stage 4 — Validate honestly

- Walk-forward validation, with purged, embargoed, date-grouped splits.
- A block bootstrap for all confidence intervals, including the distribution of drawdowns.
- The Deflated Sharpe Ratio and PBO, using the honest trial count.
- A permutation test: shuffle the signal, and confirm that the real result falls outside the null distribution.
- Stability across subsamples: by decade, volatility regime, market state, sector and size bucket. **[Practice]** Momentum that works only in one decade or one sector is not momentum.
- The one-extra-bar delay test.

Stage 5 — Deploy carefully

- Paper trade for 6–12 months. Compare realized IC and realized costs with the backtest's expectations. The gap measures the model error.
- Start at a fraction of the target size, and scale in as realized costs confirm the model.
- Instrument everything: live IC, turnover, cost per trade against prediction, factor exposures, and drawdown against the bootstrapped distribution.
- Commit to a shutdown rule *before* starting. Express it relative to the bootstrapped distribution of drawdowns, not as a round number. Decide in advance what evidence would show that the edge is gone. Otherwise the decision falls during a drawdown, the worst time to make it.

Advice for someone starting today

1. **Spend the first month on data and pipeline correctness, not on signals.** Most failed momentum research fails on data, and the failure is silent.
2. **Beat the simple benchmark or use it.** Any added complexity must beat the benchmark, net of costs and out of sample.
3. **Diversify across markets before optimizing within one.** It is the largest available improvement in the Sharpe ratio, and the least likely to be overfit.
4. **Scale everything by volatility:** signal, position and portfolio. It costs little, and it addresses momentum's single worst characteristic.
5. **Count trials honestly and deflate accordingly.** The expected best Sharpe ratio from 1,000 noise strategies on a decade of monthly data is about 1.2. A result not clearly above that level shows nothing beyond noise.
6. **Know which of the three sources the strategy harvests:** own-autocovariance, lead-lag structure, or dispersion in unconditional means. They have completely different stability and capacity, and only the first is what most people mean by "momentum."

9.4 Closing

Momentum is the most durable empirical regularity in financial markets, and at the same time one of the least understood. It has persisted through two centuries of data, 30 years of academic scrutiny, a replication crisis that eliminated most other anomalies, and its own widespread adoption. Yet researchers have not reached a consensus on why it exists. The most serious modern explanations are conditional risk, order-flow mechanics and flows from delegated management. Each would imply meaningfully different implementations.

The practical response is to build systems that work under any of the competing explanations, without waiting for the theory. Such systems diversify across markets, so that the result does not depend on a single mechanism. They scale positions by volatility, so that the crash exposure is bounded, whether it comes from behavioral over-extension or from a conditional beta. They account for costs, so that the edge survives implementation. They validate results honestly, so that the builder knows what the system does.

The variance-ratio profile remains the reference point. For a new momentum indicator, ask which horizon it operates at, what it is measured against, and what units it uses. Also ask whether that market trends at that horizon at all. The shape of its kernel matters little.

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§9 Key takeaways

1. **One mechanism:** price adjusts incompletely to information, because of frictions in diffusion, psychology, execution and the repricing of risk.
2. **One observable measure:** the variance-ratio profile. Equivalently, excess low-frequency spectral power. Equivalently again, long-horizon variance exceeding short-horizon variance. The measured quantity and the quantity that pays are the same.
3. **One master form:** $s = g\big(\sum h_k(r-b)/\mathcal{N}\big)$. Choose the horizon first, the benchmark and normalizer second, the transform third and the kernel shape last.
4. **The decision tree depends on two questions:** how many comparable assets are available, and whether market or factor neutrality is needed. The answers determine the benchmark, which is the choice that matters most.
5. **Build in stages.** Half of the stages are not about momentum: infrastructure → characterization → the simplest thing → known refinements → honest validation → careful deployment.

6. **Diversify, scale by volatility, and count trials.** If you do only three things, do those.

Appendix A: Concepts from first principles

The main text uses many concepts in passing and assumes that a quantitative practitioner has met them before. This appendix does not make that assumption. Each entry has two parts. The first part gives **the idea** in plain language from first principles. The second part gives the **formal** version, with the definition to implement or cite. The entries are grouped by subject, not by order of appearance. Each entry says where the main text relies on it.

Following the *argument* of the chapter does not require any of this. *Implementing* the chapter requires most of it.

A.1 Time series and stochastic processes

Stochastic process. *Idea:* a model of the whole population of paths the market could have produced, not of the one observed price path. Every claim such as "returns are unpredictable" is a claim about that population. The data contain only one draw from it, which is the fundamental difficulty of the entire field. *Formally:* a family $\{X_t\}_{t\in\mathbb{Z}}$ of random variables on a common probability space. The observed data $(x_1, \dots, x_T)$ are a single realization.

Stationarity. *Idea:* the statistical behavior of the process does not depend on *when* it is observed. Learning from history at all requires some version of this property. Without it, an average over the past estimates nothing in particular, because the quantity being averaged changes during the averaging. *Formally:* **strict** stationarity requires the joint distribution of $(X_t, \dots, X_{t+k})$ to be invariant to time shifts. **Weak (covariance)** stationarity, the version actually used, requires only

$$\mathbb{E}[X_t] = \mu, \qquad \mathrm{Var}(X_t) = \sigma^2 < \infty, \qquad \mathrm{Cov}(X_t, X_{t-k}) = \gamma_k$$

with all three independent of t. Prices are not stationary, since their variance grows without bound. Returns are approximately stationary over moderate spans. Volatility regimes break even that approximation. *Used in:* §4.8, §6.9, §7.2.2.

Ergodicity. *Idea:* one long path eventually samples the whole population of paths, so a time average converges to the population average. Stationarity means the population is stable. Ergodicity means a single history is enough to learn it. A market with a permanent structural break is non-ergodic: no amount of data from before the break carries information about the period after it. *Formally:* $\frac{1}{T}\sum_{t=1}^{T} X_t \to \mathbb{E}[X_t]$ almost surely as $T \to \infty$.

Autocovariance and autocorrelation. *Idea:* the extent to which a series correlates with its own value k steps back. Momentum depends on this quantity, and it is the object that all of §4 estimates. *Formally:* $\gamma_k = \mathrm{Cov}(r_t, r_{t-k})$ and $\rho_k = \gamma_k/\gamma_0 \in [-1, 1]$. The sequence $\{\rho_k\}$ is the autocorrelation function (ACF). Under independence, $\rho_k = 0$ for $k \neq 0$, and the sample estimate has a standard error of $\approx 1/\sqrt{T}$ (§4.5.4).

White noise, martingale difference, and iid — the hierarchy. *Idea:* three progressively stronger versions of "unpredictable". Researchers frequently conflate them. Most momentum studies test only the weakest. Only the strongest rules out volatility clustering. *Formally:*

- **White noise:** $\mathbb{E}[r_t] = 0$ and $\mathrm{Cov}(r_t, r_{t-k}) = 0$ for $k \neq 0$. This rules out *linear* predictability only.
- **Martingale difference sequence (MDS):** $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$. This rules out predictability of the *mean* by any function of the past, linear or not.
- **iid:** the returns are independent and identically distributed. This rules out predictability of *anything*, including volatility.

Financial returns are close to white noise, arguably close to an MDS, and clearly not iid, because volatility is predictable (§1.4). Because of this hierarchy, "returns are unforecastable" and "returns are random" are different claims.

Random walk. The random walk is the null hypothesis of the entire field, and every measure in §4 is defined against it. It matters more than any other single concept here, and it generates far more apparent structure than people expect. So it has its own section: see **A.2** below.

Moving-average process and the Wold representation. *Idea:* any well-behaved stationary series can be written as a weighted sum of its own past shocks. The weights determine the process, because they encode all of its linear memory. The incomplete-adjustment model of §1.2 has exactly this form, and its weights give the speed at which information is impounded.

Formally: an MA(q) is $r_t = \sum_{j=0}^{q} \psi_j \varepsilon_{t-j}$. **Wold's theorem** says that any covariance-stationary process with no deterministic component has an MA(∞) representation $r_t = \mu + \sum_{j\geq 0} \psi_j \varepsilon_{t-j}$ with $\sum \psi_j^2 < \infty$. Its autocorrelations are

$$\rho_k = \frac{\sum_{j\geq 0} \psi_j \psi_{j+k}}{\sum_{j\geq 0} \psi_j^2}$$

Used in: §1.2, which is where ρ_k comes from, and §9.1.

Autoregressive process. *Idea:* the complementary description. Today's value is a fraction of yesterday's value plus new noise. One parameter generates geometrically decaying memory. *Formally:* AR(1) is $r_t = \phi r_{t-1} + u_t$ with $|\phi| < 1$, giving $\rho_k = \phi^k$. Momentum is $\phi > 0$, and short-horizon reversal is $\phi < 0$. Injecting a known ϕ into synthetic data is the standard way to check that a research pipeline detects an effect it should detect (§7.7.3).

Impulse response function. *Idea:* suppose one unit of news arrives today and nothing else ever happens. The price path that follows is the impulse response. Its shape carries the whole content of the momentum explanation. A gradual rise means under-reaction. A rise followed by a fall means over-reaction and eventual reversal. *Formally:* the cumulative response $\Psi_j = \sum_{i\leq j} \psi_i$ is the fraction of a shock impounded within j periods, and ψ_j is its per-period increment. $\Psi_j \uparrow 1$ monotonically is pure under-reaction. Ψ_j overshooting 1 and coming back is over-reaction. *Used in:* §1.2; §2.6, where the propagator model is an estimated impulse response; §9.1.

Variance ratio, derived. *Idea:* compare how far the price actually moves over q periods with how far a random walk with the same one-period variance would move. A ratio greater than one means the steps reinforce each other (trend). A ratio less than one means they cancel (reversion). *Formally:* start from the variance of a sum,

$$\text{Var}\Big(\sum_{i=0}^{q-1} r_{t-i}\Big) = q\gamma_0 + 2\sum_{k=1}^{q-1}(q-k)\gamma_k$$

and divide by $q\gamma_0$. The definition of §1.2 follows:

$$\text{VR}(q) = 1 + 2\sum_{k=1}^{q-1}\left(1 - \frac{k}{q}\right)\rho_k$$

The triangular weights $(1 - k/q)$ arise because a window of q contains $q - k$ pairs of observations separated by k lags. VR is therefore a *cumulative* statistic. It aggregates all autocorrelations up to lag q. This gives it power against many small correlations, which is its strength. It cannot say which lag is responsible, which is its weakness. *Used in:* everywhere.

Long memory. *Idea:* dependence that decays so slowly that the sum of all the autocorrelations diverges. Short-memory processes decay geometrically. Long-memory processes decay as a power law, so events from long ago still matter. Order flow has long memory (§2.6), and returns do not. *Formally:* $\rho_k \sim c\,k^{-\alpha}$ with $0 < \alpha < 1$, so $\sum_k \rho_k = \infty$. Equivalently, the spectral density diverges at zero frequency.

Fractional Brownian motion and the Hurst exponent. *Idea:* a one-parameter family of processes that interpolates between "reverting", "random walk" and "trending". The parameter sets how the range of the series grows with the length of the window. It is a clean idealization but a poor description of real returns, so §4.5.2 is skeptical of it. *Formally:* fBm $B_H(t)$ has $\text{Var}(B_H(t+q) - B_H(t)) = \sigma^2 q^{2H}$, with $H \in (0,1)$. $H = \frac{1}{2}$ is a standard random walk, $H > \frac{1}{2}$ is persistent, and $H < \frac{1}{2}$ is anti-persistent. Comparing with the definition of VR gives $\text{VR}(q) \propto q^{2H-1}$. So H **is the log-log slope of the variance-ratio profile, constrained to be a straight line.** The real profile is never a straight line, since real markets revert at short horizons and trend at intermediate ones. A single H is then an average over incompatible regimes.

Jensen's inequality and the arithmetic/geometric gap. *Idea:* compounding lowers growth when volatility is high. Two assets with the same average return but different volatilities end at different wealth levels, and the more volatile one ends lower. Ranking assets on cumulative simple return therefore rewards volatility. *Formally:* for a concave function, $\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$. With $f = \ln$ and a second-order expansion,

$$\underbrace{\mathbb{E}[\ln(1+R)]}_{\text{geometric, } \approx g} \approx \underbrace{\mathbb{E}[R]}_{\text{arithmetic, } \mu} - \frac{\sigma^2}{2}$$

Used in: §4.1.2.

A.2 The random walk in markets

The random walk is the **null hypothesis against which every claim in this chapter is defined**. Momentum is a departure from it, mean reversion is the opposite departure, and the variance ratio measures the size of the departure. Understanding the random walk therefore matters for two reasons. First, it makes a claim about markets. Second, *a random walk creates a large amount of apparent structure on its own*, and most false discoveries in this field mistake that structure for signal.

The model is older than everything in §2. Louis **Bachelier** (1900) modeled Paris bond prices as what is now called Brownian motion. Einstein used the same mathematics for suspended particles five years later. The finance profession rediscovered it 60 years after Bachelier.

The random walk, in three strengths. *Idea:* "random walk" names three different claims of increasing severity, and conflating them is the most common confusion in this literature. Financial returns satisfy the weakest version, arguably satisfy the middle one, and clearly violate the strongest, because volatility is predictable even when direction is not. A test that rejects the strongest version may be detecting nothing but volatility clustering. *Formally:* write $p_t = \mu + p_{t-1} + \varepsilon_t$. The standard taxonomy (Campbell, Lo & MacKinlay, 1997) appears in the table.

	Assumption on $\{\varepsilon_t\}$	Rules out	Consistent with markets?
RW1	iid	Any dependence at all, including in volatility	**No** — volatility clusters
RW2	Independent, not identically distributed	Any dependence, allows changing variance	Closer, but still strong
RW3	Uncorrelated ($\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$)	Linear predictability only	**Approximately yes**

Momentum is the claim that even RW3 fails. This is why §4.5.1 requires the **heteroskedasticity-robust** variance-ratio statistic. The naive version tests RW1, so it rejects on volatility clustering alone, which does not indicate predictability. The taxonomy also maps onto the hierarchy in A.1: RW3 has white-noise increments, RW2 is close to an MDS, and RW1 is full independence. *Used in:* §1.4, §4.5.1.

Why the random walk is the right null: Samuelson's argument. *Idea:* unpredictability is a *consequence* of forecasting done well. It is not an assumption about investor psychology. If everyone's best estimate of tomorrow's price is already today's price, then whatever moves the price tomorrow must be something nobody could forecast. Otherwise it would already be in today's price. Competent anticipation makes price changes look random. *Formally:* if $p_t = \mathbb{E}[p_{t+1} \mid \mathcal{F}_t]$, then $\varepsilon_{t+1} = p_{t+1} - p_t$ satisfies $\mathbb{E}[\varepsilon_{t+1} \mid \mathcal{F}_t] = 0$. The increment is a martingale difference sequence by construction (Samuelson, 1965). The general no-arbitrage version is the **First Fundamental Theorem of Asset Pricing**: absence of arbitrage is equivalent to the existence of a measure $\mathbb{Q}$ under which discounted prices are martingales.

A random walk is neither necessary nor sufficient for efficiency. *Idea:* much popular discussion errs on this point. Efficiency requires prices to be a martingale *after* adjusting for the equilibrium expected return. If that expected return varies over time, as risk premia demonstrably do, then prices are predictable, and the market may still be perfectly efficient. Conversely, a price could pass every random-walk test and still be wildly mispriced relative to fundamentals. *Formally:* efficiency asserts $\mathbb{E}[r_{t+1} \mid \mathcal{F}_t] = \mu_t$, where μ_t is the required return implied by a model. A random walk asserts that $\mu_t = \mu$ is constant. The gap between these two statements is the joint hypothesis problem (§1.3). For this reason, rejecting the random walk does not by itself demonstrate inefficiency. *Used in:* §1.3, §2.2.

Moments and covariance structure. *Idea:* this is the arithmetic behind every scaling rule in the chapter. The level of a random walk has a variance that grows without bound. So prices are not stationary, and the analysis must work with returns. The *increments* are well behaved. The correlation between the price now and the price later decays only as a square root. A random walk is therefore extremely persistent in levels and entirely unpredictable in changes. *Formally:* with $p_t = p_0 + \mu t + \sum_{i \le t} \varepsilon_i$ and $\mathrm{Var}(\varepsilon) = \sigma^2$,

$$\mathbb{E}[p_t] = p_0 + \mu t, \qquad \mathrm{Var}(p_t) = \sigma^2 t, \qquad \mathrm{Cov}(p_s, p_t) = \sigma^2 \min(s,t), \qquad \mathrm{Corr}(p_s, p_t) = \sqrt{s/t}$$

for $s < t$. Increments over non-overlapping intervals are uncorrelated, and $\mathrm{Var}(p_{t+q} - p_t) = q\sigma^2$. The variance ratio tests for this linear growth of variance with the horizon.

The two scaling laws, and the most consequential piece of arithmetic here. *Idea:* over a span of T periods, the *drift* accumulates in proportion to T, while the *noise* accumulates only in proportion to $\sqrt{T}$. So signal grows faster than noise, but only slowly. The difficulty of this field follows from that one fact. It sets the annualization conventions. It sets how long a strategy must run before anyone knows whether it works. It also explains why more frequent sampling does not help. *Formally:* a signal of μT against noise of $\sigma\sqrt{T}$ gives a t-statistic

$$t = \frac{\mu T}{\sigma\sqrt{T}} = \frac{\mu}{\sigma}\sqrt{T} = \text{SR} \cdot \sqrt{T}$$

with SR the annualized Sharpe ratio and T in **years**. Three direct consequences follow:

- **Annualization.** Mean returns scale by A, volatilities by $\sqrt{A}$, and Sharpe ratios by $\sqrt{A}$ (§7.4.1).
- **Detection time.** Reaching $t = 2$ requires $T = (2/\text{SR})^2$ years: **4 years at a Sharpe ratio of 1.0, 16 years at 0.5, and 64 years at 0.25.** §8.5 puts a realistic momentum program at a Sharpe ratio of 0.4–0.8. So *a full career is barely enough to establish that such a program works*, and under the Harvey–Liu–Zhu threshold of $t > 3$ it is not enough. This is the main reason the field relies so heavily on breadth, on centuries of data, and on the Fundamental Law.
- **Frequency does not help.** Both μT and $\sigma\sqrt{T}$ depend on the calendar span, not on how finely it is sliced. Sampling 10 times more often within the same span leaves $\text{SE}(\hat{\mu}) = \sigma/\sqrt{T}$ unchanged, while cutting $\text{SE}(\hat{\sigma}) \approx \sigma/\sqrt{2n}$ by a factor of three. **Merton (1980)** derived this result: high-frequency data are valuable for the denominator of a signal and useless for the numerator. *Used in:* §6.2, §7.4.1.

Continuous-time limit: Brownian motion. *Idea:* a random walk with tiny steps becomes a continuous process, and the shape of that process does not depend on the distribution of the individual steps. This universality is why the Gaussian appears throughout finance, although nobody believes returns are Gaussian. The universality concerns *sums*, not individual returns. It also shows when to distrust the Gaussian. The convergence needs finite variance and weak dependence. Fat-tailed, dependent financial returns approach the limit slowly, especially in the tails that matter most. *Formally:* **Donsker's theorem**, the functional CLT, says that for iid ε_i with mean 0 and variance σ^2,

$$\frac{1}{\sigma\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} \varepsilon_i \implies W_t$$

a standard Wiener process. It has $W_0 = 0$, independent increments, $W_t - W_s \sim N(0, t-s)$, and continuous paths that are nowhere differentiable. The non-differentiability is the formal statement that a price has no "velocity". It underlies the rejection of the physics metaphor in §1.1.

Geometric Brownian motion, and why the analysis works in logs. *Idea:* a plain random walk in price levels allows negative prices. It also makes a $1 move equally significant for a $10 stock and a $1,000 stock. Making the *log* price the random walk fixes both problems. Prices stay positive, and returns add across time. This also exposes **variance drag**: an asset with a positive expected return still has a lower expected *growth rate*, by exactly half its variance. *Formally:* $dP_t/P_t = \mu\,dt + \sigma\,dW_t$ makes P_t lognormal, and Itô's lemma gives the log dynamics

$$d\ln P_t = \left(\mu - \tfrac{\sigma^2}{2}\right) dt + \sigma\,dW_t$$

So the median outcome grows at $\mu - \sigma^2/2$, while the mean grows at μ. This is the continuous-time form of the Jensen gap in A.1. It is the mechanism behind the mechanical tilt of cross-sectional momentum toward volatile names (§4.1.2). *Used in:* §4.1.2.

A random walk looks as if it trends, and this causes costly errors. *Idea:* this is the most important practical point in the section. Human pattern recognition badly overestimates how much structure a random walk *should* show. So chart-based trend identification produces confident findings on data with no predictability whatsoever. Three results contradict the naive intuition, and each is worth knowing well:

- **The arcsine law.** The fraction of time a driftless random walk spends above its starting point has the limiting density $f(x) = \frac{1}{\pi\sqrt{x(1-x)}}$ on $(0, 1)$. The density is **U-shaped**, so the fraction does *not* concentrate near one half. The most likely outcomes are spending almost *all* the time above the starting point, or almost all of it below. A random walk typically looks as if it has a persistent bias. The same arcsine distribution governs the timing of the maximum and the time of the last visit to the origin.
- **Sign changes are rare.** The expected number of sign changes in n steps is asymptotically $\sqrt{2n/\pi}$: about **13 in 250 steps, not 125**. Long one-sided excursions are the norm. So "this market has been in an uptrend for months" is the *expected* appearance of a market with no trend at all.
- **Drawdowns are large.** For driftless Brownian motion, the expected maximum drawdown over $[0, T]$ is $\mathbb{E}[\text{MDD}] = \sqrt{\pi/2}\,\sigma\sqrt{T} \approx 1.25\,\sigma\sqrt{T}$, and it keeps growing with the observation window. Compare any observed drawdown with this benchmark before calling it evidence of anything (§7.4.3).

The operational conclusion is to **compute what a random walk would have looked like before concluding that a market trends.** The synthetic-data check of §7.7.3 does this. For the same reason, the variance-ratio profile with confidence bands is more reliable than looking at a chart.

Barriers, first passage and the reflection principle. *Idea:* the mathematics of stops and targets. Consider exiting a position at either a profit target or a stop loss. Even with *zero* edge, the placement of the two barriers alone determines the hit rate. A tight target with a wide stop wins most of the time and makes no money. This result is the formal basis for the statement in §7.3.4 that the hit rate is nearly irrelevant. Interpret triple-barrier labeling (A.10) against it. *Formally:* the **reflection principle** gives the distribution of the running maximum,

$$\Pr\left[\max_{s\le t} W_s \ge a\right] = 2\Pr[W_t \ge a]$$

Take a driftless walk started at 0, with an upper barrier $+a$ and a lower barrier $-b$. This is the gambler's-ruin problem, and

$$\Pr[\text{hit } +a \text{ before } -b] = \frac{b}{a+b}, \qquad \mathbb{E}[\text{time to hit either}] = \frac{ab}{\sigma^2}$$

The expected P&L is $\frac{b}{a+b}\cdot a - \frac{a}{a+b}\cdot b = 0$, however the barriers are set. The win rate and the win size trade off exactly, as they must. *Used in:* §7.3.4, §4.9.

Spurious regression between random walks. *Idea:* regressing one random walk on another, completely independent one usually produces a "significant" coefficient and a high R^2. The problem gets *worse* with more data, not better. The regression residuals are themselves a random walk, so the standard errors rest on an assumption that is badly wrong. Treat any regression on price *levels* as spurious until shown otherwise. Examples are pairs relationships, "leading indicators", and macro variables regressed against prices. *Formally:* Granger & Newbold (1974) found $|t| > 2$ in roughly three quarters of regressions between independent random walks at $T = 100$. Phillips (1986) showed that the t-statistic diverges at rate $\sqrt{T}$, while R^2 converges to a non-degenerate random variable rather than to zero. The fixes are to regress differences rather than levels, or to establish cointegration first. This is also why the regression of price on time in §4.2.1 needs care. Fitting a trend line to an I(1) series reports a highly significant slope on data with no trend at all.

Unit roots, and why a test cannot settle this. *Idea:* finite samples cannot decide whether a series is a random walk or a slowly mean-reverting series. A process that reverts with a coefficient of 0.99 per day is economically large: its half-life is about 69 days, so shocks fade out fully within a year. Yet over decades the coefficient is statistically almost indistinguishable from 1.00. The low power is intrinsic. It is not a defect of any particular test. *Formally:* the model $p_t = \phi p_{t-1} + \varepsilon_t$ has a **unit root** at $\phi = 1$, where it is non-stationary and shocks are permanent. It is stationary for $|\phi| < 1$, where shocks decay with half-life $\ln 2/\ln(1/\phi)$. The augmented Dickey–Fuller test tests $H_0 : \phi = 1$, and it has notoriously low power against local alternatives $\phi = 1 - c/T$. In practice, do not expect a hypothesis test to settle whether momentum exists. Estimate the variance-ratio profile and the IC term structure, and look at *magnitudes with confidence bands*.

What actually breaks the random walk in real markets, and why most of it cannot be traded. *Idea:* the first autocorrelation a naive study finds is almost always microstructure, not momentum. Two mechanisms dominate. They have opposite signs, and neither can be exploited after costs. So assume that any autocorrelation found at short horizons is an artifact until shown otherwise. *Formally:*

- **Bid-ask bounce** induces *negative* first-order autocorrelation in the returns of transaction prices. In the model of Roll (1984), with effective spread s and no information flow, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -s^2/4$. The spread can therefore be recovered as $s = 2\sqrt{-\text{Cov}}$. Prices oscillate between bid and ask with no change in value. This is the microstructure part of the reason the classic momentum signal skips the most recent month (§1.5).
- **Stale and non-synchronous prices** induce *positive* autocorrelation, and spurious *cross*-autocorrelation that looks like a lead-lag effect. Suppose a large stock trades at 16:00:00 and a small stock last traded at 15:47. The index return then attributes part of today's news to the small stock tomorrow. Lo & MacKinlay (1990b) show that this produces both index-level autocorrelation and an apparent pattern of large caps leading small caps, with no real lead-lag effect. This affects term (B) of the profit decomposition in §4.6.2 directly. It is also why illiquid or smoothed assets show inflated Sharpe ratios (§4.3.2). *Used in:* §1.5, §4.3.2, §4.6.2, §6.5.

A.3 Estimation, inference, and the multiple-testing apparatus

Ordinary least squares and leverage. *Idea:* fit a line by minimizing squared vertical distances. The fitted slope is a *weighted average of the data*, with unequal weights. Points far from the centre of the x-range have the largest weights. So a bad print at the edge of a regression window does more damage than one in the middle. *Formally:* $\hat{\beta} = (X'X)^{-1}X'y$. The hat matrix $\mathbf{H} = X(X'X)^{-1}X'$ has diagonal entries h_{ii}, the **leverage** of observation i. The leverages sum to the number of parameters. For a regression on a time trend, h_{ii} is largest at the two ends of the window. *Used in:* §4.2.1.

Heteroskedasticity and autocorrelation. *Idea:* the *estimates* of OLS survive most violations of its assumptions, but its *standard errors* do not. If residuals cluster in volatility or are serially correlated, the usual formula counts each observation as fresh information when it is not. The resulting t-statistics are too large, often by a factor of two or three. *Formally:* Homoskedasticity means $\text{Var}(\varepsilon_t \mid X) = \sigma^2$ is constant. No autocorrelation means $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$. Financial residuals violate both.

HAC (Newey–West) standard errors. *Idea:* rebuild the standard error so that correlated observations count only once. The method adds the estimated covariances between nearby residuals back into the variance. The weights taper to zero after a chosen number of lags. *Formally:* with $g_t = x_t\hat{\varepsilon}_t$,

$$\hat{S} = \hat{\Gamma}_0 + \sum_{k=1}^{K}\left(1 - \frac{k}{K+1}\right)\left(\hat{\Gamma}_k + \hat{\Gamma}_k'\right), \qquad \hat{\Gamma}_k = \frac{1}{T}\sum_t g_t g_{t-k}'$$

and $\text{Var}(\hat{\beta}) = (X'X)^{-1}\hat{S}(X'X)^{-1}$. The Bartlett weights $(1 - k/(K+1))$ guarantee a positive semi-definite result. This is the contribution of Newey & West (1987). Choose $K \geq h - 1$ for data whose observations overlap over h periods. *Used in:* §6.3, §7.5.

Overlapping observations and effective sample size. *Idea:* a 12-month return computed every month shares 11 months of data with its neighbor. The dataset then holds far fewer independent observations than rows. So every naive significance test is overconfident. *Formally:* with h-period overlap, the effective sample size is roughly T/h, and the overlap inflates a naive t-statistic by up to $\sqrt{h}$. Hansen & Hodrick (1980) give the correction for overlapping forecast horizons. *Used in:* §6.3.

Statistical power. *Idea:* power is the probability that a test finds an effect that is genuinely there. Momentum research has low power, because real effects are small relative to noise. This produces two symmetric errors. One is believing that a null result means "no effect". The other is believing that a significant result means "real effect" after a wide search. *Formally:* power $= 1 - \Pr[\text{Type II error}]$. Power increases with the effect size, the sample size and the significance threshold. The random-walk tests of the 1950s (§2.2) had low power against exactly the alternatives that later proved real.

Bootstrap. *Idea:* instead of deriving the sampling distribution of a statistic mathematically, the bootstrap creates it by resampling the available data. Financial data require resampling *blocks* rather than individual observations. Shuffling single returns destroys the serial dependence that the strategy trades. *Formally:*

- **iid bootstrap:** draw T observations with replacement. It is invalid for time series.
- **Block bootstrap:** draw contiguous blocks of length b. Choose b larger than the memory of the signal.
- **Stationary bootstrap** (Politis & Romano, 1994): block lengths are geometric with mean $1/p$. This makes the resampled series stationary and removes the sharp sensitivity to b.
- **Circular block bootstrap:** wrap the series from end to start, so the resample draws every observation equally often.

Used in: §7.7.1.

Permutation (randomization) test. *Idea:* destroy exactly the relationship claimed, leave everything else intact, and count how often chance reproduces the result. This test makes the fewest assumptions of any available test, and researchers underuse it. *Formally:* under the null of no relation between signal and return, the labels are exchangeable. Compute the statistic on many random relabelings to obtain its null distribution. The p-value is the fraction of permuted statistics that exceed the observed one. *Used in:* §7.7.2.

Family-wise error rate vs. false discovery rate. *Idea:* a researcher can control two different quantities when running many tests. FWER controls the chance of *even one* false positive. It suits cases where a single false discovery is costly. FDR controls the *expected proportion* of discoveries that are false. It suits screening a library of candidate signals, where a few false hits among many true ones are tolerable. *Formally:* with M tests, FWER $= \Pr[\text{at least one false rejection}]$. Bonferroni controls it by testing each hypothesis at q/M, which is very conservative when the tests are correlated. FDR $= \mathbb{E}[V/R]$, false rejections over total rejections. The **Benjamini–Hochberg (1995)** procedure controls it. Sort the p-values in ascending order, and reject the largest k satisfying $p_{(k)} \le \frac{k}{M}q$. *Used in:* §7.6.3.

Expected maximum of M draws. *Idea:* the best of many pure-noise strategies looks good, although none of it is real. The maximum of random variables grows predictably with the number of draws. The growth is logarithmic, which is slow, so intuition underweights it. A thousand trials are not 10 times worse than a hundred, only about 15% worse. *Formally:* for M iid standard normals, $\mathbb{E}[\max_m Z_m] \approx \sqrt{2\ln M}$. So the expected best Sharpe ratio among M skill-free backtests is $\approx \mathrm{SE}(\mathrm{SR})\sqrt{2\ln M}$. More precisely, the maximum follows a **Gumbel** distribution. The refined approximation that the Deflated Sharpe Ratio uses is

$$\mathbb{E}[\max_m \mathrm{SR}_m] \approx \sqrt{\mathrm{Var}(\mathrm{SR}_m)}\left[(1-\gamma)Z^{-1}\left(1-\tfrac{1}{M}\right)+\gamma Z^{-1}\left(1-\tfrac{1}{Me}\right)\right]$$

The formula uses the Euler–Mascheroni constant $\gamma \approx 0.5772$, the mean of the standard Gumbel distribution. *Used in:* §7.6.

White's Reality Check and Hansen's SPA. *Idea:* a formal test of whether the best of M strategies beats the benchmark, given that the researcher picked it *because* it was best. The test resamples all M strategies *together*. It therefore accounts for the dependence among similar rules. A thousand similar momentum rules are nowhere near a thousand independent tests. *Formally:* let $f_{m,t}$ be the performance of model m over the benchmark. Test $H_0 : \max_m \mathbb{E}[f_m] \le 0$ using $V = \max_m \sqrt{T}\,\bar{f}_m$, against a bootstrap distribution of $V^* = \max_m \sqrt{T}(\bar{f}^*_m - \bar{f}_m)$. **The SPA test of Hansen (2005)** studentizes each $\bar{f}_m$ and down-weights models that are clearly inferior. That removes the Reality Check's conservatism when the candidate set contains many bad models. **Romano & Wolf (2005)** extend this to identify *which* models survive. *Used in:* §7.6.1.

Probabilistic and Deflated Sharpe Ratio. *Idea:* convert an observed Sharpe ratio into a probability that the true Sharpe ratio beats a stated benchmark. The conversion corrects for non-normal returns and for the selection of this strategy from many. The deflated version sets the benchmark to the level that selection alone would have produced. *Formally:* the Probabilistic Sharpe Ratio against a threshold SR_0 is

$$\mathrm{PSR}(\mathrm{SR}_0) = Z\left(\frac{(\widehat{\mathrm{SR}}-\mathrm{SR}_0)\sqrt{T-1}}{\sqrt{1-\hat{\gamma}_3\widehat{\mathrm{SR}}+\frac{\hat{\gamma}_4-1}{4}\widehat{\mathrm{SR}}^2}}\right)$$

with $\hat{\gamma}_3$ the skewness and $\hat{\gamma}_4$ the kurtosis. The **DSR** is the PSR evaluated at $\mathrm{SR}_0 = \mathbb{E}[\max_m \mathrm{SR}_m]$, from the entry above. Negative skew and fat tails reduce the PSR. This is correct, because they make the realized Sharpe ratio a less reliable estimate. *Used in:* §7.6.2.

Probability of backtest overfitting (PBO). *Idea:* instead of asking whether a particular strategy is overfit, ask whether the *selection procedure* overfits. Split the data many ways. In each in-sample half, choose the best configuration and record its out-of-sample rank. If the winners land below the median about half the time, the procedure is no better than picking at random. *Formally:* the method is combinatorially symmetric cross-validation (Bailey, Borwein, López de Prado & Zhu, 2017). Partition the data into S blocks, form all $\binom{S}{S/2}$ train/test splits, and estimate PBO $= \Pr[\text{IS-best configuration ranks below median OOS}]$. A value above about 0.5 is disqualifying. *Used in:* §7.6.2.

Cross-validation, purging and embargo. *Idea:* ordinary k-fold cross-validation is invalid on financial data for two reasons. It trains on the future to predict the past. Also, adjacent observations share overlapping label windows, so nearly identical rows end up on both sides of the split. Purging deletes training rows whose outcome window touches the test period. The embargo deletes a further buffer of rows, to handle residual serial correlation. *Formally:* for a label spanning $[t, t+h]$, purge

any training observation whose label window intersects the test span. Then embargo a further δ observations after the test block. **CPCV** forms all $\binom{K}{k}$ choices of k test groups from K, which yields a *distribution* of backtest paths rather than one number. *Used in:* §7.2.4.

Clustered standard errors and Fama–MacBeth. *Idea:* 500 stocks on the same day are not 500 independent observations, because they mostly move together. Treating them as independent inflates t-statistics enormously. Two standard fixes exist. One clusters the standard errors by date. The other runs one cross-sectional regression per period and draws inferences from the resulting time series of coefficients. *Formally:* Fama–MacBeth estimates $\hat{\beta}_t$ cross-sectionally at each t. It then reports $\bar{\beta} = \frac{1}{T}\sum_t \hat{\beta}_t$, with SE $= \text{sd}(\hat{\beta}_t)/\sqrt{T}$. Apply a HAC correction to the standard error if the $\hat{\beta}_t$ are autocorrelated, as they are with overlapping momentum windows. *Used in:* §7.5.

Robust statistics: MAD and winsorization. *Idea:* one bad print should not be able to set the signal. The mean and the standard deviation have a breakdown point of zero. A single arbitrary observation can move them arbitrarily far. The median and the median absolute deviation tolerate bad values in up to half the sample. Winsorizing, which clips values to a percentile, keeps the observation and its direction while bounding its influence. Deleting the observation silently biases the sample. *Formally:* MAD $=$ median$_i\,|x_i - \text{median}(x)|$, and $1.4826 \cdot$ MAD is a consistent estimator of σ for Gaussian data. The constant is $1/Z^{-1}(0.75)$. The robust z-score is $(x - \text{median})/(1.4826\,\text{MAD})$. *Used in:* §4.3.3, §6.6.

Orthant probability (the arcsine identity). *Idea:* the exact translation between the correlation of a signal with returns and the frequency of getting the direction right. This explains why a hit rate of 51.6% indicates a good signal and a hit rate of 70% indicates a bug. *Formally:* for a bivariate normal pair with correlation ρ,

$$\Pr[\text{sign}(X) = \text{sign}(Y)] = \frac{1}{2} + \frac{\arcsin\rho}{\pi}$$

so the hit rate is $\approx \frac{1}{2} + \rho/\pi$ for small ρ. *Used in:* §7.3.4.

Welford's algorithm and catastrophic cancellation. *Idea:* the textbook variance formula $\mathbb{E}[X^2] - \mathbb{E}[X]^2$ subtracts two large, nearly equal numbers. In floating point, that subtraction destroys precision. This is a real bug in long-running streaming code, where it can produce negative variances. Welford's update keeps a running mean and a running sum of squared deviations from it, and avoids the subtraction. *Formally:* with $\bar{x}_n$ the running mean and $M_{2,n} = \sum_{i\le n}(x_i - \bar{x}_n)^2$,

$$\bar{x}_n = \bar{x}_{n-1} + \frac{x_n - \bar{x}_{n-1}}{n}, \qquad M_{2,n} = M_{2,n-1} + (x_n - \bar{x}_{n-1})(x_n - \bar{x}_n)$$

and $\hat{\sigma}^2 = M_{2,n}/(n-1)$. *Used in:* §4.2.1, §4.3.2.

Monotonic deque for rolling extremes. *Idea:* a naive rolling maximum over a window of n costs $O(n)$ per bar, but a queue reduces the cost to constant amortized time. Keep a queue of candidates in decreasing order, and discard any element that a newer, larger element makes irrelevant. The queue then pushes and pops each observation exactly once. This makes every channel, Donchian and stochastic indicator an $O(1)$ amortized computation. *Formally:* maintain a deque of indices whose values decrease monotonically. Pop from the back while the incoming value exceeds the back. Pop from the front when its index leaves the window. The front is always the window maximum. The total work is $O(T)$ over T bars. *Used in:* §4.4.2, §4.4.3.

A.4 Volatility and its estimation

Conditional variance and volatility clustering. *Idea:* volatility is a state that moves, not a constant to estimate once. Unlike returns, it moves *predictably*. Turbulent days follow turbulent days. This is the most reliable regularity in financial data. Every normalization in §4.3 uses this predictability to help with the much harder problem of direction. *Formally:* $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$. Empirically, Corr$(|r_t|, |r_{t-1}|) \approx 0.2$–$0.4$ and decays slowly, while Corr$(r_t, r_{t-1}) \approx 0$. *Used in:* §1.4, §4.3.

EWMA (RiskMetrics) volatility. *Idea:* estimate today's variance as a decaying average of past squared returns. Recent observations get the largest weights, and old ones lose weight gradually instead of dropping out at once. The estimator has one parameter, $O(1)$ state and no fitting. *Formally:* $\hat{\sigma}_t^2 = \lambda\hat{\sigma}_{t-1}^2 + (1-\lambda)r_t^2$, with $\lambda = 0.94$ the RiskMetrics daily convention, a half-life of about 11 days. It is the GARCH model below with $\omega = 0$ and $\alpha + \beta = 1$. It is therefore an *integrated* process with no long-run level to revert to. *Used in:* §4.3.1.

GARCH(1,1). *Idea:* the same decaying average, with a pull toward a long-run level. Variance reverts toward that level, so forecasts at long horizons converge to the unconditional variance. This matters when forecasting volatility over the life of a position rather than nowcasting it. *Formally:*

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2, \qquad \alpha + \beta < 1$$

with unconditional variance $\omega/(1-\alpha-\beta)$ and a shock half-life of $\ln 2 / \ln(1/(\alpha+\beta))$. Typical daily equity estimates have $\alpha + \beta \approx 0.97$–$0.99$, which is a half-life of weeks to months. *Used in:* §4.3.1.

Estimator efficiency and range-based estimators. *Idea:* a close-to-close return records where a price ended, not how far it traveled. The bar's high and low contain much more information about volatility than the close-to-close return does. So range-based estimators reach a given precision with a fraction of the observations. That matters because a short volatility window keeps the estimate responsive. *Formally:* efficiency is the ratio of the variances of two unbiased estimators. Relative to close-to-close:

- **Parkinson (1980)** is about 5× as efficient: $\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} (\ln \mathrm{Hi}_t / \mathrm{Lo}_t)^2$.
- **Garman–Klass (1980)** adds the open and close. It is roughly 7× as efficient, at the cost of assuming no drift and no jumps.
- **Yang–Zhang (2000)** combines an overnight component, an open-to-close component, and the Rogers–Satchell term, which does not depend on drift. That is why it handles opening gaps, and why it is often the best single choice for daily bars.

Used in: §4.3.1.

Realized variance and microstructure noise. *Idea:* with intraday data, summing squared intraday returns measures a day's volatility almost exactly. In theory, the estimate improves without limit as sampling gets faster. In practice, at very fine scales the sum measures bid-ask bounce rather than price, and the estimator diverges. Five-minute sampling is the standard compromise. *Formally:* $\mathrm{RV}_t = \sum_{i=1}^{n} r_{t,i}^2 \to \int_t^{t+1} \sigma_s^2 \, ds$ as $n \to \infty$ under a pure diffusion. With additive noise $\tilde{p} = p + u$, $\mathbb{E}[\mathrm{RV}] = \mathrm{IV} + 2n \, \mathrm{Var}(u)$, so the bias grows linearly with the sampling frequency. Noise-robust alternatives include two-scale estimators and realized kernels. *Used in:* §4.3.1, §6.2.

Volatility targeting. *Idea:* hold risk constant, not notional. If volatility doubles, halve the position. Volatility is persistent and forecastable, so this gives a much more stable risk profile than a fixed position size. It also reduces exposure in the states where momentum crashes. *Formally:* $w_t = w_t^{\text{signal}} \cdot \sigma^* / \hat{\sigma}_t$ at the asset level. At the portfolio level, $W_t = w_t \cdot \sigma_p^* / \hat{\sigma}_{p,t}$, using a forecast of portfolio volatility from the covariance matrix. It is a *feedback rule*. When many funds run it at once, volatility triggers correlated deleveraging. *Used in:* §6.7, §8.2.

A.5 State-space models and filtering

State-space model. *Idea:* separate the quantity of interest from the quantity observed. Posit a hidden state that follows simple dynamics, and observations equal to the state plus noise. Almost every smoother in technical analysis implicitly solves this problem. Writing the model down explicitly lets the analyst derive the right smoother instead of guessing it. *Formally:* a linear Gaussian state-space model is

$$x_t = F x_{t-1} + w_t, \quad w_t \sim N(0, Q); \qquad y_t = H x_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, R)$$

with x_t the unobserved state and y_t the observation. *Used in:* §4.7.1.

Kalman filter. *Idea:* the filter runs a two-step loop. First, *predict* the state using the dynamics. Then *update* that prediction with the new observation, weighting the two by their reliability. The weight is the gain. The gain rises when the state is volatile relative to the observation noise. So the filter automatically becomes more responsive when the signal is strong relative to the noise. Its output includes a variance, which measures the reliability of the estimate. *Formally:*

$$\hat{x}_{t|t-1} = F \hat{x}_{t-1|t-1}, \qquad P_{t|t-1} = F P_{t-1|t-1} F' + Q$$

$$K_t = P_{t|t-1} H' \left(H P_{t|t-1} H' + R \right)^{-1}$$

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t \left(y_t - H \hat{x}_{t|t-1} \right), \qquad P_{t|t} = (I - K_t H) P_{t|t-1}$$

The term $y_t - H\hat{x}_{t|t-1}$ is the **innovation**, the part of the new observation that the prediction could not anticipate. Winsorizing the innovation is the cheapest way to make the filter robust to jumps. *Used in:* §4.7.1.

Filtering, prediction and smoothing, and why the difference is a backtest bug. *Idea:* three different questions about the same state. Filtering asks what the state is *now*, given everything up to now. A trader can act only on the filtered estimate. Smoothing asks what the state was *at an earlier time*, given all data, including what happened afterwards. It produces clean charts. In a backtest, it causes severe look-ahead bias. *Formally:* the filtered distribution is $\Pr[x_t \mid \mathcal{F}_t]$, the predicted one is $\Pr[x_{t+k} \mid \mathcal{F}_t]$, and the smoothed one is $\Pr[x_t \mid \mathcal{F}_T]$ with $T > t$. **Only filtered quantities are admissible in a signal.** *Used in:* §4.7.2, §6.10.

Local level and local linear trend models. *Idea:* these are the two simplest useful state-space models. The local level model says that a true price follows a random walk beneath the noisy observed price. The local linear trend model adds a slope that also follows a random walk. It therefore estimates both the price level and the speed at which it moves. This is a momentum signal with an uncertainty estimate attached. *Formally:* the local level model is $\mu_t = \mu_{t-1} + \eta_t$, $y_t = \mu_t + \varepsilon_t$. The local linear trend adds $\mu_t = \mu_{t-1} + \beta_{t-1} + \eta_t$ and $\beta_t = \beta_{t-1} + \zeta_t$, that is, $F = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $H = \left(\begin{smallmatrix} 1 & 0 \end{smallmatrix}\right)$. *Used in:* §4.7.1.

Steady state and the EWMA equivalence. *Idea:* on a time-invariant model, the Kalman gain stops changing after enough observations. The uncertainty settles at a level where the noise that the dynamics add balances the information that observations supply. For the local level model, the resulting fixed-gain recursion *is* an exponentially weighted moving average. This gives a theoretical justification for a century of practitioner smoothing. It also replaces the arbitrary choice of decay with an estimable quantity, the signal-to-noise ratio. *Formally:* let $q = \sigma_\eta^2/\sigma_\varepsilon^2$ be the **signal-to-noise ratio**. The steady-state prior variance P^- is the positive root of $(P^-)^2 - \sigma_\eta^2 P^- - \sigma_\eta^2\sigma_\varepsilon^2 = 0$. The corresponding gain $K = P^-/(P^- + \sigma_\varepsilon^2)$ simplifies to

$$K = \frac{\sqrt{q^2 + 4q} - q}{2}$$

after which $\hat{\mu}_t = \hat{\mu}_{t-1} + K(y_t - \hat{\mu}_{t-1})$. This is an EWMA with $\alpha = K$. A larger q means a signal that moves fast relative to its noise. It gives a larger gain and a shorter effective lookback. *Used in:* §4.1.4, §4.7.1, §5.5.

Hidden Markov and Markov-switching models. *Idea:* posit a small number of unobserved regimes instead of one set of parameters. Each regime has its own mean and volatility. A matrix of probabilities governs switching between regimes. The model never observes the regime. It infers a probability distribution over regimes, and the trader acts on that distribution. *Formally:* the latent state is $S_t \in \{1..K\}$, with transition matrix $\Pi_{jk} = \Pr[S_t = k \mid S_{t-1} = j]$ and $r_t \mid S_t = k \sim N(\mu_k, \sigma_k^2)$. The **forward algorithm** produces filtered probabilities recursively:

$$\Pr[S_t = k \mid \mathcal{F}_t] \;\propto\; f(r_t \mid S_t = k) \sum_j \Pi_{jk} \Pr[S_{t-1} = j \mid \mathcal{F}_{t-1}]$$

EM (Baum–Welch) fits the parameters. It alternates between computing state probabilities given the parameters and re-estimating the parameters given the state probabilities. It converges only to a local optimum, and it is sensitive to initialization and to label switching. *Used in:* §4.7.2.

Bayesian online change-point detection. *Idea:* maintain a probability distribution over how long the current regime has been running. Each new observation either extends the current run or resets it to zero. The posterior over the run length shows how mature the regime is and how likely a break just occurred. It is the natural formalization of the trend life cycle of §1.6. *Formally:* let ρ_t be the run length and $H(\rho)$ the hazard rate, the prior probability of a change given the run so far. The recursion is

$$\Pr[\rho_t, \mathcal{F}_t] = \sum_{\rho_{t-1}} \Pr[\rho_{t-1}, \mathcal{F}_{t-1}]\; \pi(r_t \mid \rho_{t-1}) \begin{cases} H(\rho_{t-1}) & \rho_t = 0 \\ 1 - H(\rho_{t-1}) & \rho_t = \rho_{t-1} + 1 \end{cases}$$

The cost is $O(t)$ per step unless the algorithm prunes low-probability run lengths (Adams & MacKay, 2007). *Used in:* §4.7.3.

A.6 Signal processing

Linear filter and convolution. *Idea:* a filter produces an output series as a weighted sum of the input's recent past. Every moving average, crossover, EWMA and regression slope in §4 is a filter. They differ only in their weights. *Formally:* $y_t = \sum_{k\geq 0} h_k x_{t-k}$, with $\{h_k\}$ the **impulse response**, the output produced by a single unit spike at the input. A filter is **causal** if $h_k = 0$ for $k < 0$, so that it uses no future data. *Used in:* §4.1.5, §4.8.

Transfer function, gain and phase lag. *Idea:* given a pure oscillation as input, a filter returns the same oscillation, scaled by a factor and shifted in time. Repeating this at every frequency characterizes the filter completely. The scale factor is the gain, and the shift is the lag. This shows what a smoother *actually* does, which can differ from what its parameter name suggests. *Formally:* the transfer function is the discrete-time Fourier transform of the impulse response,

$$H(\omega) = \sum_k h_k e^{-i\omega k}, \qquad \text{gain} = |H(\omega)|, \qquad \text{phase lag} = -\frac{\arg H(\omega)}{\omega}$$

For an n-bar SMA, $H(\omega) = \frac{1}{n}\frac{\sin(n\omega/2)}{\sin(\omega/2)}e^{-i\omega(n-1)/2}$. The lag is a constant $(n-1)/2$ bars at every frequency. The magnitude is a **Dirichlet kernel** with **sidelobes**. Sidelobes are secondary peaks of alternating sign that let some high-frequency energy through with its sign inverted. This is the formal reason an SMA occasionally gives unexpected results. *Used in:* §4.8.

Low-pass, high-pass and band-pass. *Idea:* a moving average keeps slow movements and suppresses fast ones, so it is a low-pass filter. Subtracting a moving average from the price does the opposite and gives a high-pass filter. Subtracting a slow average from a fast one keeps an intermediate band and suppresses both extremes, so it is a band-pass filter. A crossover is exactly this. So a crossover has a *characteristic horizon* set by both spans, not just the fast one. *Formally:* the crossover kernel of §4.1.5 has $H(0) = 0$, because both averages contain a constant drift equally and the difference removes it. It has $|H(\omega)| \to 0$ as $\omega \to \pi$, with a peak in between.

Spectral density and the Wiener–Khinchin relation. *Idea:* spectral density carries the same information as the autocorrelation function. It shows how much of the series' variance lies at each speed of oscillation. Momentum is excess variance at slow speeds. This view helps because filters are simple in frequency space and complicated in time. *Formally:* $f_r(\omega) = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty}\gamma_k e^{-i\omega k}$, with the inverse relation $\gamma_k = \int_{-\pi}^{\pi} f_r(\omega)e^{i\omega k}d\omega$. White noise has a flat density, $f_r(\omega) = \sigma^2/2\pi$. At zero frequency, $2\pi f_r(0) = \sum_k \gamma_k = \gamma_0(1 + 2\sum_{k\geq 1}\rho_k)$. This gives the identity that closes §4.8:

$$\lim_{q\to\infty} \mathrm{VR}(q) = \frac{2\pi f_r(0)}{\sigma_r^2}$$

Filtering multiplies spectra, so $f_y(\omega) = |H(\omega)|^2 f_x(\omega)$. Reasoning about filters is therefore much easier in the frequency domain than in the time domain. *Used in:* §4.8, §5.5.

Two-sided filters and look-ahead. *Idea:* the smoothers with the cleanest output are symmetric in time. They use as much data after each point as before, so they have no lag. A trader cannot use them, because at time t they require data from after t. This is the most common way a look-ahead bug enters a research pipeline. The code raises no error, and the chart looks good. *Formally:* any h_k with support on $k < 0$ is non-causal. This class includes `filtfilt`-style forward-backward filtering, centred moving averages, the Hodrick–Prescott filter, and Kalman *smoothed* states. *Used in:* §4.8, §6.10.

Hodrick–Prescott and ℓ_1 trend filtering. *Idea:* extract a trend as the curve that stays close to the data while being as smooth as possible, with a parameter controlling the trade-off. Penalizing squared second differences (HP) gives a smoothly curving trend. Penalizing absolute second differences (ℓ_1) gives a piecewise-linear trend with sharp kinks, which is closer to how practitioners actually draw trendlines. Both are two-sided as usually implemented. *Formally:*

$$\text{HP:}\ \min_{\tau}\sum_t (p_t-\tau_t)^2 + \eta\sum_t\left[(\tau_{t+1}-\tau_t)-(\tau_t-\tau_{t-1})\right]^2$$

$$\ell_1:\ \min_{\tau}\sum_t (p_t-\tau_t)^2 + \eta\sum_t\left|(\tau_{t+1}-\tau_t)-(\tau_t-\tau_{t-1})\right|$$

Hamilton (2018) gives the definitive critique of the HP filter. It shows that the filter creates dynamics that come from the filter rather than from the data, and that its behavior at the endpoints is unreliable. For ℓ_1, see Kim, Koh, Boyd & Gorinevsky (2009). *Used in:* §4.8.

Wavelet multiresolution analysis. *Idea:* Fourier analysis shows which frequencies are present, but not *when* they occur. That is useless for a series whose behavior changes over time. Wavelets localize in both time and frequency. They decompose a series into components at successive scales, which is a natural decomposition of momentum across horizons. The limitation is at the right-hand edge. Near the most recent observation, the only point where a trader can act, the filter has too little data. *Formally:* successive convolution with scaling and wavelet filters yields detail coefficients at dyadic scales 2^j, which sum back to the original series. Use undecimated, causal variants with explicit boundary handling. *Used in:* §4.8, §4.9.

Slutsky–Yule effect. *Idea:* averaging and differencing *create* apparent cycles in data that have none. A moving average of pure noise oscillates, although the oscillation means nothing. Any claim to have discovered a "market cycle" must first rule out that the smoothing used to find it created the cycle. *Formally:* if x_t is white noise, the filtered series $y_t = \sum h_k x_{t-k}$ has spectral density $|H(\omega)|^2\sigma^2/2\pi$. The density peaks wherever $|H|$ peaks. So the *output* has a spectral peak with no corresponding structure in the input. *Used in:* §4.8.

A.7 Portfolio theory and performance measurement

Mean–variance optimization. *Idea:* an investor who cares about expected return, dislikes variance, and trades the two off linearly should hold a position in each asset. The position should equal the asset's expected return divided by its variance. All position sizing in this chapter is a special case of this rule. The formula also explains why the units of the signal matter so much. Expected return and expected Sharpe ratio imply different powers of volatility in the answer. *Formally:* maximizing $\mathbb{E}[w'r] - \frac{\gamma}{2}w'\Sigma w$ gives $w^* = \frac{1}{\gamma}\Sigma^{-1}\mathbb{E}[r]$. For a single asset, $w^* \propto \mu/\sigma^2 = \mathrm{SR}/\sigma$. *Used in:* §4.0, §6.7.

Sharpe ratio and its sampling distribution. *Idea:* excess return per unit of volatility. It is the standard scale-free measure of a strategy's quality, and its *uncertainty* is far larger than most people assume. It also makes two assumptions that are wrong for trend following. One is that returns are iid, so that the ratio can be annualized by $\sqrt{A}$. The other is that upside and downside deviations are equally undesirable. *Formally:* $\mathrm{SR} = (\mathbb{E}[R_p] - R_f)/\operatorname{sd}(R_p)$, annualized as $\mathrm{SR}\sqrt{A}$ under iid returns, with

$$\mathrm{SE}(\widehat{\mathrm{SR}}) \approx \sqrt{\frac{1 + \mathrm{SR}^2/2}{T}}$$

(Lo, 2002). With autocorrelated returns, the correct multi-period scaling replaces $\sqrt{q}$ by $q/\sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\rho_k}$. This replacement is *smaller* than $\sqrt{q}$ when $\rho_k > 0$. So smoothed or illiquid strategies overstate their Sharpe ratio under naive annualization. *Used in:* §7.4.1.

Skewness and kurtosis. *Idea:* the third and fourth moments describe the shape that the Sharpe ratio ignores. Positive skew means many small losses and rare large gains. Trend-following returns typically have positive skew. This is a *desirable* property, yet the Sharpe ratio penalizes it, because the large gains inflate the denominator. Excess kurtosis means fat tails in both directions. *Formally:* $\gamma_3 = \mathbb{E}[(X-\mu)^3]/\sigma^3$ and $\gamma_4 = \mathbb{E}[(X-\mu)^4]/\sigma^4$, which are 0 and 3 for a Gaussian. *Used in:* §7.4.2, §7.6.2.

Sortino ratio. *Idea:* the Sharpe ratio with the upside removed from the risk measure, so that large gains do not penalize a strategy. It suits positively skewed strategies. It is noisier than the Sharpe ratio, because it estimates the denominator from the losing observations only. *Formally:* $(\mathbb{E}[R_p] - \tau)/\sqrt{\mathbb{E}[\min(R_p - \tau, 0)^2]}$ for a target τ. *Used in:* §7.4.2.

Drawdown as an extreme-value statistic. *Idea:* the maximum drawdown is a single realized extreme from a single path. It estimates no stable quantity. It grows mechanically with the length of observation, so two identical strategies observed over different spans report different MDDs for purely statistical reasons. Use it for operational limits, never for comparison. *Formally:* $\mathrm{MDD} = \max_t(1 - W_t/\max_{s\le t} W_s)$. For a driftless random walk, the expected maximum drawdown grows like $\sqrt{T}$. With positive drift, it grows like $\log T$. To compare strategies, use the *bootstrapped distribution* of drawdowns. *Used in:* §7.4.3.

Information coefficient. *Idea:* the correlation between predictions and outcomes. It is the cleanest measure of a signal's raw forecasting power. Its typical size in liquid markets is 0.02 to 0.05. Every practitioner should know this number, because it turns the intuition of "mostly noise" into a workable design target. *Formally:* $\mathrm{IC}_t = \mathrm{Corr}_i(s_{i,t}, r_{i,t+1:t+h})$, computed across assets each period. Prefer the Spearman (rank) version for fat-tailed data. The **IC-IR** is $\overline{\mathrm{IC}}/\operatorname{sd}(\mathrm{IC}_t)$, and its t-statistic is $\text{IC-IR}\sqrt{T}$. *Used in:* §7.3.1.

Fundamental Law of Active Management. *Idea:* skill and breadth substitute for each other. A weak signal applied independently to many assets can produce the same information ratio as a strong signal applied to a few. The key caveat is that the bets must be independent, since correlated bets do not count separately. The refinement adds a third term, the fraction of the theoretical edge that survives constraints and costs. *Formally:* IR $\approx$ IC$\sqrt{\text{breadth}}$ (Grinold, 1989), refined to IR $\approx$ TC $\cdot$ IC $\cdot$ $\sqrt{\text{breadth}}$ (Clarke, de Silva & Thorley, 2002). The **transfer coefficient** TC is the correlation between the unconstrained optimal portfolio and the one actually held. Typical realized TC is 0.3–0.6, so implementation, not prediction, loses most theoretical alpha. *Used in:* §7.3.1.

Factor models, alpha and beta. *Idea:* decompose a return into the part that exposure to known common risks explains and the part it does not. Alpha is always defined *relative to a specified model*. So returns alone cannot resolve the momentum debate (§1.3). For the same reason, a researcher must regress any new momentum signal against UMD before claiming that it is novel. *Formally:* $R_{p,t} - R_{f,t} = \alpha + \sum_k \beta_k F_{k,t} + \varepsilon_t$. The standard factors are **MKT** (market excess return), **SMB** (small minus big), **HML** (high minus low book-to-market) and **UMD/WML** (winners minus losers, the momentum factor of Carhart, 1997). The Fama–French five-factor model adds RMW and CMA. *Used in:* §4.6.4, §7.4.4.

Conditional beta. *Idea:* an exposure that changes with the state of the world. A strategy can have an average market beta of zero while it is reliably long the market in calm periods and sharply short it in panics. This is momentum's actual risk profile. An unconditional regression reports it as "market neutral". *Formally:* $\beta_t = \beta_0 + \beta_1 \mathbb{1}\{\text{bear}\} + \beta_2 \widehat{\sigma}_{m,t} + \ldots$, estimated with interaction terms or with subsamples by state. Daniel & Moskowitz (2016) show that momentum's conditional beta turns sharply negative after market declines. *Used in:* §1.6, §7.4.4.

Convexity, straddles and the lookback straddle. *Idea:* a payoff is convex when large moves in *either* direction help. Option positions have convex payoffs, and empirically so do trend-following returns. The reason is mechanical. A trend follower increases exposure as a move extends, which replicates the delta profile of a long option. The best available model of a trend program's payoff is a portfolio of options that pay the largest move within a period. *Formally:* a **straddle** is a long call plus a long put at the same strike, paying $|S_T - K|$. A **lookback straddle** pays $\max_{t \le T} S_t - \min_{t \le T} S_t$, the full range. Fung & Hsieh (2001) show that portfolios of lookback straddles replicate CTA returns well. Dao et al. (2017) derive the same convexity from the variance difference $\text{VR}(q) - 1$. *Used in:* §1.2, §2.3, §5.5.

Aim portfolio and no-trade bands. *Idea:* with trading costs, the optimal portfolio differs from the one the signal implies today. It is a partial step from the current portfolio toward a weighted average of the portfolios that the signal implies now and later. Under proportional costs, the optimum is a no-trade region instead. Do not trade until the drift from target is large enough to justify the cost. *Formally:* under quadratic costs, Gârleanu & Pedersen (2013) show that the optimal policy is $w_t = (1-\theta) w_{t-1} + \theta\, \text{aim}_t$. Here aim_t is a discounted average of expected future optimal portfolios, and the ratio of cost to risk sets θ. Under proportional costs, the optimum is a band. Trade only to the nearest edge of a no-trade region around the target. *Used in:* §6.4, §6.8.

A.8 Market microstructure and trading costs

Limit order book. *Idea:* the market is a queue of resting buy and sell orders at each price. A trade happens when an incoming order crosses the spread and consumes resting liquidity. The order moves the price by an amount that depends on how much liquidity it consumes. Order flow moves prices. This is the mechanical source of momentum at short horizons. *Formally:* the book is a set of (price, quantity) pairs on each side. The **spread** is the best ask minus the best bid. **Depth** is the quantity available at or near the top. The **mid** is the average of the best bid and ask.

Metaorder and child orders. *Idea:* an institution that wants to buy a million shares cannot buy them at once without paying a very high price. So it splits the parent decision into hundreds of small child orders, executed over hours or days. This creates persistent, one-directional pressure on price for as long as the execution lasts. This is a mechanical source of return autocorrelation, and it requires no psychology. *Formally:* an institution executes a metaorder of total size Q over a period at a **participation rate** $\phi = Q/(V \cdot \text{duration})$, where V is the average daily volume (**ADV**). Institutional practice caps ϕ at 5–20%. *Used in:* §1.3, §2.6, §6.8.

Kyle's lambda. *Idea:* the first formal model in which price impact is the *mechanism of price discovery* rather than a friction. A market maker who cannot tell informed from uninformed flow must move the price in proportion to net order flow, because flow is evidence about value. Impact therefore carries information about value. *Formally:* in Kyle (1985), the equilibrium pricing rule is linear, $\Delta p = \lambda\, \Omega$. Here Ω is net order flow, and λ measures illiquidity, the inverse of market depth. *Used in:* §2.6.

Adverse selection (Glosten–Milgrom). *Idea:* a market maker loses to informed traders and must recoup those losses from uninformed ones. The bid-ask spread is exactly that compensation. So spreads widen when information asymmetry rises,

and the spread is not an arbitrary fee. *Formally:* the bid and ask are conditional expectations of value given the direction of the incoming order: ask $= \mathbb{E}[V \mid \text{buy}]$ and bid $= \mathbb{E}[V \mid \text{sell}]$. The spread is the resulting gap. *Used in:* §2.6.

Square-root law of market impact. *Idea:* the most robust quantitative regularity in trading is that impact grows with the *square root* of size, not linearly. Trading four times as much costs twice as much per share. This concavity makes large-scale systematic trading possible. It turns a strategy's capacity from an opinion into a calculation. *Formally:*

$$\Delta p \approx Y\sigma\sqrt{Q/V}, \qquad Y \approx 0.5\text{–}1$$

with Q and V in matching units, σ the daily return volatility, and Δp a relative price move. The leading explanation is **latent liquidity**. Most trading intentions never appear in the book. The density of latent orders near the current price vanishes linearly, and integrating that density gives a square root. *Used in:* §1.3, §2.6, §6.8.

Capacity. *Idea:* capacity is the size at which a strategy's own impact consumes its entire edge. Impact is concave and edge is linear in size, so the two cross at a well-defined point. That point depends on the *square* of the ratio of edge to volatility. So a modestly better signal supports a disproportionately larger size. *Formally:* setting $Y\sigma\sqrt{Q/V} = \alpha$ gives $Q^* \approx V(\alpha/Y\sigma)^2$. *Used in:* §6.8.

Order-flow long memory and the propagator model. *Idea:* order flow is predictable but prices are not, and the propagator model resolves this paradox. The signs of successive market orders are strongly and persistently autocorrelated, out to thousands of trades. Order splitting causes this autocorrelation. Yet prices remain nearly unpredictable. Each trade's impact *decays* over time in exactly the way that cancels the predictability of the flow. Liquidity providers thus enforce efficiency. The practical lesson is that **predictable flow does not imply predictable price**. *Formally:* the trade signs ϵ_t have $\text{Corr}(\epsilon_t, \epsilon_{t-k}) \sim k^{-\alpha}$ with $\alpha < 1$, which is long memory, with a Hurst exponent of about 0.6–0.8. The propagator model writes

$$p_t = \sum_{k<t} G(t-k)\,\epsilon_k f(v_k) + \text{noise}$$

with a decaying kernel G that offsets the flow autocorrelation, so that prices remain close to a martingale (Bouchaud, Gefen, Potters & Wyart, 2004). *Used in:* §1.3, §2.6.

Order flow imbalance. *Idea:* at horizons under a minute, the net imbalance between buying and selling pressure at the top of the book explains almost all price changes. The effect called "intraday momentum" is really flow prediction, and it has little in common with the 12-month effect. *Formally:* OFI aggregates signed changes in the sizes of the bid and ask queues over an interval. The regression $\Delta p_t = \beta\,\text{OFI}_t + \varepsilon_t$ achieves a high R^2 at short horizons (Cont, Kukanov & Stoikov, 2014). *Used in:* §2.6.

Cost decomposition. *Idea:* the total cost of trading has three parts. The first is an unavoidable immediate component, crossing the spread. The second depends on size, which is impact. The third is explicit charges. Only impact is under the trader's control, through sizing and patience, and only impact binds at scale. *Formally:* cost per unit of notional $= \frac{1}{2}\text{spread} + Y\sigma\sqrt{Q/V} + \text{fees, borrow, financing.}$ **Implementation shortfall** measures the whole cost empirically, as the gap between the price at the decision and the average realized fill. *Used in:* §6.8.

A.9 Asset pricing and behavioral concepts

The three forms of the Efficient Market Hypothesis. *Idea:* a graded claim about which information is already in the price. The weak form says prices reflect past prices. The semi-strong form says prices reflect all public information. The strong form says prices also reflect private information. Momentum uses past prices alone, so it challenges the weak form. That is why Jegadeesh & Titman (1993) was so consequential. *Formally:* prices reflect an information set $\mathcal{I}$ if $\mathbb{E}[r_{t+1} \mid \mathcal{I}_t]$ equals the equilibrium expected return. $\mathcal{I}$ is past prices for the weak form, public information for the semi-strong form, and all information for the strong form. *Used in:* §1.3, §2.2.

The joint hypothesis problem. *Idea:* no test can examine market efficiency on its own. Efficiency says prices are right *given* the correct model of what returns should be. So any rejection applies to efficiency and the model jointly. Returns alone can never show whether the market or the risk model was wrong. The thirty-year momentum debate is an instance of this problem. *Formally:* a test of $\mathbb{E}[r_{t+1} \mid \mathcal{I}_t] = f(\text{risk}; \theta)$ jointly tests efficiency and the specification f (Fama, 1970). *Used in:* §1.3, §2.2.

Grossman–Stiglitz. *Idea:* perfectly efficient prices cannot exist in equilibrium. If prices already revealed everything, nobody would pay to gather information, and then prices could not reveal anything. So equilibrium requires prices to be slightly

inefficient, by just enough to pay for the research that keeps them nearly efficient. Momentum exists in that gap. The claim that "the anomaly should have been arbitraged away" is therefore not an argument on its own. *Formally:* in Grossman & Stiglitz (1980), no informationally efficient equilibrium exists when information is costly. The cost of acquiring information sets the equilibrium level of noise. *Used in:* §1.3.

Risk premium vs. alpha. *Idea:* two economically opposite explanations of the same positive average return. A risk premium compensates for bearing an exposure that hurts at the worst times. It should persist, and an investor should not lever it carelessly. Alpha is a mispricing, and it should decay as investors exploit it. Momentum has candidate explanations of both kinds. So do not assume that momentum is all alpha. *Formally:* under a factor model, the premium is $\beta'\mathbb{E}[F]$ and the alpha is the intercept. The distinction depends entirely on which factors are in the model. *Used in:* §1.3, §8.3.

Post-earnings-announcement drift. *Idea:* this drift is the cleanest natural experiment in under-reaction. After an earnings surprise, which is public, dated and unambiguous news, prices keep drifting in the direction of the surprise for weeks. So prices demonstrably do not absorb information instantly. *Formally:* portfolios sorted on standardized unexpected earnings earn significant abnormal returns over the following 60 days (Bernard & Thomas, 1989, 1990). *Used in:* §1.3.

Prospect theory, mental accounting and the disposition effect. *Idea:* investors evaluate outcomes as gains and losses relative to a reference point, rather than as levels of wealth. They are risk-averse over gains and risk-seeking over losses, and they track each position in a separate mental account. The behavioral consequence is the **disposition effect**, selling winners too early and holding losers too long. The market consequence is momentum. Selling pressure above the aggregate cost basis slows the price's adjustment to good news. This behavioral explanation has the strongest independent confirmation. It predicts something *other than returns*, the relation to unrealized capital gains, and that prediction holds. *Formally:* the value function $v(x)$ is concave for $x > 0$, convex for $x < 0$, and steeper for losses. Grinblatt & Han (2005) build a turnover-weighted reference price and show that it subsumes much of momentum. *Used in:* §1.3, §2.5.

Conservatism and representativeness. *Idea:* two opposed biases of judgement combine to generate momentum followed by reversal. Conservatism means updating too little on each new piece of evidence. It causes under-reaction at a short horizon. Representativeness means treating a short run of similar outcomes as evidence of a pattern. It causes over-extrapolation at a longer horizon. Barberis, Shleifer & Vishny (1998) show that a regime-switching model of beliefs with both biases produces the observed impulse response. *Used in:* §1.3.

Overconfidence and biased self-attribution. *Idea:* investors overweight their own private analysis. They treat confirming public news as proof of their skill and dismiss disconfirming news as noise. Confidence therefore *rises* on confirmation and barely falls on contradiction. This drives continued over-reaction before the eventual correction (Daniel, Hirshleifer & Subrahmanyam, 1998). *Used in:* §1.3.

Anchoring. *Idea:* people judge a value by adjusting from a salient reference number, and they adjust too little. The 52-week high is such a reference. Near it, investors are reluctant to bid higher regardless of the news. So prices absorb good news slowly and drift through the anchor over time. This implies a specific prediction, which George & Hwang (2004) confirmed. Proximity to the 52-week high predicts returns without using any return path. *Used in:* §4.4.3.

Limits to arbitrage and the flows of delegated management. *Idea:* correcting even an obvious mispricing requires capital that is willing to bear interim losses. Delegated managers face redemptions exactly when their positions move against them. So capital flows *out* of underperforming strategies and *into* outperforming ones, which amplifies price moves instead of damping them. This produces momentum from fully rational agents who face an agency friction. *Formally:* Vayanos & Woolley (2013) model fund flows that respond to past performance. The model generates momentum at short horizons and reversal at long ones. *Used in:* §1.3, §8.6.

Instrumented PCA (IPCA). *Idea:* instead of assuming that factor loadings are constant, let each asset's exposures be functions of its observable characteristics. A stock that has recently risen may thereby *become* a higher-beta stock. If past returns predict future betas, then part of what looks like momentum alpha compensates a risk exposure that the characteristic already signaled. This is currently the most serious rational challenge to momentum. *Formally:* $r_{i,t+1} = \beta(z_{i,t})' f_{t+1} + \varepsilon_{i,t+1}$, with $\beta(z) = \Gamma' z$ estimated jointly with the latent factors f (Kelly, Moskowitz & Pruitt, 2021). *Used in:* §1.3, §8.6.

Point-in-time data, survivorship and delisting returns. *Idea:* researchers who tidy a database usually delete the past. A universe that contains only companies that still exist omits exactly the names that momentum's short leg would have held. Prices adjusted with today's split factors let a price-level filter use future information. Reconstructing what was *knowable at the time* is tedious work, and most silent backtest failures occur here. *Formally:* a point-in-time database stores each fact with the date it became known, so a query "as of t" returns the vintage available at t. **Delisting returns** record the terminal value received when a security stops trading, which is often large and negative. Omitting them biases every long-history study upward. *Used in:* §6.10.

Corporate-action adjustment and the futures roll. *Idea:* a price series is a construction, not a natural object. Splits and dividends make the raw series discontinuous. Analysts must splice expiring futures contracts into a continuous series that does not exist in the market. Both operations can create fake momentum. *Formally:* **back-adjustment** subtracts each roll

gap from all prior history. It preserves differences but destroys price levels, and over long histories it can produce negative prices. **Ratio adjustment** multiplies by the ratio at each gap, which preserves positive prices and percentage returns. It is generally preferable. Either way, **back-adjusted price levels are not real prices**, so any measure based on levels, such as channel position, the 52-week high or price filters, needs care. *Used in:* §6.5, §6.10.

A.10 Machine learning in a low-signal setting

Bias–variance trade-off. *Idea:* prediction error decomposes into two parts. One comes from a model too rigid to capture the truth. The other comes from a model so flexible that it fits the noise. In finance the signal is so small that the second part dominates almost immediately. So heavy regularization and few features usually beat complex models. *Formally:* $\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2_{\text{irreducible}}$.

Regularization: ridge and lasso. *Idea:* penalize the size of the fitted coefficients, so the model cannot fit noise. Ridge shrinks all coefficients smoothly toward zero. It is the right default when predictors are correlated, as momentum features always are. Lasso can set coefficients exactly to zero, which selects features. That is attractive, but the selection is unstable when predictors are collinear. *Formally:* ridge minimizes $\|y - X\beta\|^2 + \eta\|\beta\|_2^2$, with closed form $\hat{\beta} = (X'X + \eta I)^{-1}X'y$. Lasso uses $\eta\|\beta\|_1$ instead. *Used in:* §4.9, §2.7.

Double descent and the "virtue of complexity". *Idea:* in the classical view, test error falls and then rises as the model gains parameters. The modern observation is that going *past* the point where the model fits the training data exactly can make test error fall again. Among the many perfect fits, the regularizer selects a well-behaved one. Kelly, Malamud & Zhou (2024) argue that this holds in return prediction. That contradicts decades of orthodoxy that favored parsimony. The claim is **[Contested]**, and it is consequential if true. *Formally:* test error, as a function of the ratio of parameters to samples P/T, peaks near the interpolation threshold $P/T = 1$. Under ridge regularization, it can decline for $P/T \gg 1$. *Used in:* §2.7, §4.9.

Tree ensembles. *Idea:* a single decision tree is a set of nested if-then splits. It has high variance, but it captures interactions and nonlinearity automatically. Averaging many decorrelated trees, a random forest, reduces the variance. Fitting trees in sequence to the errors of the previous ensemble, gradient boosting, reduces the bias. Boosted trees on a modest set of well-motivated features are a standard model in applied return prediction. *Formally:* a random forest averages trees grown on bootstrap samples with random subsets of features. Gradient boosting fits $F_m = F_{m-1} + \nu h_m$, where h_m approximates the negative gradient of the loss at F_{m-1} and ν is a learning rate. *Used in:* §4.9.

Sequence models: LSTM, attention and Transformers. *Idea:* instead of choosing the kernel that weights past returns, let the model learn it. A recurrent network carries a hidden state forward through time, with learned gates that control what the state retains. An attention mechanism instead computes, for each output, a set of learned weights over all input positions. That is a *data-dependent* kernel, the natural generalization of everything in §4.1.5. The cost is a very large number of parameters relative to a very small effective sample. *Formally:* attention computes $\text{softmax}(QK'/\sqrt{d})V$ for learned query, key and value projections (Lim, Zohren & Roberts, 2019; Wood et al., 2021). *Used in:* §4.9.

Triple-barrier labeling and meta-labeling. *Idea:* the usual label, the return over the next h days, does not match what any trader does. A real position ends when it hits a profit target, a stop or a time limit, whichever comes first. So the label should record which one. **Meta-labeling** then splits the problem in two. A primary model decides the direction, and a secondary model decides whether to act and at what size. This lets the analyst optimize sizing separately from prediction, which is a useful separation. *Formally:* label $y_t \in \{+1, -1, 0\}$ by the barrier touched first. The barriers are the upper barrier ($+a\hat{\sigma}_t$), the lower barrier ($-b\hat{\sigma}_t$), and the vertical barrier, a time limit (López de Prado, 2018). *Used in:* §4.9, §7.3.4.

Sample uniqueness and effective sample size. *Idea:* with overlapping labels and hundreds of correlated assets, the number of rows in a training set greatly overstates the independent information it holds. This overstatement, not the choice of model, is the binding constraint on machine learning in finance. No architecture fixes it. *Formally:* the uniqueness of observation i is the average, over the bars its label spans, of the reciprocal of the number of labels covering each bar. Weighting samples by uniqueness, or bootstrapping by blocks of dates, restores approximately correct inference. *Used in:* §4.9.

Path signatures. *Idea:* the path signature is a principled basis for functions of a *path*, rather than of its endpoints. It collects iterated integrals along the path. Its low-order terms recover displacement, and the next terms capture the *order in which* moves happened. It is the natural formal answer to the weakness of §4.1.1, blindness to the path. Practitioners underuse it relative to its elegance. *Formally:* for a path $X : [0, T] \to \mathbb{R}^d$, the signature is the collection $S(X)^{i_1 \cdots i_k} = \int_{0<t_1<\cdots<t_k<T} dX^{i_1}_{t_1} \cdots dX^{i_k}_{t_k}$, truncated at order k. Its dimension grows like d^k (Lyons; Levin, Lyons & Ni, 2013). *Used in:* §4.9.

Fractional differentiation. *Idea:* differencing a price series to make it stationary destroys almost all of its memory, leaving returns, which have almost no memory. Fractional differencing takes a *non-integer* difference. It removes just enough non-stationarity to pass a statistical test while keeping as much memory as possible. *Formally:* Expand $(1-B)^d$ as a binomial series with $d \in (0,1)$, apply it to log prices, and truncate at a weight threshold. Choose the smallest d that passes an ADF test (López de Prado, 2018). *Used in:* §3.2.

A.11 Glossary of symbols

The table lists every recurring symbol, with the sections where it is used.

Symbol	Meaning	Where
$P_t,\ p_t$	Price; log price, $p_t = \ln P_t$	throughout
$R_t,\ r_t$	Simple return; log return, $r_t = p_t - p_{t-1}$	throughout
$r_{a:b}$	Cumulative log return over $(a, b]$, $= p_b - p_a$	throughout
$\text{Hi}_t,\ \text{Lo}_t$	Bar high; bar low	§4.3.1, §4.4
$\mu,\ \sigma$	Unconditional mean return; volatility	throughout
$\hat{\sigma}_t$	Estimate at t of per-bar return volatility	§4.3, §6.7
σ^*	Volatility target	§4.6.1, §6.7
$\gamma_k,\ \rho_k$	Lag-k autocovariance; autocorrelation	§1.2, A.1
δ_t	Innovation to fundamental value	§1.2
$\Psi_j,\ \psi_j$	Cumulative; per-period impulse response	§1.2, §9.1
$\mathcal{F}_t$	Information set available at t	throughout
$L,\ S,\ H$	Lookback; skip; holding period, in bars	§4.1.1
q	Aggregation horizon of a variance ratio	§1.2, §4.5.1
$N,\ T$	Number of assets; number of bars	throughout
A	Bars per year (252 daily, 12 monthly)	§4.3.2, §7.4.1
$\text{VR}(q)$	Variance ratio at horizon q	§1.2, §4.5.1
$f_r(\omega)$	Spectral density of returns	§4.8
$H(\omega)$	Filter transfer function	§4.8
h_k	Filter kernel / impulse response weights	§4.1.5, §5.2
$b_{i,t}$	Benchmark subtracted before measuring	§5.2, §5.4
$\mathcal{N}_{i,t}$	Normalizer	§5.2
$g(\cdot)$	Signal-to-position transform	§5.2
$s_{i,t},\ w_{i,t}$	Signal; portfolio weight	throughout
$\lambda,\ \alpha$	EWMA decay; smoothing constant, $\alpha = 1 - \lambda$	§4.1.4
$n_f,\ n_s$	Fast; slow moving-average span	§4.1.5
$\hat{b}_t$	Rolling regression slope	§4.2.1
$\hat{\sigma}_\varepsilon$	Regression residual standard deviation	§4.2.2
Γ	Lag-1 cross-autocovariance matrix	§4.6.2
σ_μ^2	Cross-sectional dispersion of mean returns	§4.6.2
$x_t,\ F,\ H,\ K_t,\ P_{t\|t}$	State; transition; observation matrix; Kalman gain; state covariance	§4.7.1, A.5
$S_t,\ \Pi$	Latent regime; transition matrix	§4.7.2
IC, IR, TC	Information coefficient; information ratio; transfer coefficient	§7.3.1
SR, DSR	Sharpe ratio; Deflated Sharpe Ratio	§7.4.1, §7.6.2
M	Number of strategy configurations tried	§7.6
$Z(\cdot),\ Z^{-1}(\cdot)$	Standard normal CDF; its inverse	§7.6.2
$\gamma_3,\ \gamma_4$	Skewness; kurtosis	§7.6.2
$Q,\ V,\ Y$	Order size; daily volume; impact coefficient	§1.3, §6.8
Q^*	Capacity	§6.8
$\mathbb{1}\{\cdot\}$	Indicator function	throughout

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